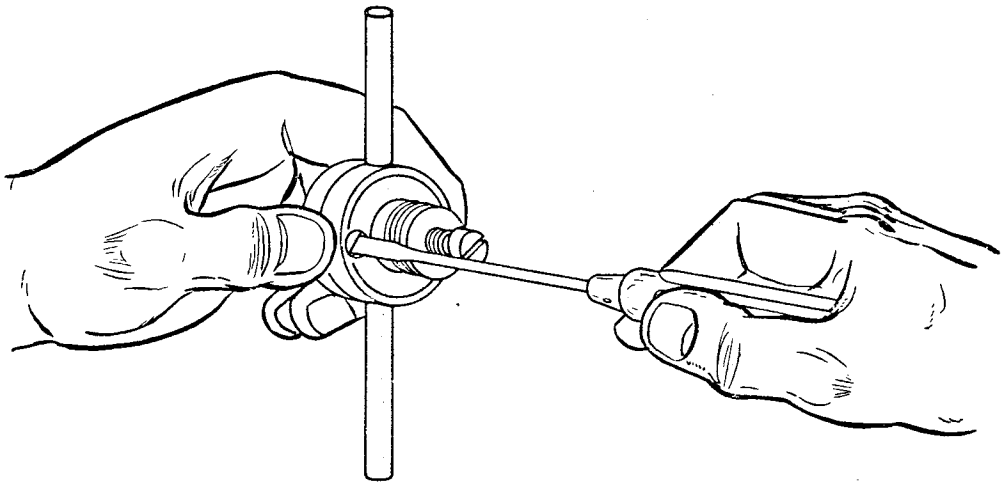
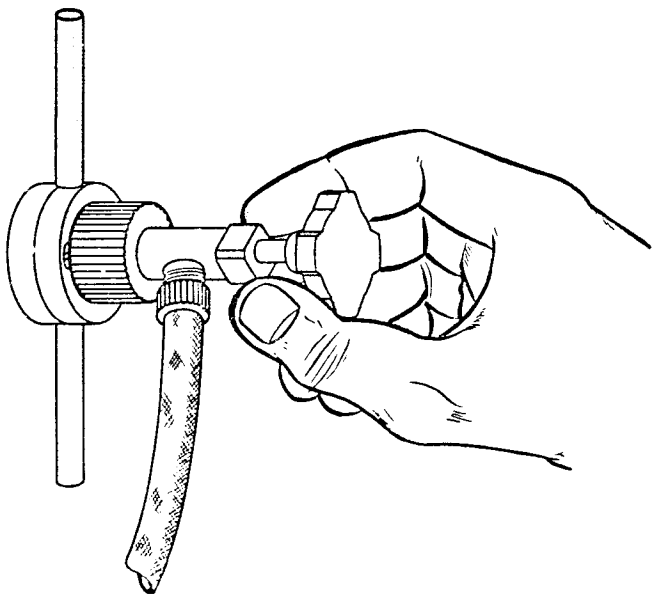
TECHNICIAN'S HANDBOOK FOR
COMPRESSION REFRIGERATORS

PART A

SERVICING AND REPAIR TECHNIQUES





This publication is one of a series of fifteen booklets that have been produced by the Expanded Programme on Immunization. They all aim to improve the standards of refrigerator and cold room maintenance and repair.

The titles of the booklets are:

- User's handbook for compression refrigerators.
- How to look after a compression refrigerator.

- User's handbook for kerosene and electric refrigerators.
- How to look after a kerosene refrigerator.

- User's handbook for gas and electric refrigerators.
- How to look after a gas refrigerator.

- User's handbook for vaccine cold stores.
- How to look after a cold store.

Each pair of booklets is a one day course for people who use these types of refrigerators or cold rooms.

Technician's handbook for compression refrigerators:

- Part A - Servicing and repair techniques.
- Part B - Faults and fault finding.
- Part C - Repair work.
- Part D - How to keep stocks of spare parts.
- Part E - Task sheets and progress tests.
- Part F - Instructor's handbook.
- Part G - Manufacturer's spare parts lists.

The last seven titles are course materials for a 10 working day course for repair technicians.

It is hoped that these books will be copied, translated, reprinted and used to improve the standards of vaccine care.

EXPANDED PROGRAMME ON IMMUNIZATION, Geneva.
Revised September 1985.

PART A

SERVICING AND REPAIR TECHNIQUES

Introduction

This course comprises 7 modules:

Part A	describes servicing and repair techniques.
Part B	explains the faults that occur and how to find the correct fault.
Part C	describes how to carry out repair work.
Part D	contains a series of case studies that describe ordering and keeping stocks of spare parts for vaccine refrigerators.
Part E	is a series of practical exercises and progress tests.
Part F	is an instructor's handbook which gives information for the course manager on how to organise and run a course.
Part G	is a collection of manufacturers' spare parts brochures.

Parts A, B and C have been written in the order that you will need the information. You must be able to carry out the techniques in part A to find the faults in part B. You must find the fault correctly in part B before starting repair work in Part C. Part E, "Task Sheets and Progress Tests" gives the order of work for the course, using Parts A, B and C.

For each task sheet always work in the following order:

1. Read the title of the task sheet.
2. Find the tools and materials you need.
3. Read the handbook reference on the task sheet.
4. Read the action described on the task sheet.
5. Ask your instructor to explain anything that you do not understand.
6. When you are quite sure what to do, start work.

Always work in the exact order shown. Follow the instructions and examine the drawings carefully. The course has been carefully prepared so that each stage of the work follows easily from the previous stage. It is therefore important that you fully understand each stage - if you do not understand ask someone to help you. By following the instructions carefully, you will be able to carry out all repair work on compression refrigerators with confidence and success.

Part D (the stocking spare parts booklet) contains its own exercises, and can be used separately from Parts A, B and C.

PART A

SERVICING AND REPAIR TECHNIQUES

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Section 1: Pipe work for refrigerators

This section of the book explains what you need to know about refrigerator pipes. It tells you what sort of pipe to use, and how to bend and join pipes properly.

Make sure that you know how to carry out all of this work before starting to replace pipes fitted to the refrigerator.

Properties of refrigerator pipe

1. All refrigerator work uses a special type of soft, clean copper pipe.

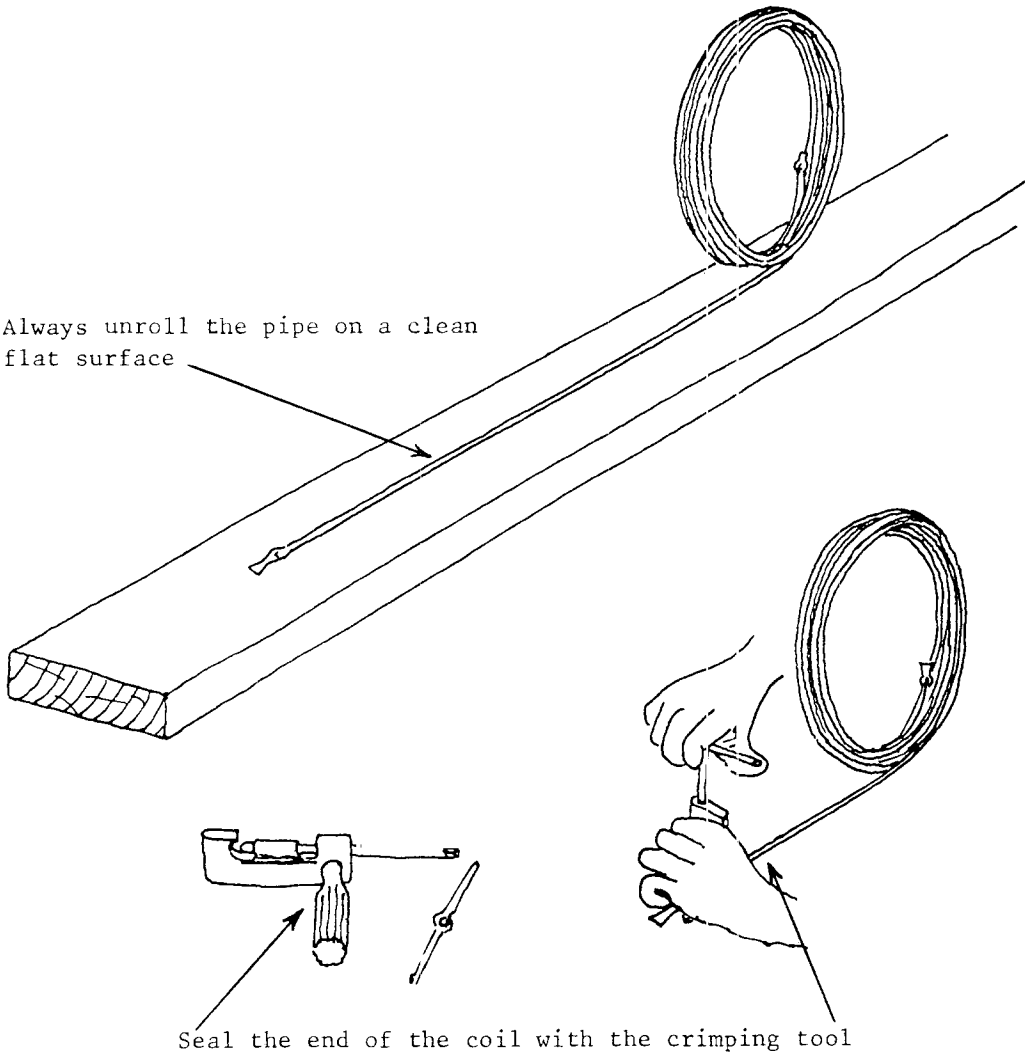
2. The pipe that you must use is supplied in coils or rolls. These are usually 6m (about 20 feet) long.

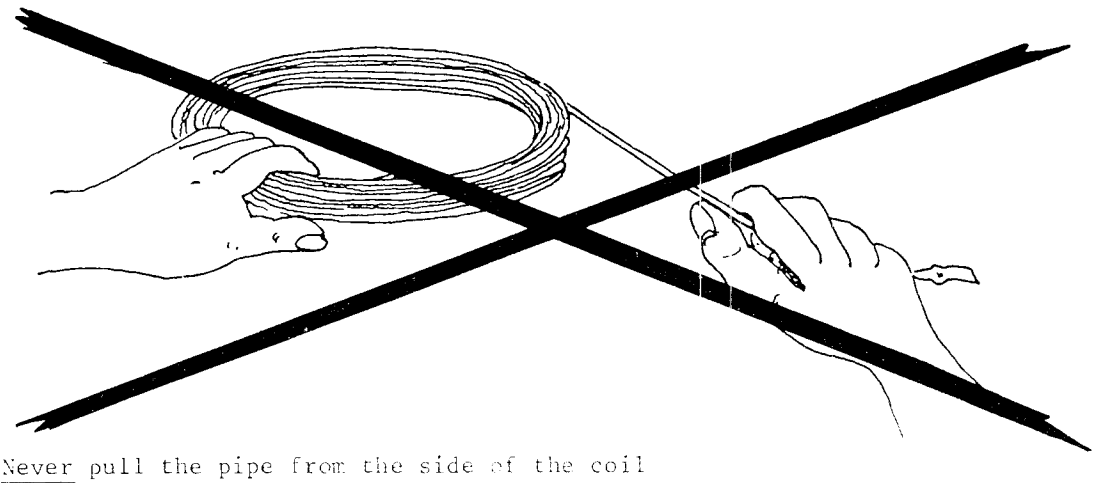
3. All dirt and moisture has been removed from the inside of this pipe. The ends have then been sealed to keep the inside clean.

4. The pipe is soft, so that it can be bent and joined easily. Materials that are soft are easily damaged. Handle the pipe carefully.

5. Refrigerator pipe is always measured by its outside diameter. The usual sizes are 1/4 inch (6mm) and 5/16 inch (8mm).

6. Only use refrigeration quality pipe. Any other pipe will damage the refrigerator cooling system.





Never pull the pipe from the side of the coil

How to handle the pipe

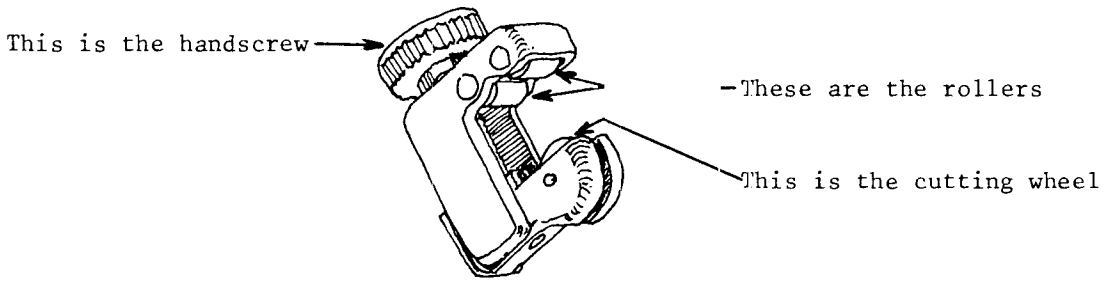
1. To remove a piece of pipe from the coil, unroll the pipe (see drawing).

2. Never pull the pipe sideways from the coil (see drawing). This will damage the pipe.

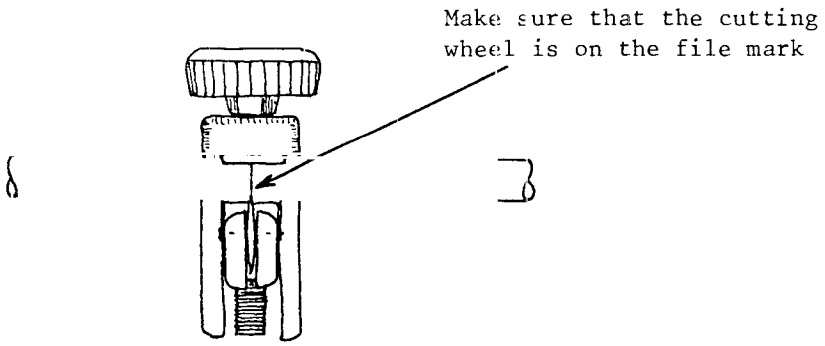
3. Cut off the pipe and then seal the open end (see drawing). This prevents dirt contaminating the rest of the coil.

4. Roll the end of the pipe back into the coil.

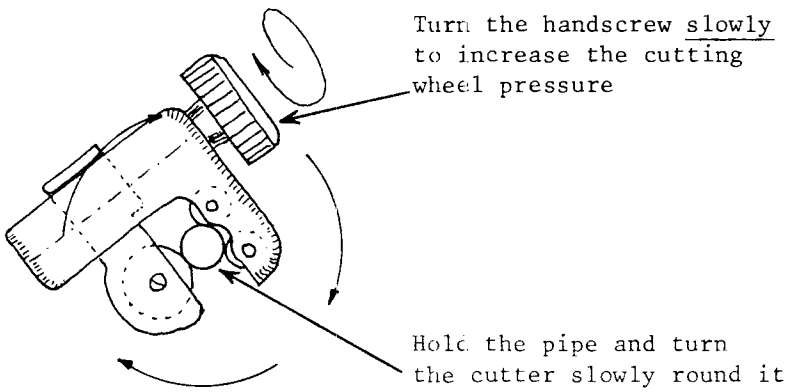
5. Take care not to dent or kink the soft copper pipe.



The pipe cutter



The pipe cutter in position on the pipe



Using the pipe cutter

How to cut the pipe

1. Unroll the length of pipe that you need and an extra 1cm (1/2 inch)(to give enough space for working).

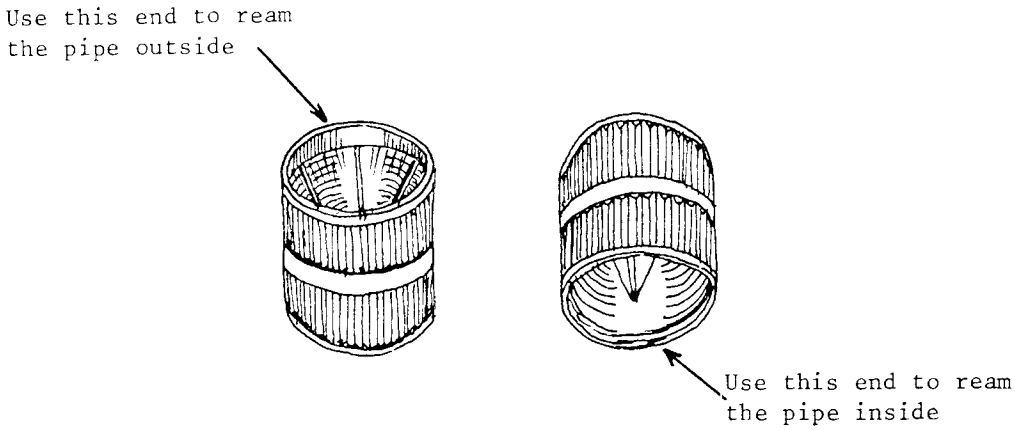
2. Carefully measure the length of pipe needed and mark the outside (where you are going to cut) with the edge of a file.

3. Examine the pipe cutter and compare it with the drawing. Make sure that you understand how it works before starting to cut the pipe.

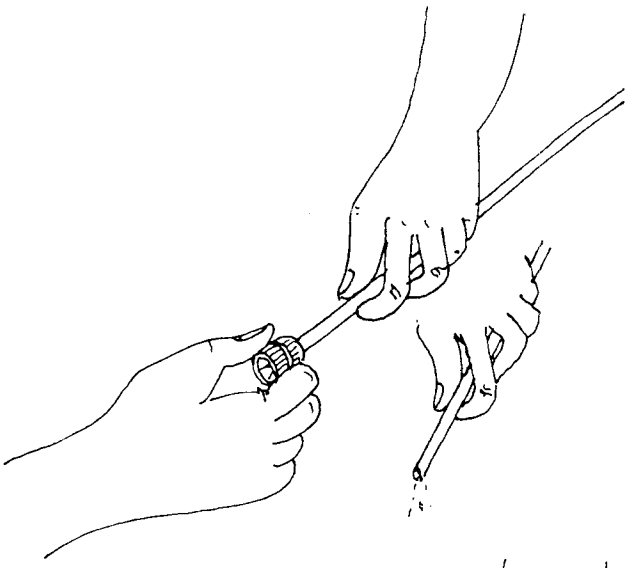
4. Place the pipe between the bottom rollers and the top cutting wheel (see drawing).

5. Position the pipe so that the cutting wheel is in line with the mark that you made with the file.

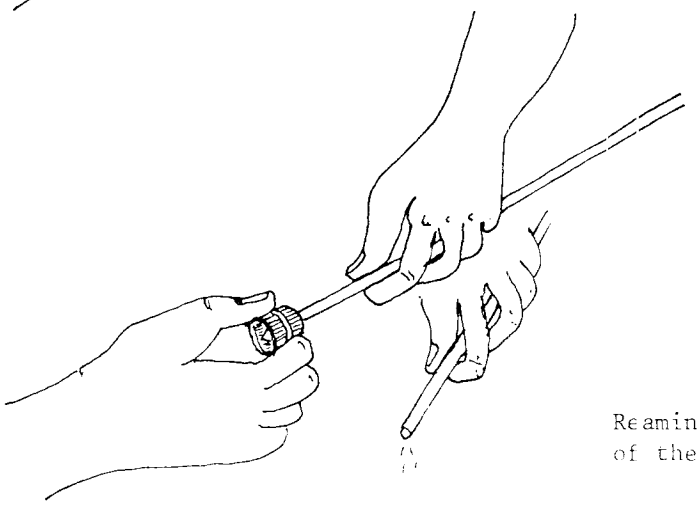
6. Tighten the handscrew until the cutting wheel just touches the outside of the pipe.



The pipe reaming tool



Reaming the inside
of the pipe



Reaming the outside
of the pipe

7. Turn the pipe cutter slowly around the pipe so that the cutting wheel cuts gradually into the outside.

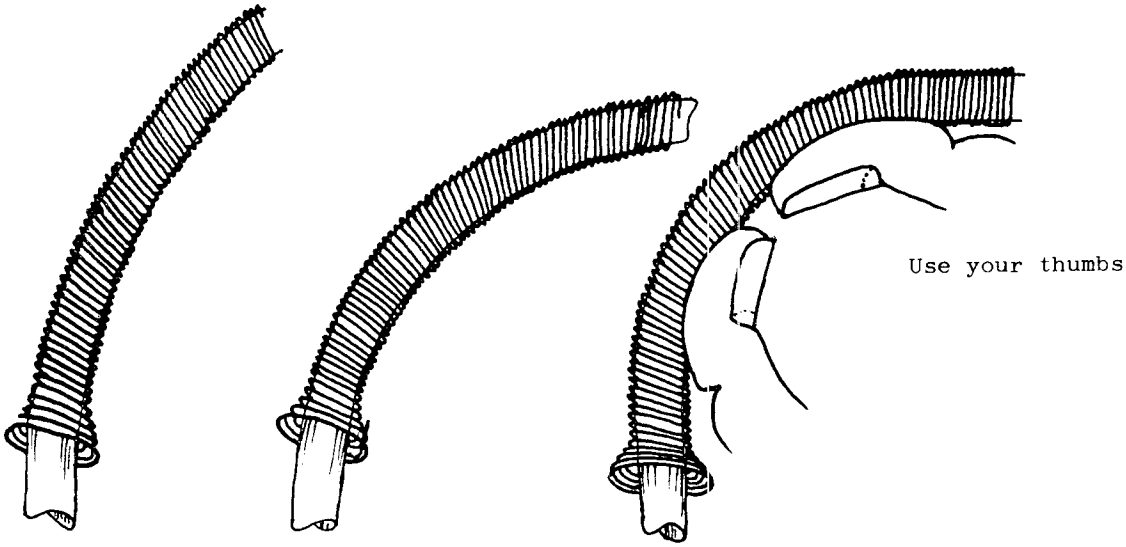
8. Turn the handscrew to increase the pressure of the cutting wheel and then turn the cutter around the pipe again (see drawing).

9. Continue the cutting by gradually increasing the pressure of the cutting wheel. Work slowly and carefully. Do not use too much pressure or you may damage the pipe.

10. When the cut is complete seal the open end and roll the pipe back into a coil (see page 4).

11. To remove all rough edges from the end of the pipe, use the reaming tool (see drawing).

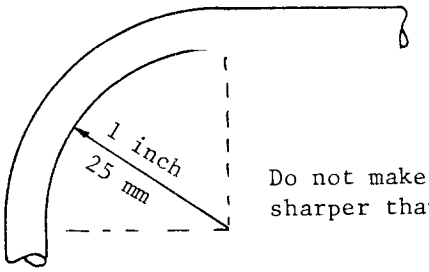
12. Hold the open end of the pipe to be reamed down (so that pieces of copper will not get inside the pipe). Turn the reaming tool (see drawing) until all rough edges have been removed from the inside of the pipe.



1. Bend the pipe a short way

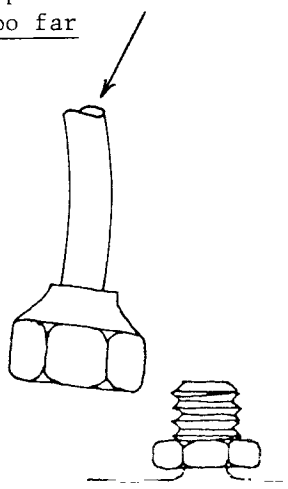
2. Bend the pipe a bit further

3. Continue until you have completed the bend

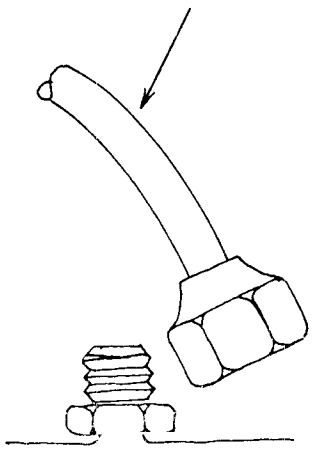


Do not make a bend sharper than this

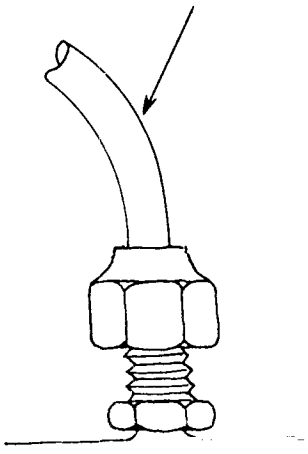
This pipe has been bent <u>too far</u>



This pipe has <u>not</u> been bent <u>far enough</u>



This pipe has been bent properly



How to bend the pipe

1. Refrigerator pipe is soft and can be bent by hand. Because it is soft it can be easily damaged when bending.

2. If the pipe is hard and will not bend easily, it must be annealed before use (how to anneal is explained on page 19).

3. Push a bending spring onto the outside of the pipe (see drawing). This helps the pipe to bend evenly.

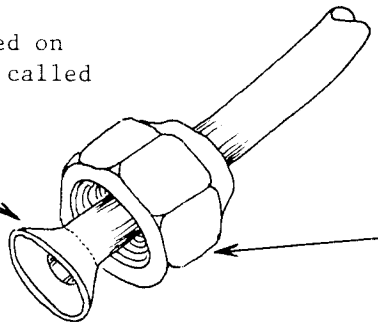
4. Bend the pipe a little at a time, using your thumbs (see drawing). Do not try to complete the bend with one movement. This will cause the pipe to buckle.

5. Do not make a sharp bend in the pipe. If the pipe diameter is 1/4 inch (6 mm), the sharpest bend that you can make without risk of damaging the pipe is 1 inch (2.5mm) radius (see drawing).

6. Pipe that cracks, splits, wrinkles or flattens during bending must <u>not</u> be used.

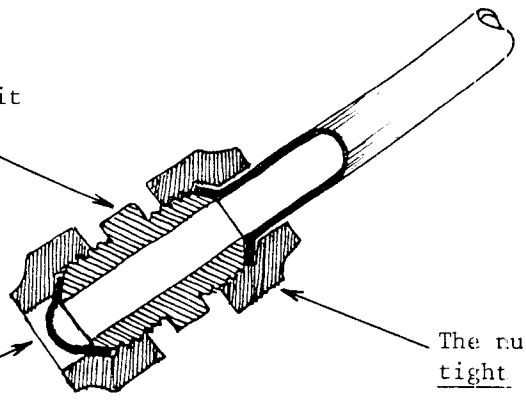
7. Bend the pipe so that it fits onto the connection easily (see drawing).

This cone, that is formed on the end of the pipe, is called a 'flare'.



This is the flare nut. It must be put on the pipe before flaring.

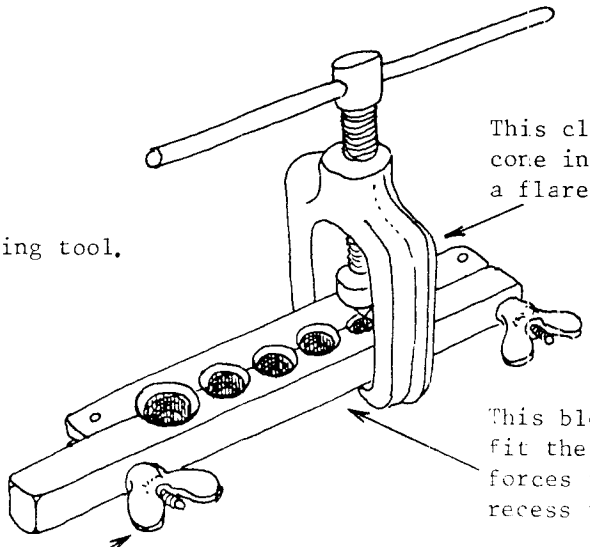
This is a flare fitting (the drawing shows how it would look inside).



The nut holds the flare tight against the fitting.

This is a bonnet - used with a flare nut to seal the end of a fitting.

This is a pipe flaring tool.



This clamp forces a metal cone into the pipe to make a flare.

This block has holes to fit the pipe. The cone forces the pipe into a recess to make the flare.

These hand nuts hold the pipe tight in the block.

How to make a flare on the end of the pipe

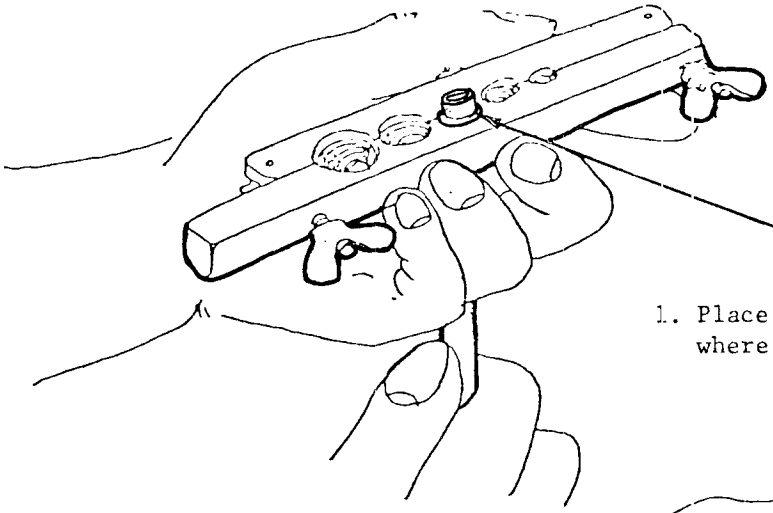
1. Refrigerator pipes are sometimes joined to fittings by making a flared connection.

2. The end of the pipe is opened out to form a cone (see drawing).

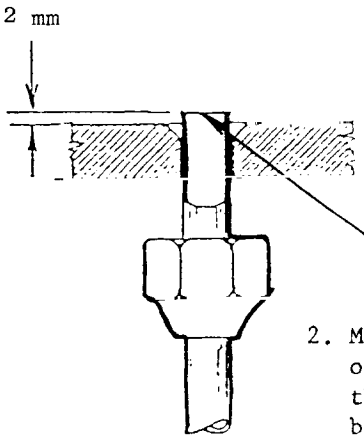
3. Always place the special flare nut on the pipe first before flaring.

4. Examine the pipe flaring tool and compare it with the bottom drawing opposite. Make sure that you understand how it works before starting to flare the end of a pipe.

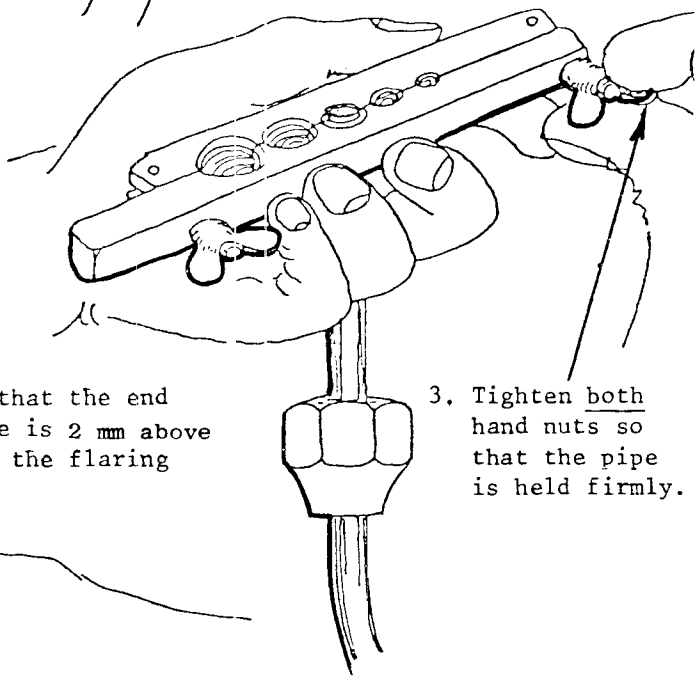
5. Make sure that the end of the pipe is free of rough edges before flaring.



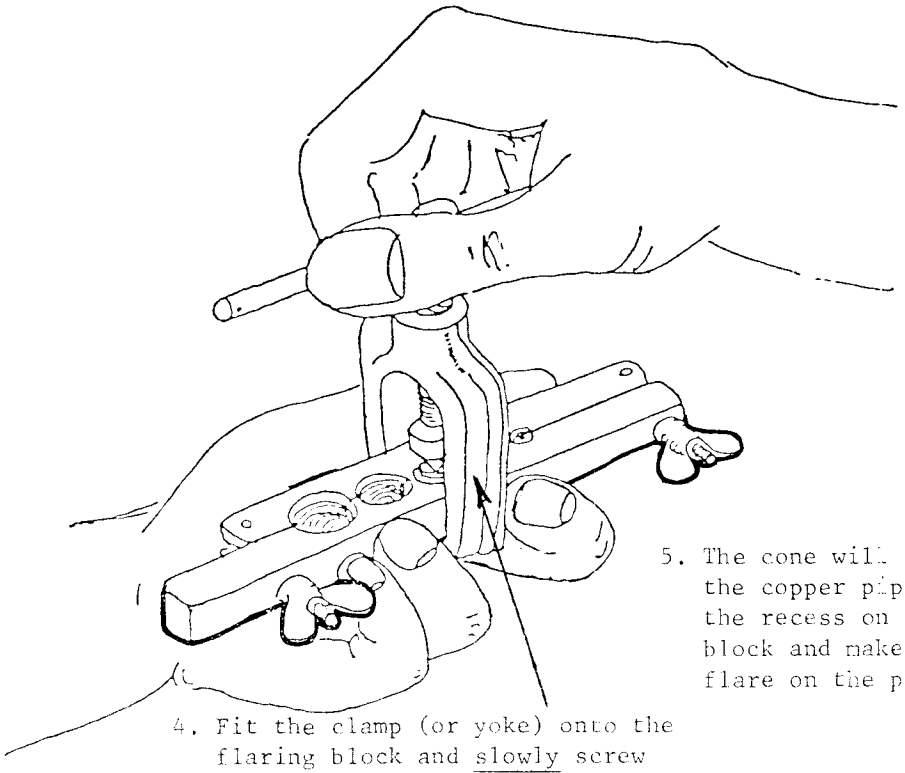
1. Place the pipe into a hole where it fits properly.



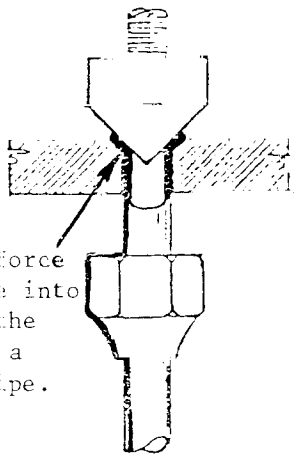
2. Make sure that the end of the pipe is 2 mm above the top of the flaring block.



3. Tighten <u>both</u> hand nuts so that the pipe is held firmly.



4. Fit the clamp (or yoke) onto the flaring block and <u>slowly</u> screw the cone into the end of the pipe.



5. The cone will force the copper pipe into the recess on the block and make a flare on the pipe.

6. Place the pipe in the tool (see drawing 1). Make sure that you have:

   (a) placed the flare nut on the pipe;
   (b) chosen the correct size hole in the flaring tool to fit the pipe; there are 5 holes to fit different sizes of pipe.

7. If the pipe is 1/4 inch (6mm) in diameter, position the pipe so that the end is at least 2mm above the top of the flaring block (see drawing 2). (This distance is calculated as "pipe diameter divided by 3"; in this case, 6mm divided by 3 = 2mm).

8. Tighten the nuts at each end of the flaring block (see drawing 3).

9. Fit the yoke to the flaring block (see drawing 4).

10. Oil the cone and slowly screw it into the end of the pipe.

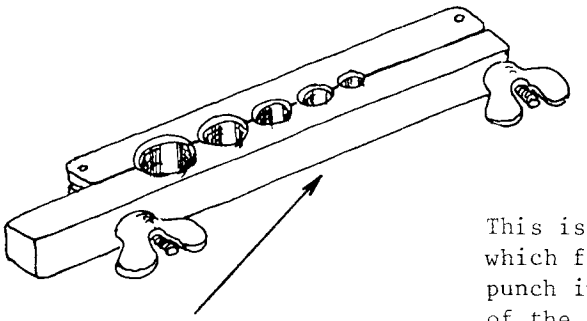
11. The end of the pipe will be formed into a flare (see drawing 5).

12. Remove the flared pipe from the block.

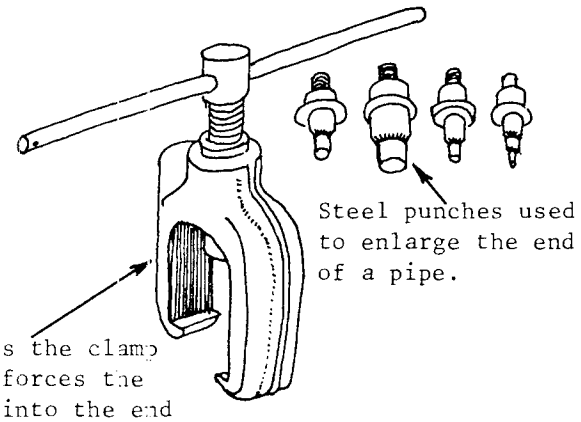
13. Examine the flare. If it has cracked, the cone was screwed down too quickly.

14. Make sure that the flare is the correct size. It should just fit inside the flare nut. If it is too losse, cut off the flare and start again at instruction 5. At instruction 7, use 3 mm instead of 2 mm. Repeat until the flare is the correct size for the flare nut - not too loose and not too tight.

Swaging the end of the pipe.

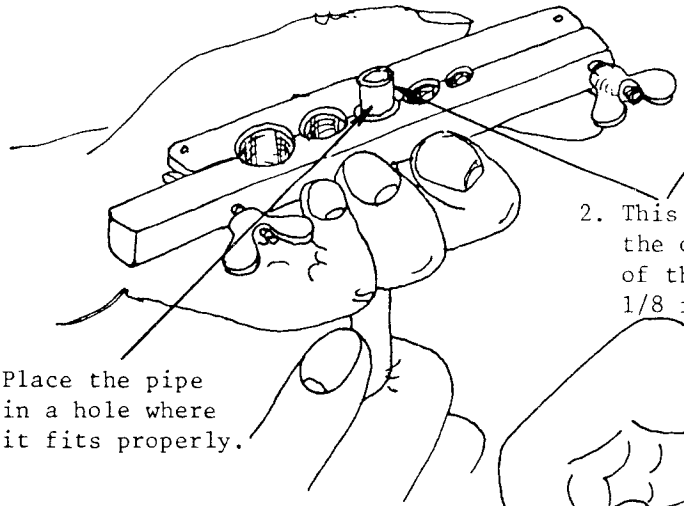


This is the swaging block. It has holes to fit the pipe.

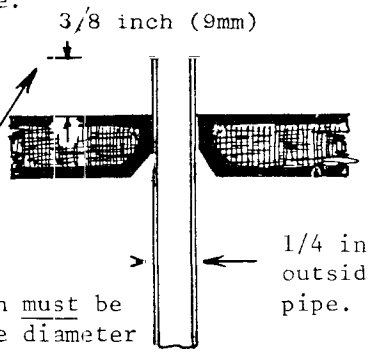


Steel punches used to enlarge the end of a pipe.

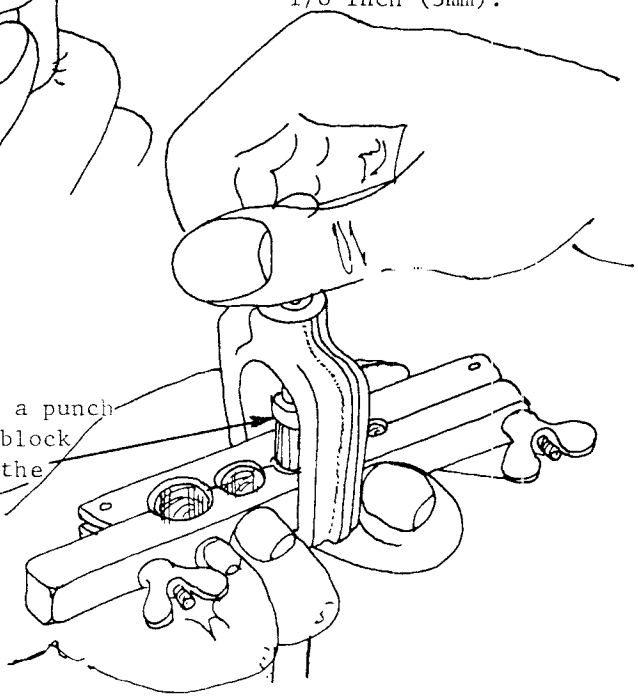
This is the clamp which forces the punch into the end of the tube.



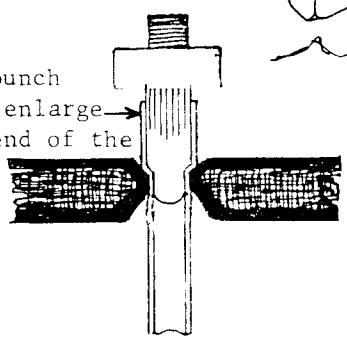
1. Place the pipe in a hole where it fits properly.



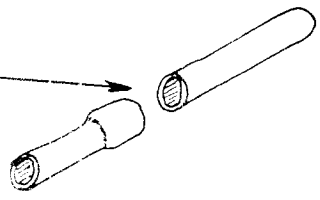
2. This length must be the outside diameter of the pipe plus 1/8 inch (3mm).



3. Fit the clamp and a punch onto the swaging block and slowly force the punch into the end of the pipe.



4. The punch will enlarge the end of the pipe.



5. One pipe will now fit inside the other.

How to swage the end of the pipe

1. To join two pieces of pipe of the same size together one must be made larger so that the other fits inside.

2. Making the end of a pipe larger is called swaging.

3. The tool used for swaging is shown in the drawings opposite. Examine the pipe swaging tool and compare it with the drawings. Make sure that you understand how it works before starting to swage the end of a pipe.

4. Place the pipe in the tool (see drawing 1). Make sure that you have chosen the correct size hole to fit the pipe.

5. Position the pipe so that the end is "the outside diameter of the pipe plus 1/8 inch (3mm)" above the top of the swaging block (see drawing 2). For 1/4 inch (6mm) pipe this will be 1/4 inch (6mm) + 1/8 inch (3mm) = 3/8 inch (9mm), as shown.

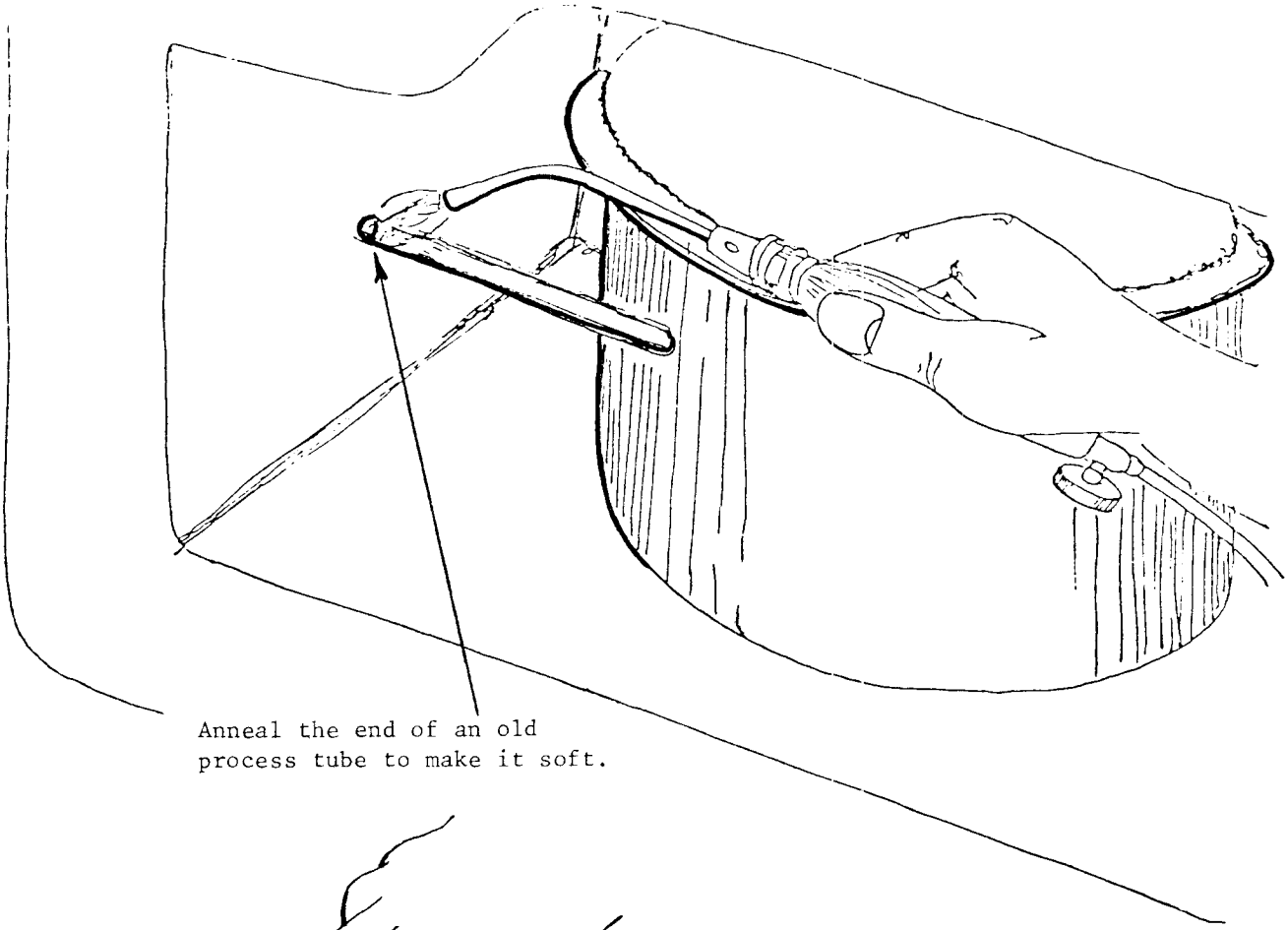
6. Tighten the nuts at each end of the swaging block.

7. Choose a punch of the right size to swage out the end of the pipe so that another piece fits inside. Oil the punch. Fit the clamp and punch onto the swaging block. Force the punch into the pipe (see drawing 3).

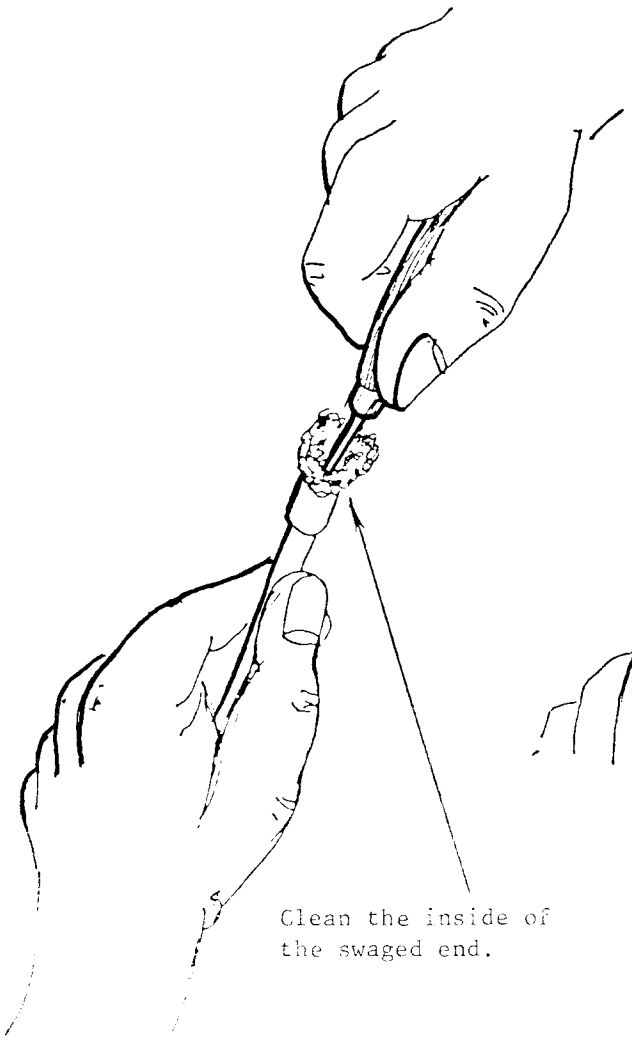
8. The punch will open out the end of the pipe (see drawing 4).

9. Remove the pipe from the swaging block. Another piece of pipe can be fitted into the swaged end (see drawing 5).

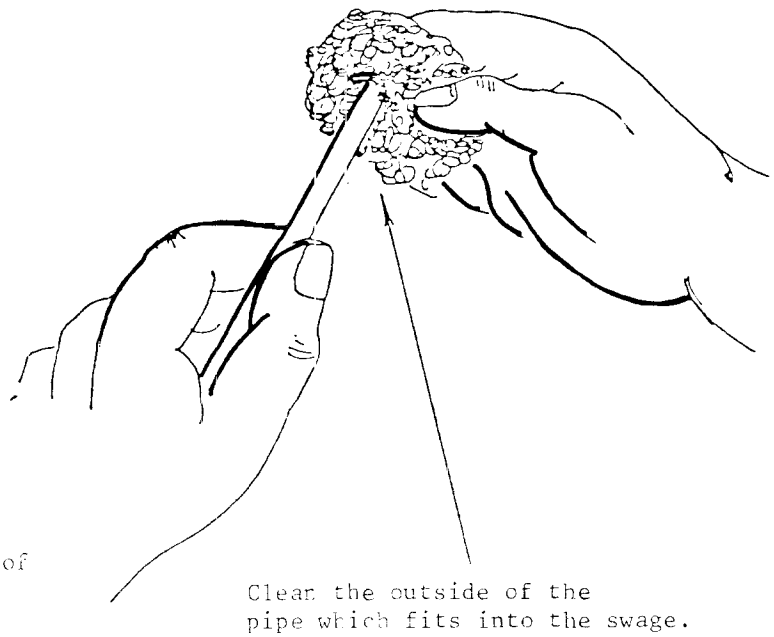
10. If the pipe does not fit easily into the swaged end, use the reaming tool (shown on page£8) to clean the inside of the swage.



Anneal the end of an old process tube to make it soft.



Clean the inside of the swaged end.



Clean the outside of the pipe which fits into the swage.

Section 2: Brazing and soldering

Preparation work

1. To join together two pieces of pipe by brazing, one pipe must first be swaged out at the end (page 17).

2. Old pipe is hard and cannot be swaged properly. It will crack or split. It must first be annealed.

How to anneal pipe

1. Use the blow torch (explained on page 23) to heat the pipe until it is a dull red colour (see drawing). Do not overheat. If the pipe becomes bright red it is too hot.

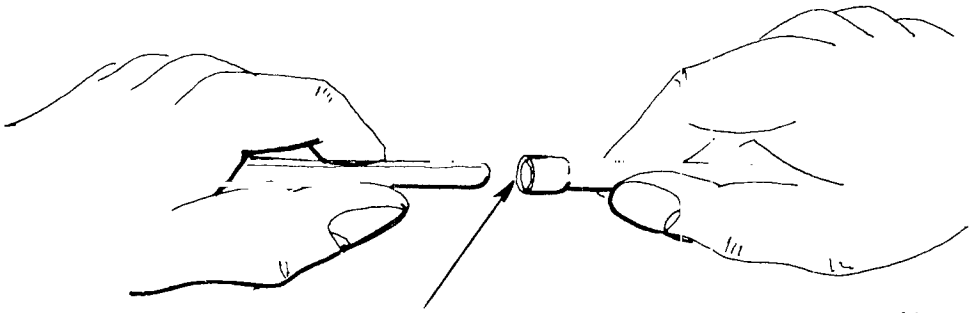
2. Allow the pipe to cool slowly to room temperature. It will now be soft and can be swaged easily.

3. If you have not used a blow torch before, practice by annealing pipe. You will then be able to braze with the torch more successfully.

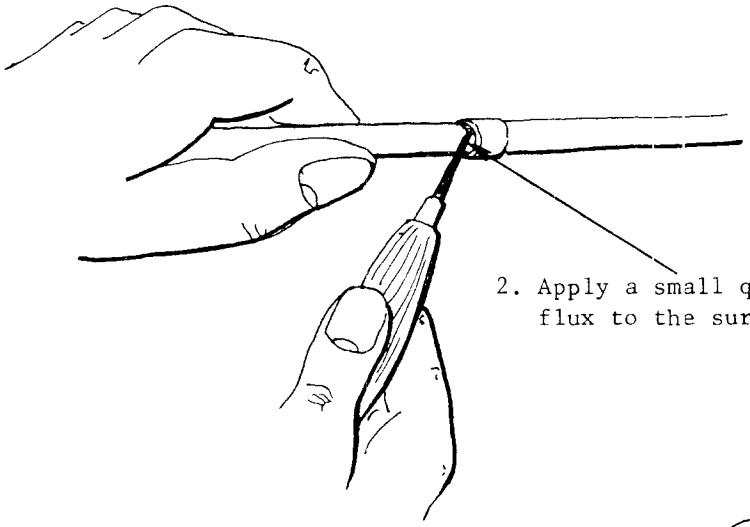
Cleaning the pipe

1. The surfaces of the pipe which are to be joined by brazing must first be cleaned.

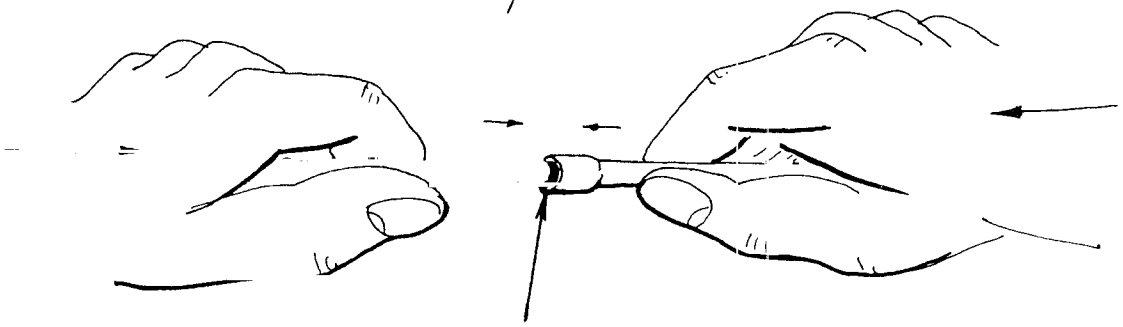
2. Use steel wool to clean these surfaces (see drawing). Do not use emery cloth or other abrasive materials. Make sure that no steel wool is left inside the pipe.



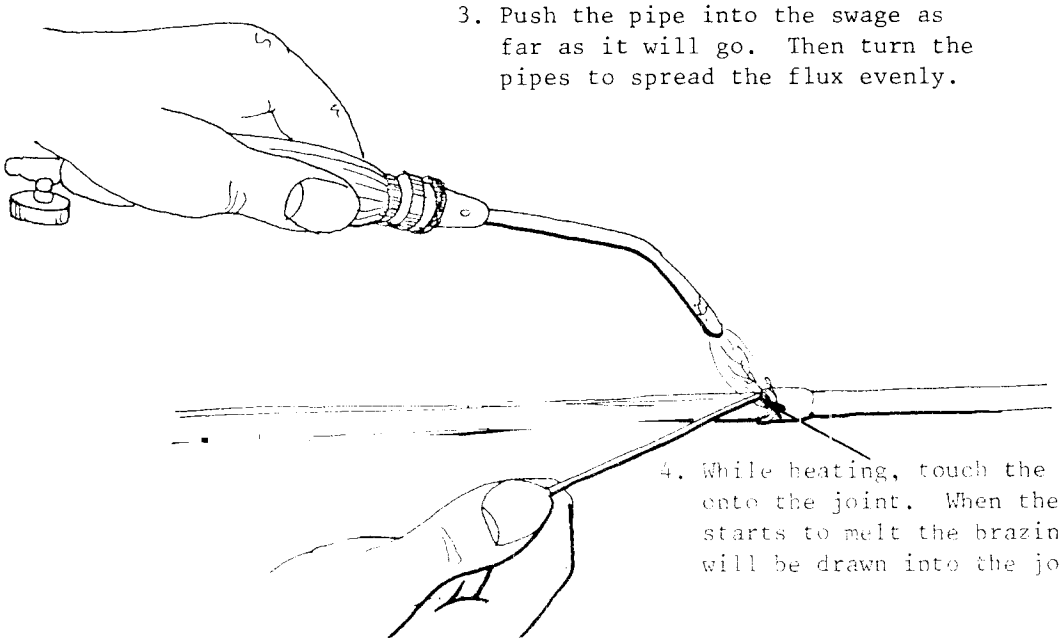
1. Make sure the pipe fits the swage easily.



2. Apply a small quantity of flux to the surfaces.



3. Push the pipe into the swage as far as it will go. Then turn the pipes to spread the flux evenly.



4. While heating, touch the brazing rod onto the joint. When the brazing rod starts to melt the brazing material will be drawn into the joint.

How to braze a swaged joint

1. Make sure that the pipe enters the swage properly. It must be an "easy" fit. If it is a tight fit there will not be sufficient space for the brazing material. If it is a loose fit the joint will be weak (see drawing 1).

2. Mix a little flux powder with spirit (or water) to make a stiff paste.

3. Apply a small amount of flux to the surfaces to be joined. Only a small quantity is necessary. Too much will contaminate the system (see drawing 2).

   Note: If "self-fluxing" brazing rods are being used this stage of the work is not necessary.

4. Push the end of one pipe into the swage of the other as far as possible. Turn the pipes to spread the flux (see drawing 3).

5. Heat the joint with the blow torch. Use a flame of the right size. (How to find the right size of flame is explained on page 23).

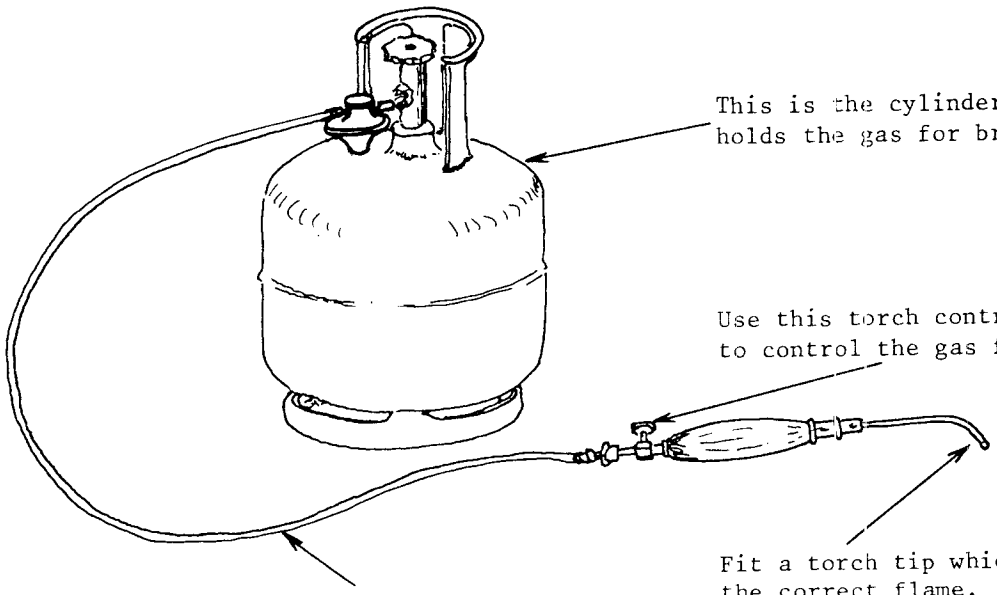
6. Wait until the joint is dull red. When the brazing rod starts to melt the joint is at the right temperature. This temperature is found by touching the brazing rod onto the joint from time to time while heating (see drawing 4). Do not overheat the joint. Do not leave the brazing rod in the flame. The brazing rod must be melted by the heat of the joint and not by the flame.

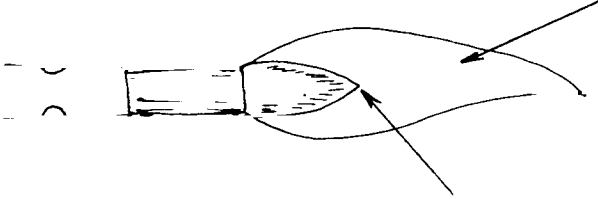
7. When the right temperature has been reached touch the joint with the brazing rod at several points.

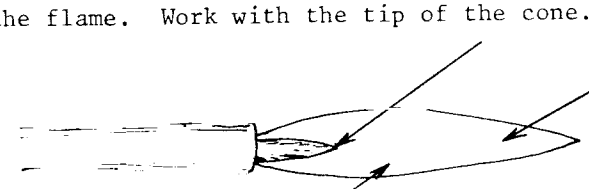
8. Molten brazing rod will be drawn into the joint and fill the space between the outside of one pipe and the swaged end of the other. When a complete ring of brazing material can be seen at the end of the swage remove the blow torch.

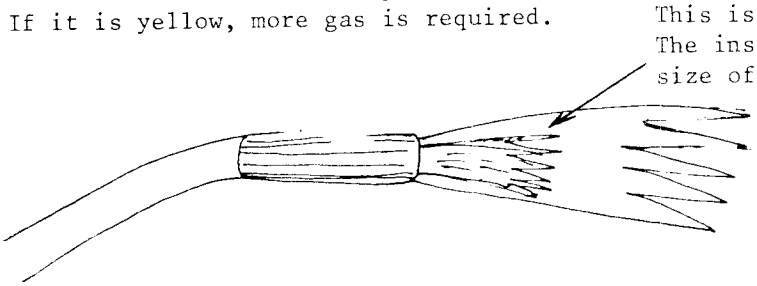
9. Allow the joint to cool. Do not move either pipe until the brazing material has hardened.

   Note: Always test brazed joints for leaks when there is refrigerant R12 in the system again. Weak joints can easily leak and are common problems.


This is the cylinder that holds the gas for brazing.
Use this torch control valve to control the gas flow.
Fit a torch tip which gives the correct flame.
Check the connections for leaks at each end of this hose with soapy water.


This is an acetylene flame suitable for pipe brazing.
The bright cone is the hottest part of the flame. Work with the tip of the cone.


This is an acetylene flame suitable for capillary tube brazing.
The outer flame should be pale blue. If it is yellow, more gas is required.


This is a propane turbo torch flame. The instructions will tell you what size of flame to use.

How to use a brazing torch

Acetylene torch (Danger. Acetylene is very inflammable. Do not allow anyone to smoke while you are brazing.)

1. Connect the torch with a flexible hose to the gas regulating valve of the acetylene cylinder. Make sure that all of the connections are tight to prevent gas leakage (see drawing). Check all connections for leaks with soapy water before lighting the torch.

2. Open the cylinder valve one turn only. Open the regulating valve fully. Open the torch control valve just enough to give a flow of gas. Light the escaping gas at the tip of the torch.

3. Adjust the torch control valve to get the correct flame. The flame should be blue. It should have a sharp bright cone in the middle with a pale outer flame (see drawing). If the flame is yellow more gas is required. Open the control valve.

4. The size of the torch tip or nozzle determines the size of the inner cone. Use a cone size that gives the required amount of heat.

Propane turbo torch (Danger. Propane is very inflammable. Do not allow anyone to smoke while you are brazing.)

1. This gives a smaller outside flame. The tip of the inner cone is much hotter than an acetylene flame of the same size (see drawing). Always work with a smaller flame than acetylene.

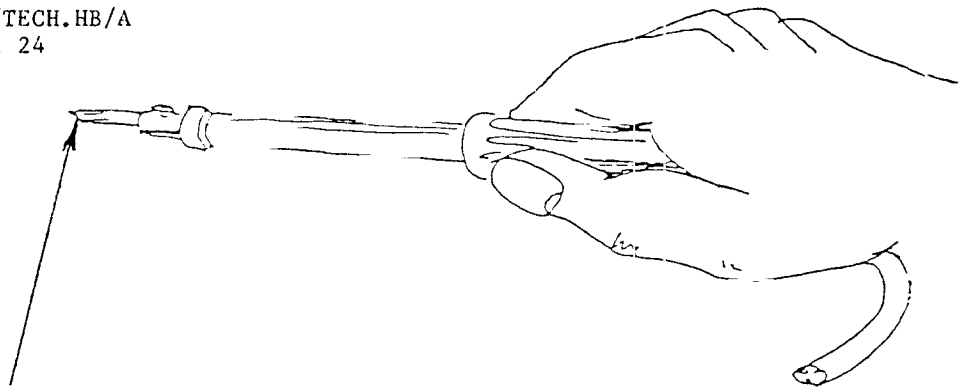
2. Connect, adjust and use this torch in the same way as acetylene described above. Check all connections for gas leaks with soapy water before lighting the torch.

3. Follow exactly the instructions supplied with the torch.

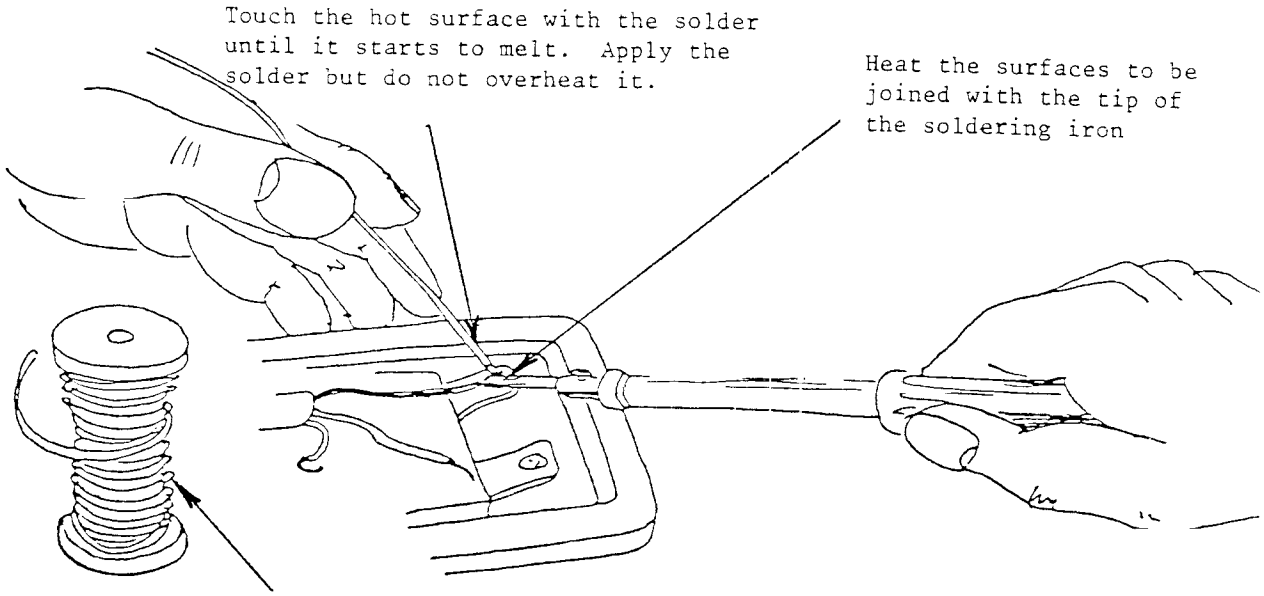
Kerosene blow lamp

1. This gives a larger flame than a propane or acetylene torch.

2. When lighting, follow carefully the instructions supplied with the torch. Wherever possible light the blow lamp in an open space for safety.

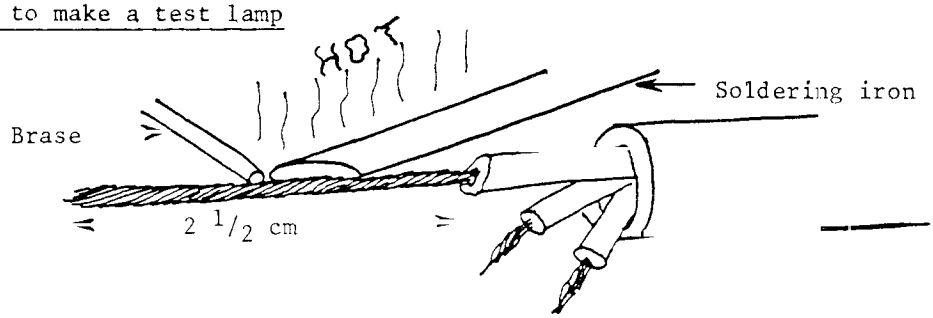


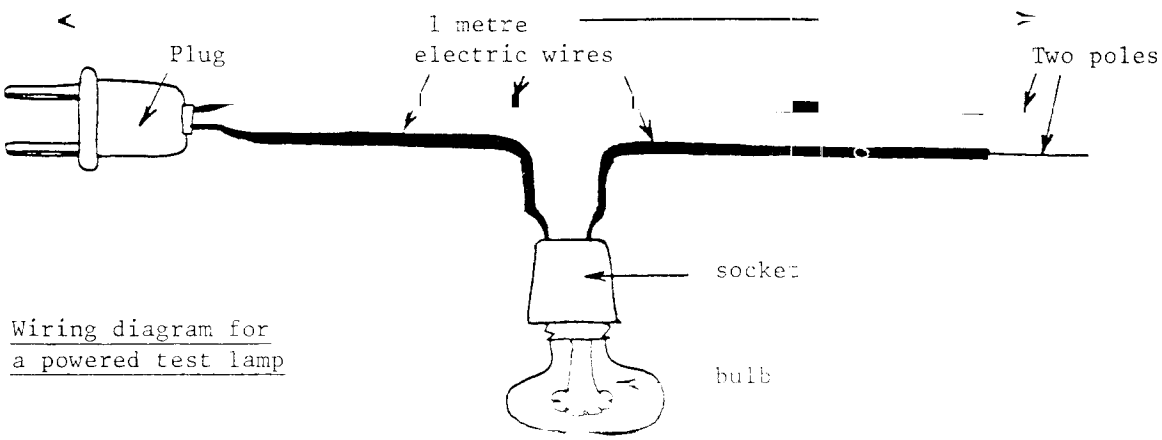
The tip of the soldering iron must be cleaned and then tinned. To tin the tip, apply a little solder when it is hot. Then quickly wipe the solder over the working surface of the tip with a damp (not wet) cloth.



Use cored solder. The flux that is necessary for a strong joint is inside the solder as a core. If you cannot get cored solder you must use a flux (see brazing, pages 20 & 21).

How to make a test lamp





Wiring diagram for a powered test lamp

How to solder electrical connections

1. Joining materials by brazing requires high temperatures but gives a very strong joint. Electrical connections do not need the strength of a brazed joint. The high temperature would also cause damage. Electrical connections are therefore soft soldered at a lower temperature.

2. Use the electric soldering iron for making all electrical connections. Prepare the tip of the iron for work as follows:

   (a) plug the iron into a socket until the tip gets hot;
   (b) remove all dirt and pieces of old solder with a wire brush;
   (c) "tin" the tip by allowing a small amount of solder to melt on it, then quickly spread and wipe the solder with a damp cloth.

3. Prepare the surfaces to be joined as follows:

   (a) remove all dirt, rust and paint with steel wool;
   (b) remove all oil or grease with spirit;
   (c) do not touch the surfaces after they have been cleaned.

4. Heat the two surfaces to be joined with the soldering iron and then "tin" each of them in turn (see 2(c) above). Both surfaces must be coated with a thin film of solder.

5. Solder the two surfaces together as follows:

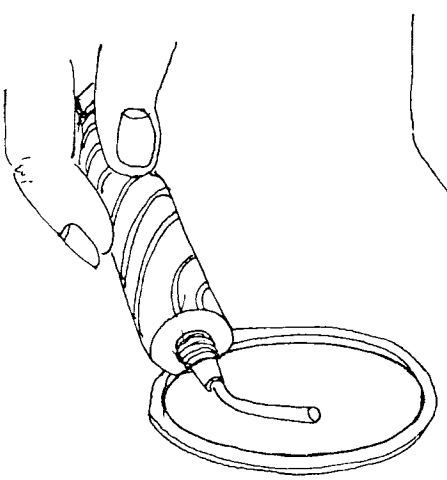
   (a) make sure that they are in the correct position and, if necessary, held together; they must not move while soldering;
   (b) heat both surfaces with the soldering iron and, from time to time, touch the hot surface with the solder (see drawing);
   (c) when the solder starts to melt allow sufficient to flow over and around the surfaces to give a strong joint.

6. Allow the surfaces to cool. Do not move either until the solder is hard.

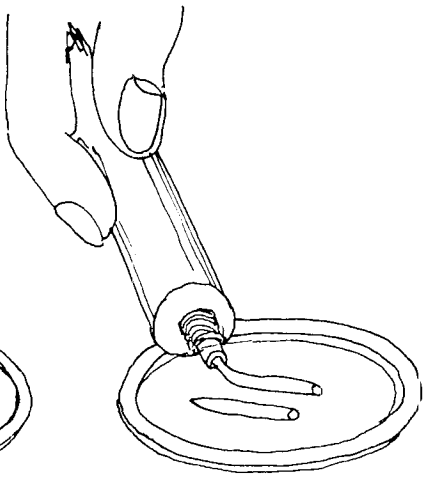
How to make a powered test lamp

1. Make two test lamp probes by "tinning" with solder pieces of wire stripped of insulation (see drawing).

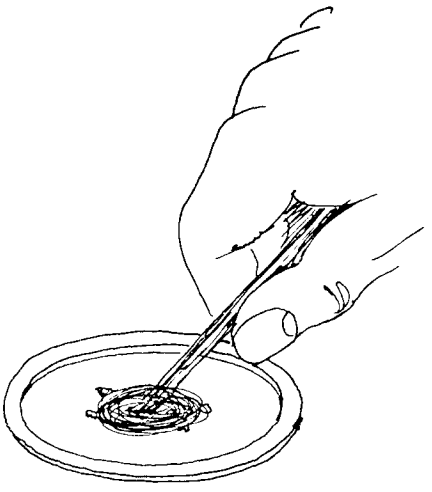
2. Make up a powered test lamp as shown in the bottom drawing opposite. For a simple (non-powered) test lamp, leave out the plug.



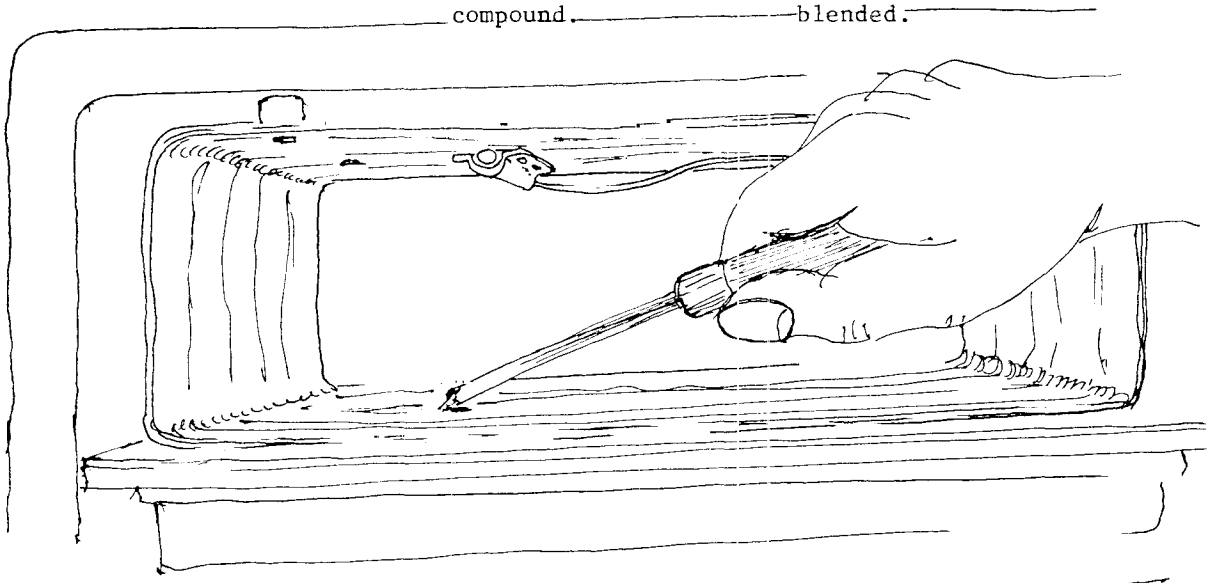
Squeeze a small amount of the compound onto a clean tin lid.



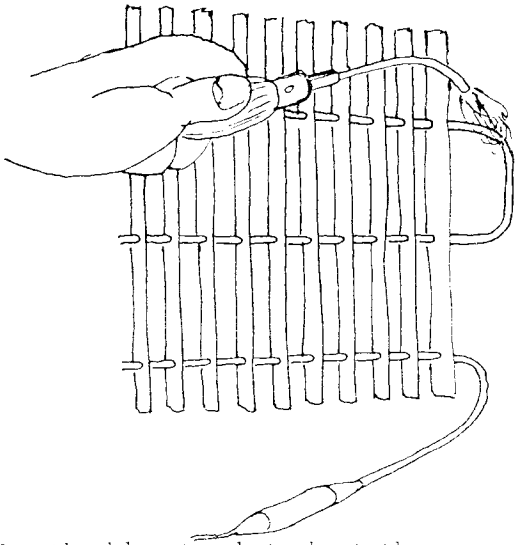
Add exactly the same amount of the second compound.



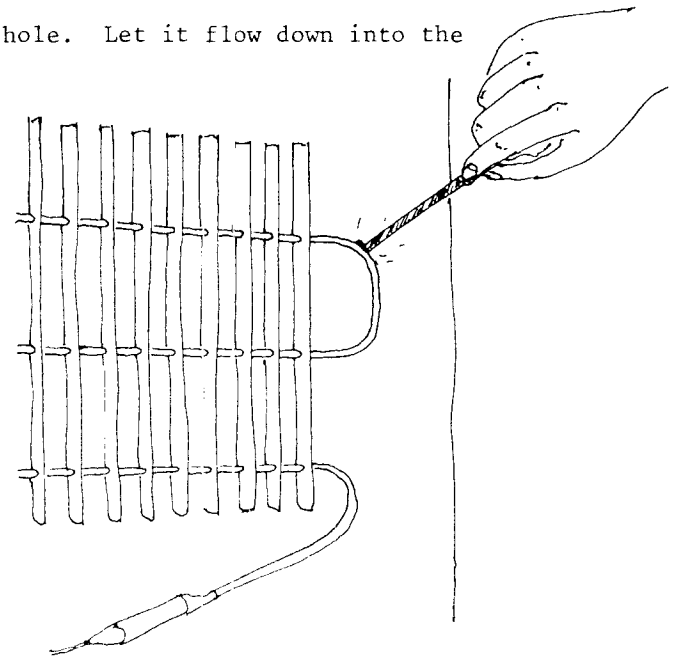
Mix the two compounds until the colour of each has blended.



Apply the mixture immediately to the hole. Let it flow down into the hole to completely seal it.



Use the blow torch to heat the surface around the leak.



When the surface is hot enough to melt the stick, remove the torch. Rub the stick onto the leak.

Section 3: Temporary repairs

Low pressure leaks

These can be repaired as follows:

1. Allow all refrigerant R12 gas to escape. You cannot seal a leak while there is still pressure in the system.

2. Clean the surface (see page 25, paragraph 3) around the leak.

3. Mix thoroughly equal quantities from each tube of the epoxy resin repair kit. Use a tin lid and an old screwdriver for mixing.

4. When the two compounds are completely mixed (the colour is uniform throughout) apply the mixture to the hole where the leak is occurring (see drawing).

5. Use the compounds immediately after mixing. Hardening starts straight away.

6. Wait at least 10 minutes for the epoxy resin to harden before filling the system with gas again. Wait several hours, to get a good, permanent repair).

High pressure leaks

Repair these as follows:

1. Allow the gas to escape and then prepare the surface (see page 25, paragraph 3).

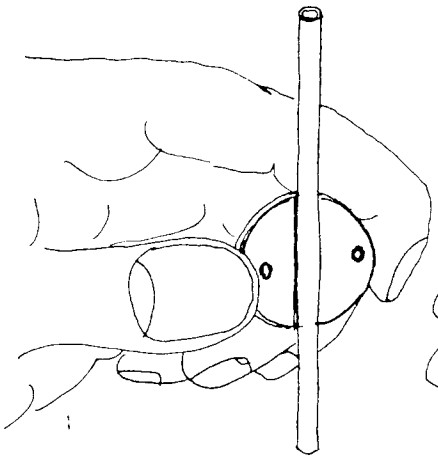
2. Heat the surface where the leak is occurring with the blow torch (see drawing).

3. Touch the hot surface, from time to time, with the end of the heat seal stick. The correct temperature has been reached when the end of the stick melts. Do not heat the stick directly - it may become too liquid and flow inside the pipe.

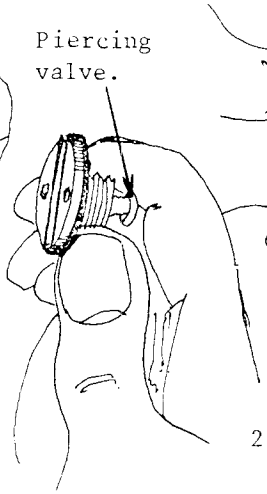
4. Remove the blow torch. Rub the end of the stick onto the hot surface. Let the melting stick seal the hole (see drawing).

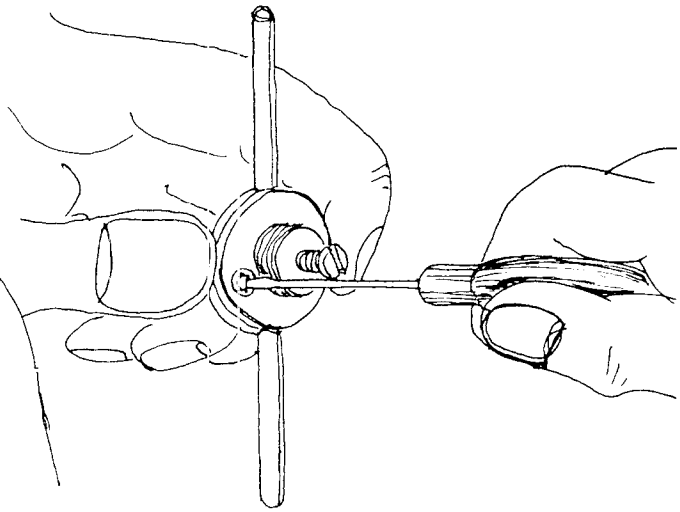
5. Wait at least 10 minutes for the seal to harden.

Note: The heat seal stick can not make a permanent repair of a high pressure leak. Always braze to make a permanent repair.

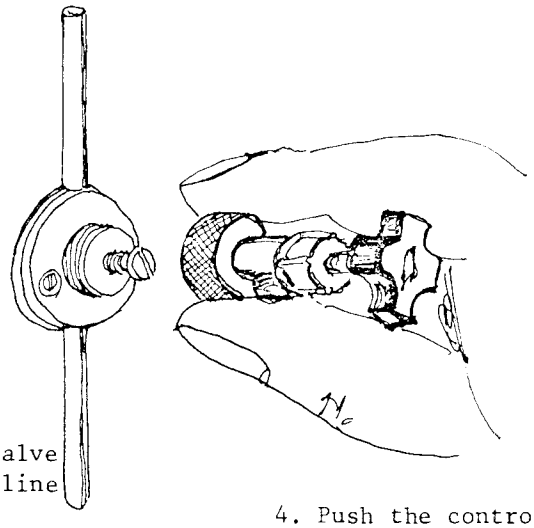


1. Fit the line tap valve to a straight length of pipe.

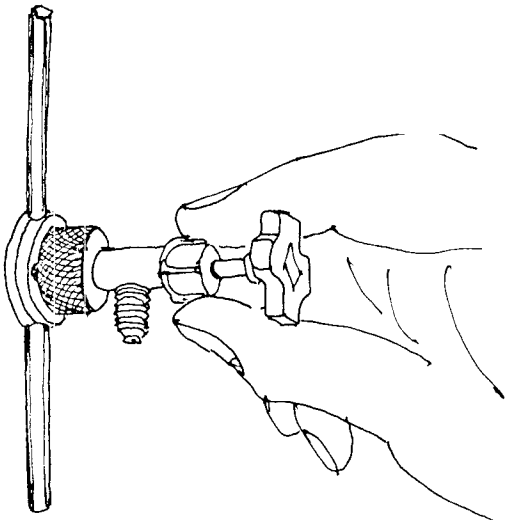




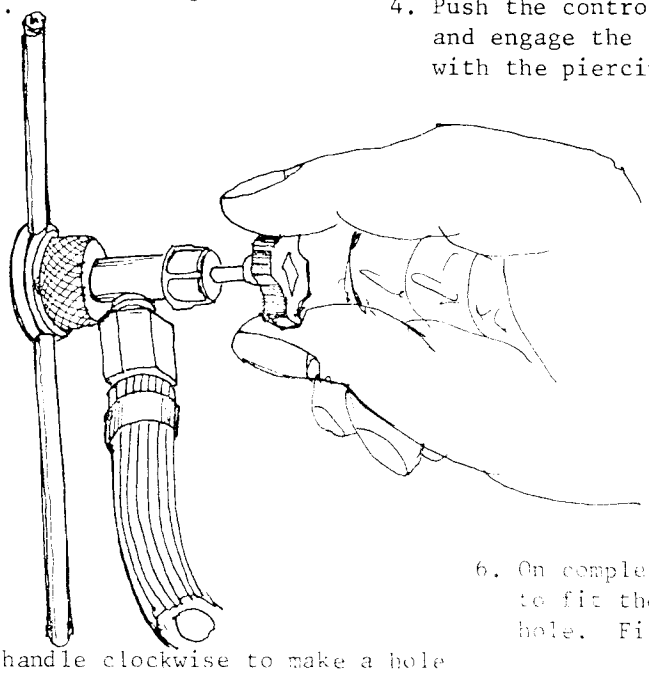
2. Make sure the piercing valve is screwed fully out before tightening the screws.



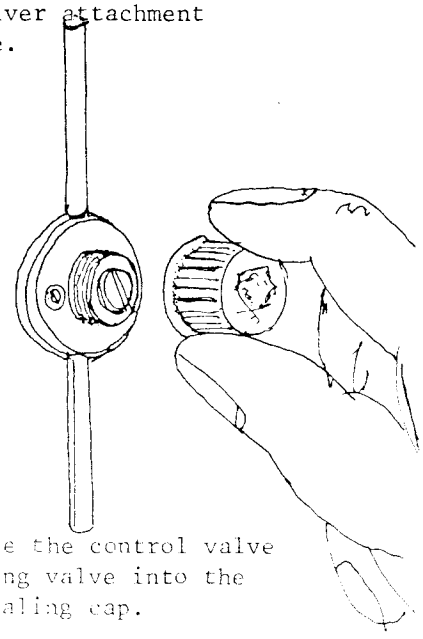
3. Fit the control valve tightly onto the line tap valve.



4. Push the control valve, handle down and engage the screwdriver attachment with the piercing valve.



5. Turn the handle clockwise to make a hole in the pipe. Turn the handle anti-clockwise to connect to the system.



6. On completion, use the control valve to fit the piercing valve into the hole. Fit the sealing cap.

Section 4: Making connections to the system

How to connect a line tap valve

1. The line tap valve can be used to connect into the compressor suction and delivery pipes.

2. Find a length of pipe that is straight, round and has no dents or scratches. Remove any dirt or paint with steel wool.

3. Select a valve that fits the pipe properly. Put a little refrigerant R12 oil on the pipe. Make sure that the piercing valve is screwed fully out (see drawing 1).

4. Tighten the clamping screws evenly. Make sure the valve is securely fixed to the pipe (see drawing 2).

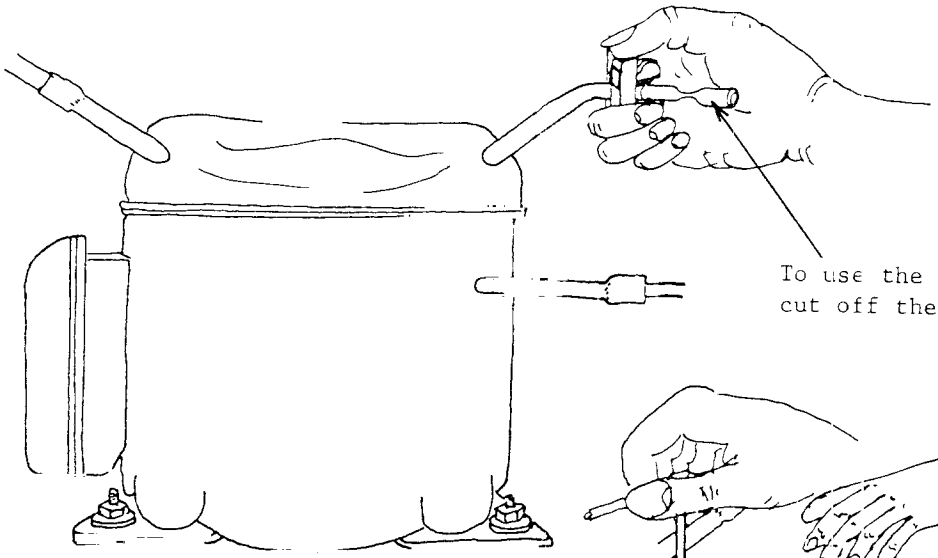
5. Fit the control valve onto the top of the line tap valve. Screw it down tight to prevent leaks (see drawing 3).

6. Push the handle of the control valve down and turn it until the screwdriver attachment fits to the slot in the piercing valve (see drawing 4).

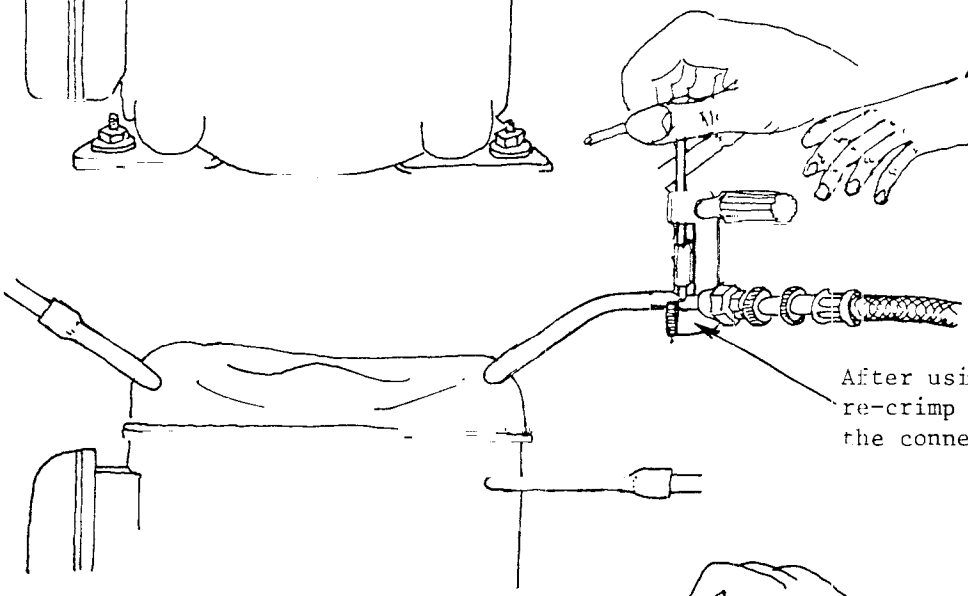
7. Connect a charging pipe to the control valve outlet. Turn the control valve handle clockwise until the piercing screw has made a hole in the pipe.

   Turn the control valve anti-clockwise to remove the screw from the hole. You now have a connection from the charging pipe to the system (see drawing 5).

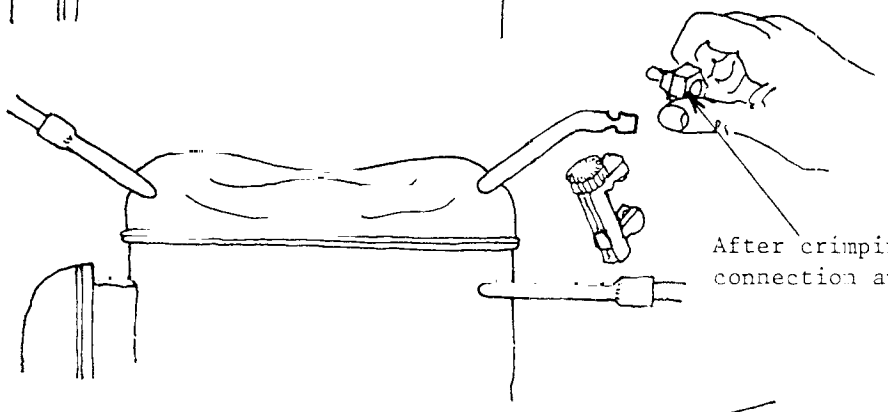
8. On completion of work, turn the control valve fully clockwise so that the piercing valve seals the hole in the pipe. Remove the control valve and replace it with the sealing cap (see drawing 6).



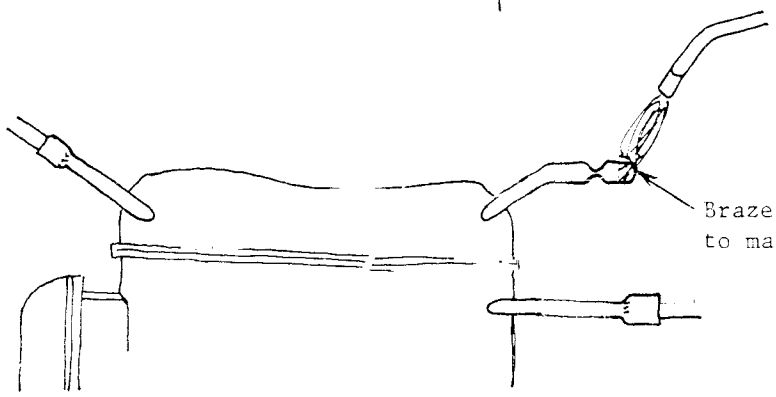
To use the process tube cut off the crimped end.



After using the process tube, re-crimp it before removing the connection.



After crimping, cut off the connection and use it again.



Braze the end of the tube to make a better seal.

How to connect to the compressor process tube

1. The compressor process tube is provided so that a connection can be made to the system without using a line tap valve.

2. The line tap valve usually has to be left on the pipe to seal the hole. Line tap valves are not always available, so you must connect to the process tube if possible.

3. The process tube is sealed by the compressor manufacturer by a "crimp" or by a sealing cap. To connect to it you must first remove the crimp with the pipe cutter (see drawing), or remove the sealing cap with pliers.

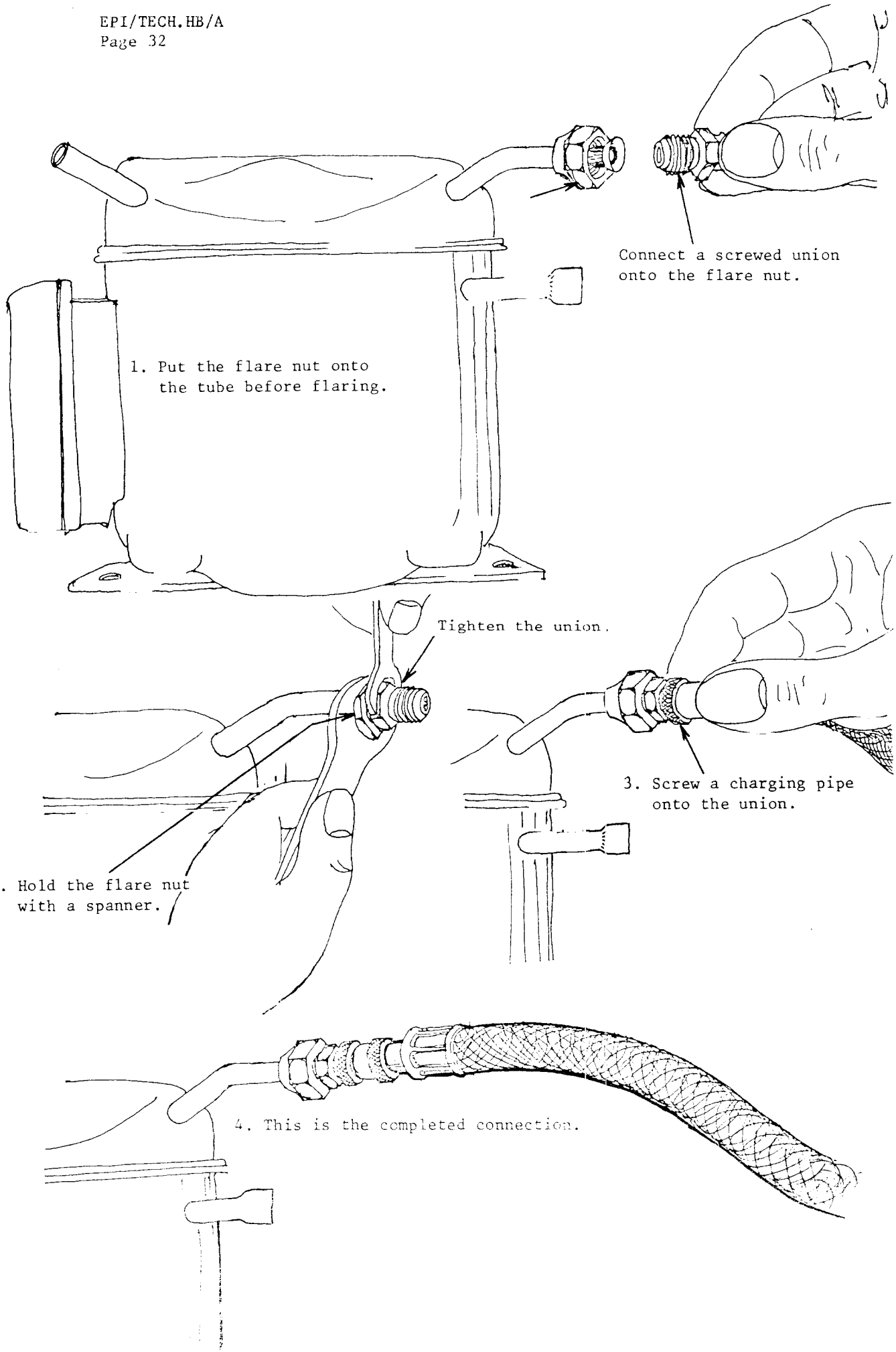
4. After the process tube has been opened, remove rough edges from the end of the pipe with the pipe reamer, if necessary. Tilt the compressor slightly so that metal pieces do not get inside the process tube.

5. There are three different methods of connecting to the open end of the process tube (see pages 32 to 37). Use the method most suitable for the job and the materials that you have.

6. Always protect the compressor when brazing on the process tube. Use a heat shield to protect the compressor from the flame. Put a wet cloth on the compressor to help keep it cool. Keep the cloth as wet as possible.

7. On completion of the work carry out the following:

   (a) re-crimp the process tube (see drawing);
   (b) remove the connection that has been made;
   (c) make the crimped seal more effective by brazing the end of the process tube (see drawing).


Connect a screwed union onto the flare nut.
1. Put the flare nut onto the tube before flaring.
Tighten the union.
3. Screw a charging pipe onto the union.
2. Hold the flare nut with a spanner.
4. This is the completed connection.

Process tube 'flared' connection

1. This method is recommended when:

   (a) you do not have a quick coupler (pages 36 and 37);
   (b) you do not want to make a brazed joint (pages 34 and 35).

2. Fit a 1/4 inch flare nut onto the process tube.

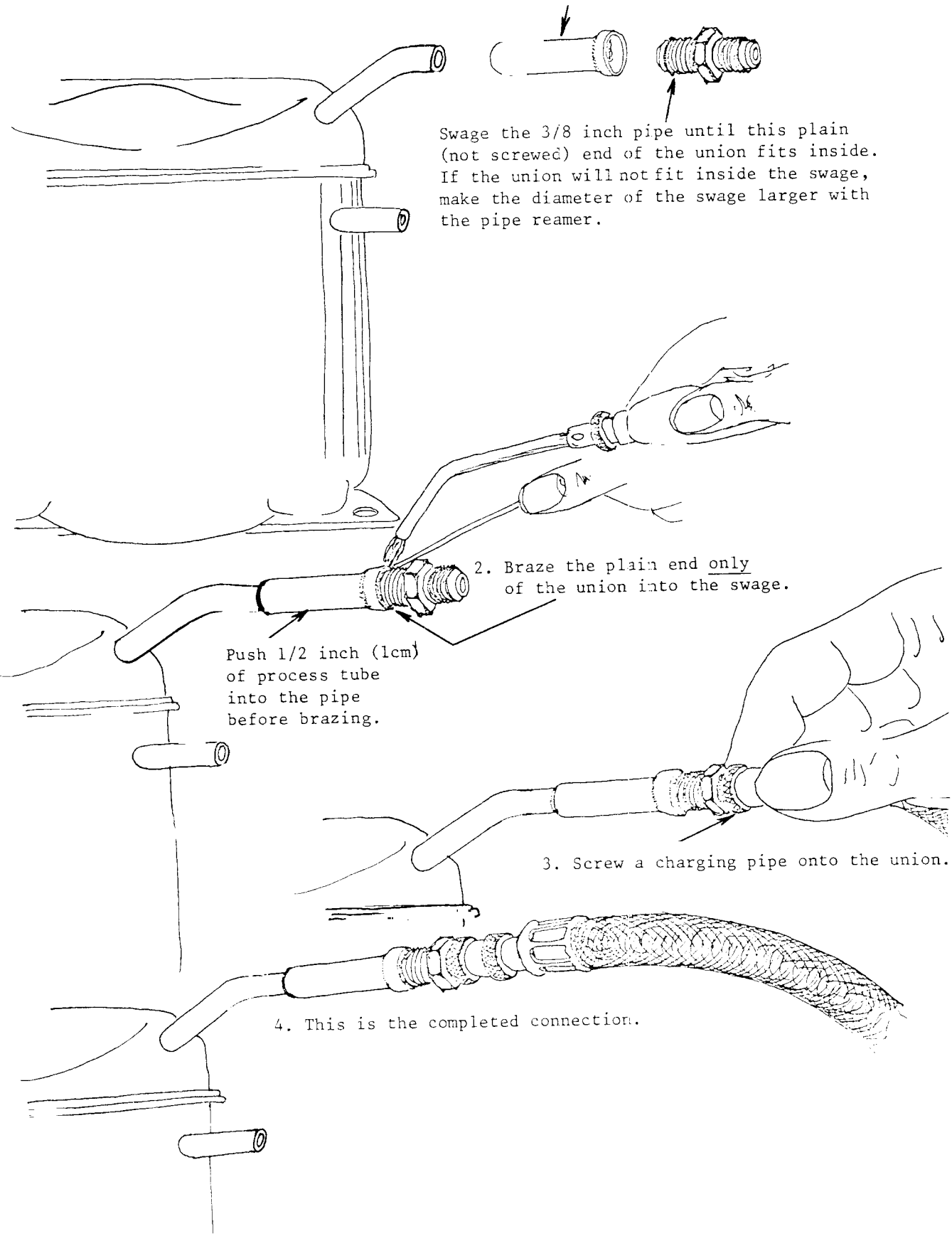
3. Flare the end of the tube (see pages 12 to 15).
   If the pipe is old and hard anneal it first (see page 19).

4. Connect a 1/4 inch screwed union onto the flared end of the pipe (see drawing 1).

5. Hold the flare nut with one spanner and tighten the screwed union into it with another spanner (see drawing 2).

6. Screw a charging pipe tightly onto the other end of the union (see drawing 3).

7. You now have a good connection onto the process tube (see drawing 4).


Swage the 3/8 inch pipe until this plain (not screwed) end of the union fits inside. If the union will not fit inside the swage, make the diameter of the swage larger with the pipe reamer.
2. Braze the plain end only of the union into the swage.
Push 1/2 inch (1cm) of process tube into the pipe before brazing.
3. Screw a charging pipe onto the union.
4. This is the completed connection.

Process tube 'brazed' connection

1. This method is recommended when you want to make a strong, permanent joint.

2. Take a short length (2 inches - 5 cm - is sufficient) of 3/8 inch (9mm) copper pipe. Swage out one end (see pages 16 and 17) until the plain end of the union fits inside. (Use the pipe reamer, if necessary, to make the inside of the swage larger.)

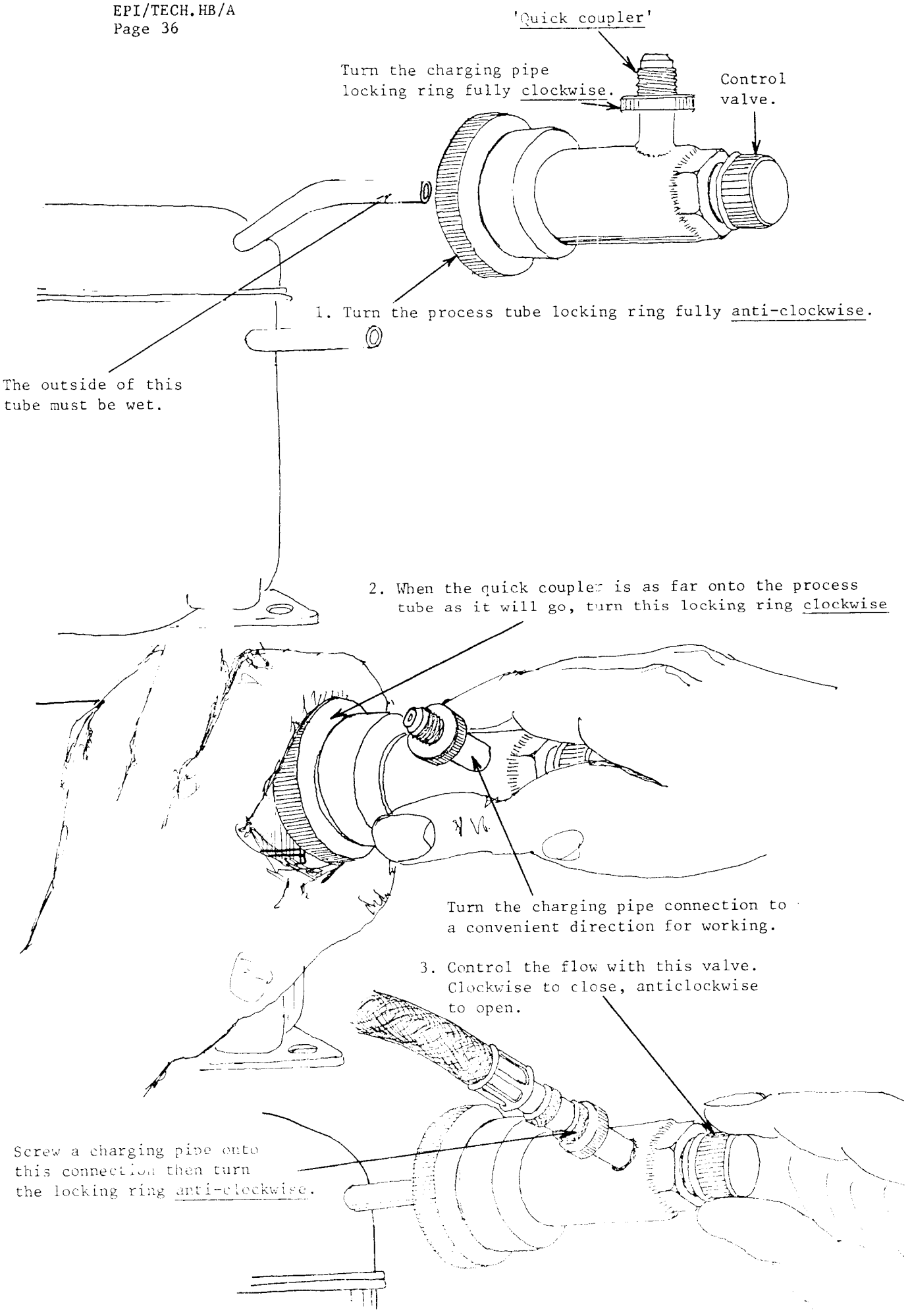
   Do not swage to the diameter of the union threads. The threads do not fit into the swage (see drawing 1).

3. Fit the plain end of the union into the swage and braze this joint (see drawing 2).

4. Fit the 3/8 inch (9mm) pipe onto the process tube and make a brazed joint (see drawing 2). If it fits loosely before brazing, squeeze it a bit. Protect the compressor from heat when brazing (saee page 31, paragraph 6).

5. Screw a charging pipe tightly onto the other end of the union (see drawing 3).

6. You now have a good connection onto the process tube (see drawing 4).


'Quick coupler'
Turn the charging pipe locking ring fully clockwise.
Control valve.
1. Turn the process tube locking ring fully anti-clockwise.
The outside of this tube must be wet.
2. When the quick coupler is as far onto the process tube as it will go, turn this locking ring clockwise
Turn the charging pipe connection to a convenient direction for working.
3. Control the flow with this valve. Clockwise to close, anticlockwise to open.
Screw a charging pipe onto this connection then turn the locking ring anti-clockwise.

Process tube 'quick coupler' connection

1. This method is recommended when you want to make a quick temporary connection.

2. Examine the quick coupler and compare it with drawing 1. Before using it make sure that you know:

   (a) how it fits onto the process tube;
   (b) how the control valve works;
   (c) how the charging pipe locking ring works.

3. Unscrew the process tube locking ring fully (see drawing 1).

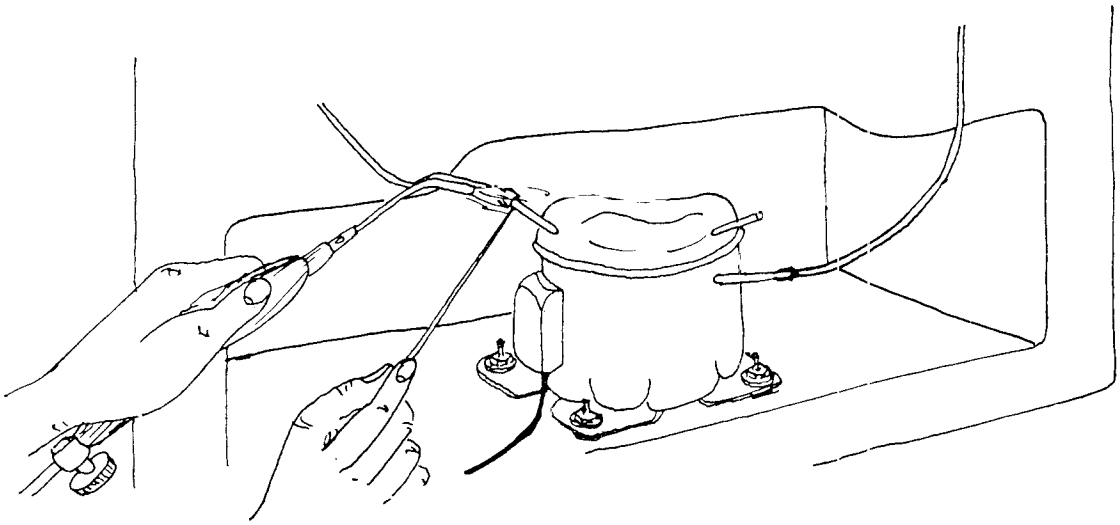
4. Wet the outside of the process tube with a little water.

5. Push the quick coupler firmly onto the process tube as far as it will go. Turn the charging pipe connection to a convenient direction for working. Tighten the process tube locking ring until the quick coupler is firmly fixed on the process tube (see drawing 2).

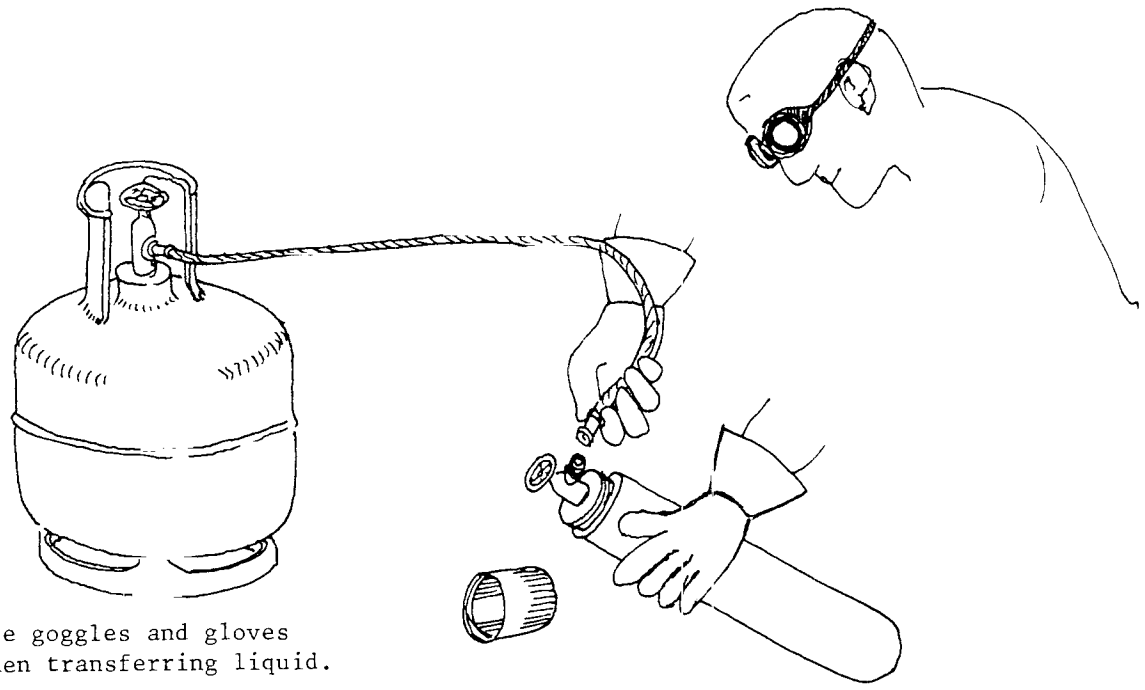
6. Screw a charging pipe firmly onto the outlet connection. Turn the charging pipe locking ring anti-clockwise until it holds the charging pipe securely (see drawing 3).

7. Control the flow with the control valve at the end. Turn clockwise to close the flow from the process tube to the charging pipe. Anti-clockwise to open (see drawing 3).

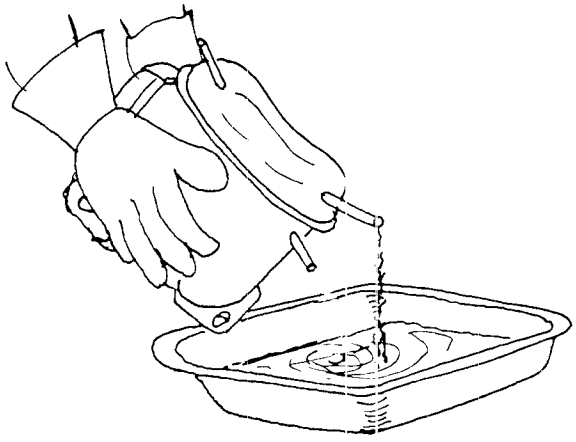
8. Some types of quick coupler do not have a control valve. If your are using this type, you can connect a line valve (shut-off valve) with the charging pipe to control the flow.



1. Make sure there is good ventilation when you are brazing.



2. Use goggles and gloves when transferring liquid.



3. Remove old compressor oil carefully.

Section 5: Refrigerant R12 safety precautions

1. Do not work where there is a high concentration of refrigerant R12 gas. If a leak occurs, or you have to release the refrigerant R12 from the system, open a window or door before starting work.

2. Make sure that the refrigerant R12 gas does not come into contact with a very high temperature such as a blow torch flame or electric element.

   Do not smoke when refrigerant R12 gas is in the surrounding air.
   Do not use the brazing torch when refrigerant R12 is in the air.

3. Do not let liquid refrigerant R12 get onto your skin. Take extra care that it does not get into your eyes.

4. When transferring liquid refrigerant R12 (from a supply cylinder into a service cylinder) wear gloves and goggles in case a sudden leak occurs (see drawing 2).

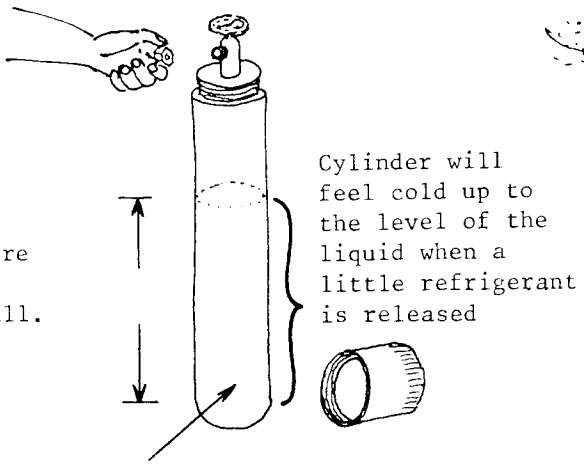
5. Check that the fittings on top of the supply cylinder and service cylinder are tight, and test them for leaks.

6. When the supply cylinder or service cylinder is not in use, fit a flare nut and bonnet to prevent any leakage.

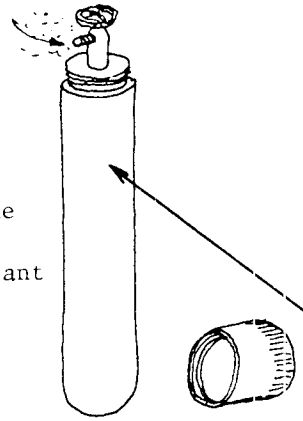
7. The refrigerant R12 inside the system is at high pressure. Work carefully when you are dismantling. The sudden escape of refrigerant R12 is dangerous.

8. The compressor oil may have acid in it. Do not let old oil touch the skin when you are removing a faulty compressor (see drawing 3).

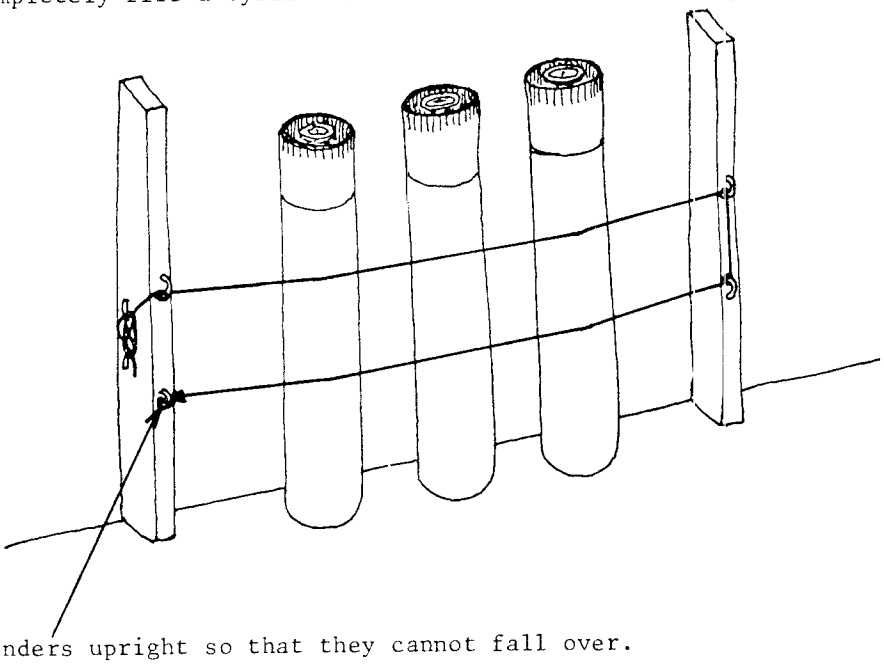
9. When charging a refrigerator make sure that you use the correct refrigerant R12. The refrigerant R12 type is shown on the manufacturer's data plate. A different refrigerant R12 may have a higher pressure. If the system is not strong enough a sudden escape of high pressure liquid and gas may occur.



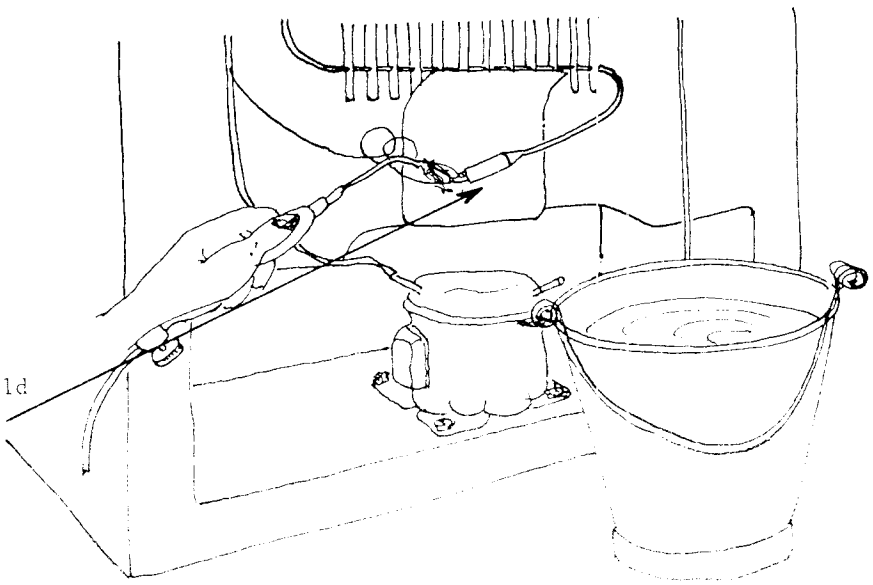
1. Do not completely fill a cylinder.



Do not completely empty a cylinder.



2. Hold cylinders upright so that they cannot fall over.



3. Use a heat shield when brazing.

10. Do not completely fill a refrigerant R12 cylinder. There must be room for the liquid to expand. If it gets hot it may explode.
Work safely. Do not let the cylinder become more than 3/4 full (see drawing 1). To check the liquid level, stand the cylinder upright and release a little refrigerant R12. The outside of the cylinder will feel cold up to the level of the liquid inside. When filling a cylinder, check with a weighing scale. Never exceed the weight of charge shown on the cylinder.

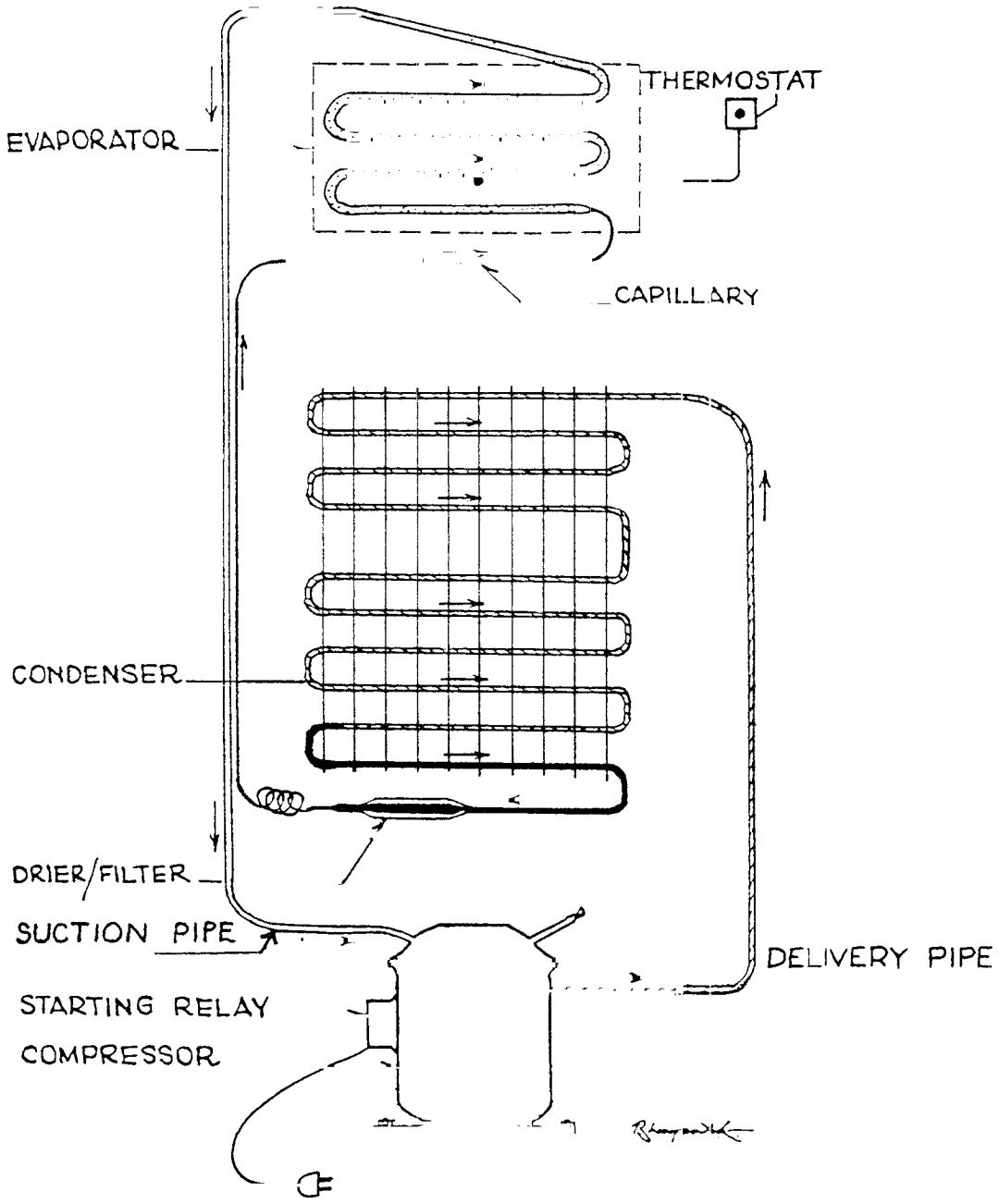
11. Do not completely empty a cylinder. Always keep a small quantity of refrigerant R12 inside to stop air and dirt getting in (see drawing 1).

12. Store cylinders in a cool space. Do not expose them to direct sunlight. Handle them carefully. Do not let them fall over (see drawing 2).

13. Do not cut or drill into a refrigeration system until all of the refrigerant R12 has been removed (see page 55).

14. When brazing or leak testing take care not to cause a fire by allowing the flame to touch an inflammable surface. Use a heat shield and keep a bucket of water near in case a fire starts (see drawing 3).

15. The manufacturer's data plate will tell you the pressure to which the system has been tested.
Do not exceed this pressure at any time.



Without oil cooling

How the refrigerator cooling system works

1. The cooling system of a refrigerator is a closed circuit of tubes. Inside this circuit is a refrigerant R12 that is pumped round the system.

2. There are different types of refrigerant R12 but the type usually used in refrigerators and freezers is called refrigerant R12.

3. The purpose of the refrigerant R12 is to move the heat from inside the refrigerator to the outside.

4. Look at the diagram opposite and follow the refrigerant R12 round the circuit on your refrigerator.

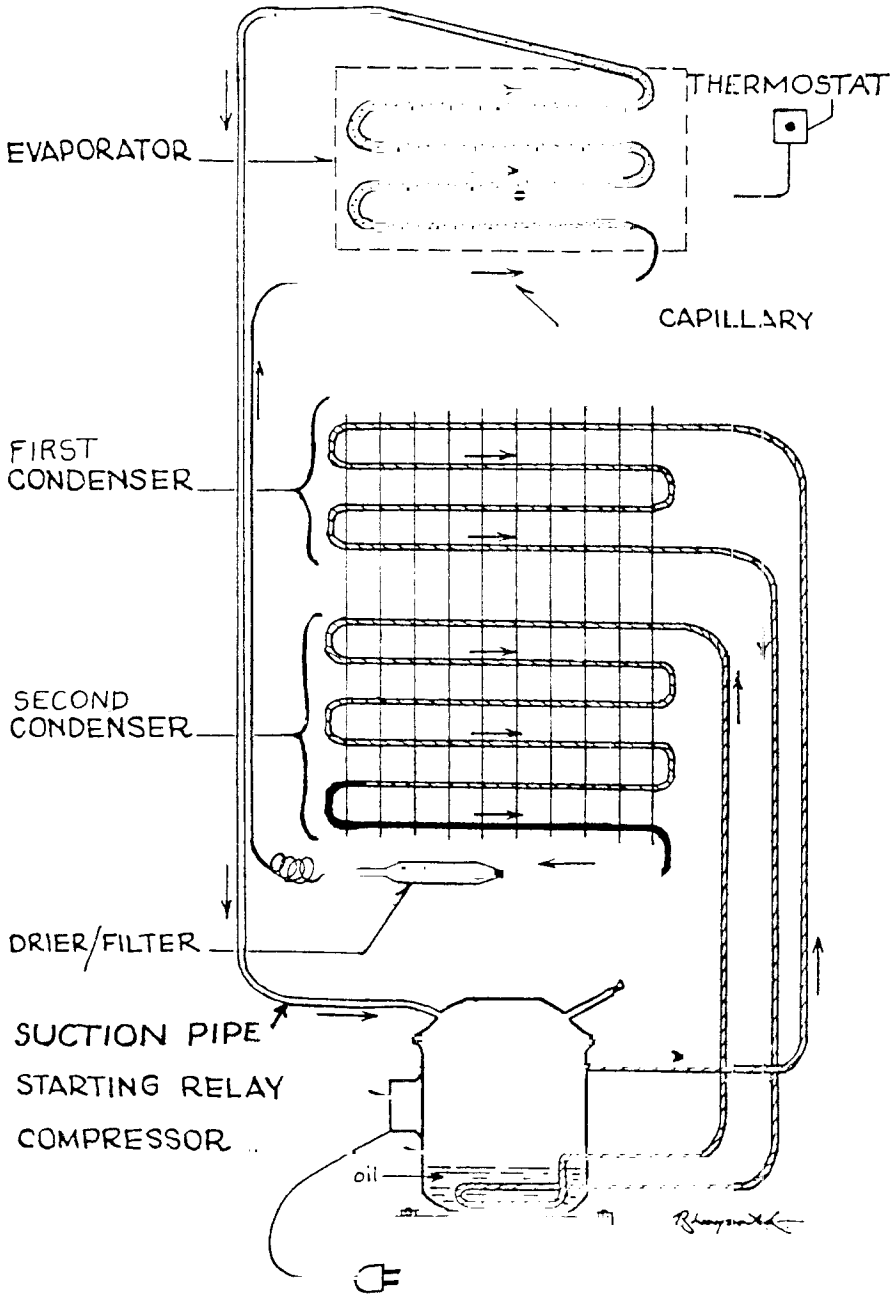
5. Start at the COMPRESSOR. The compressor pumps the refrigerant R12 gas through the DELIVERY PIPE up to the top of the refrigerator into the CONDENSOR.

6. The delivery pipe is hot when the refrigerator is running. A bicycle pump gets hot at the end when you pump up a tyre. This is because the air in the pump is under pressure. The condensor is used to cool this hot gas. At the bottom of the condensor the hot gas has condensed into a cool liquid. When a refrigerator is running properly a condensor is hot at the tope and cooler at the bottom.

7. The condensor is used to cool this hot gas. At the bottom of the condensor the hot gas has condensed into a cool liquid. When a refrigerator is running properly a condensor is hot at the top and cooler at the bottom.

8. The liquid is then filtered in the FILTER DRIER and is pumped into the CAPILLARY TUBE.

9. The capillary tube is long and thin. This means that the liquid sprays into the EVAPORATOR. When this liquid sprays it makes the evaporator cold. If you suddenly take a valve off a bicycle tyre and let the air out in a rush - this air feels cold. This is because the air is expanding quickly.



With oil cooling

10. The refrigerant R12, which has now changed back into a gas is sucked back into the compressor along the SUCTION PIPE.

11. In some refrigerators the capillary tube runs inside the suction pipe. In other types the capillary tube is wrapped round the outside of the suction pipe. This is done so that the cold gas coming out of the evaporator cools the liquid going into the evaporator.

12. The cycle now starts again. The heat that has been taken out of the inside of the refrigerator is given off into the air through the condensor.

Some refrigerators have two condensors. One for cooling the refrigerant R12 as described above; the other for cooling the oil in the compressor.

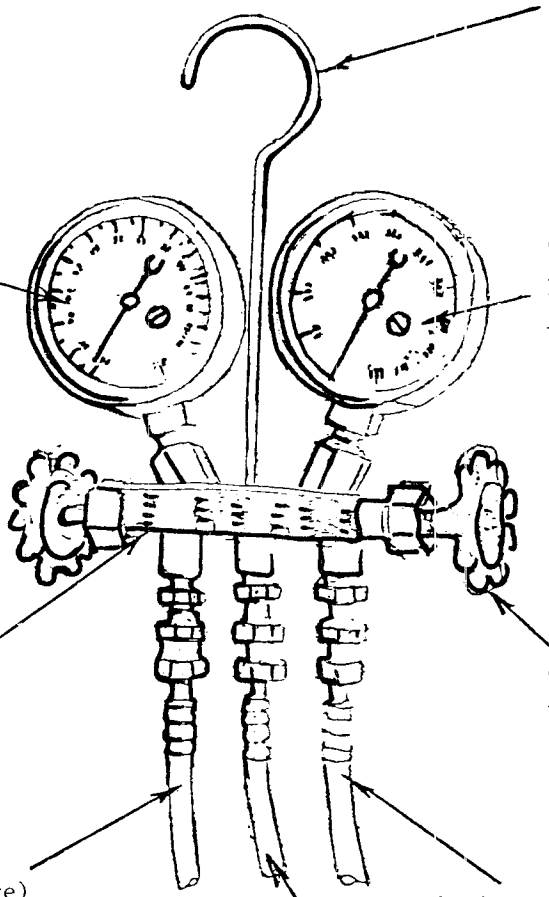
1. Look at the drawing on page 44. As before the gas is pumped out of the compressor into the condensor where it cools down and turns into a liquid.

2. This cool liquid then goes back into the compressor to cool the oil in the compressor. Like a car radiator.

3. After this liquid has taken on the heat of the oil it must be cooled again. It passes therefore into a second condensor to cool it again.

4. From this second condensor the cool liquid passes into the filter drier. The remainder of the circuit is the same as described above.

This is the support hook. Use it to hang the analyser so that you can see the gauges when you are working.



This is the low pressure gauge. It is blue.

This is the high pressure gauge. It is red.

This is the manifold bar.

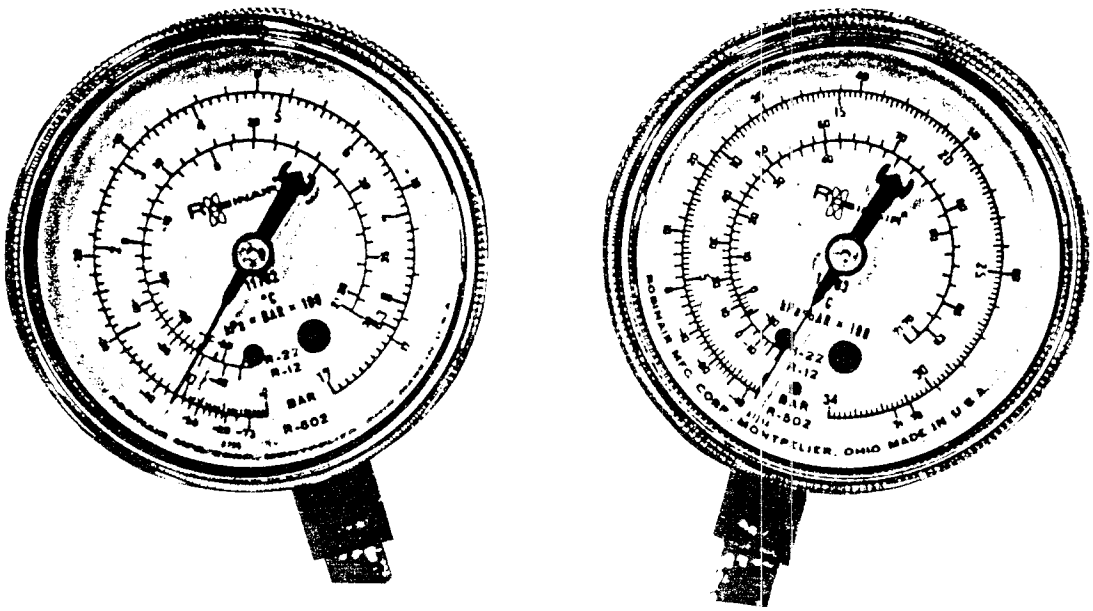
There is a hand control valve at each end of the manifold.

This is the blue (low pressure) flexible hose.

This is the red (high pressure) flexible hose.

This is the yellow (charging) flexible hose.

This is the front of a system analyser.



Section 6: How to use the system analyser

To assemble the analyser:

1. Compare the parts with the top drawing opposite. You must have the following:

   (a) a manifold with a control valve at each end;
   (b) a support hook;
   (c) a low pressure gauge (graduated from a 30 inch vacuum to 250 psi or -1 vacuum to 9 bar);
   (d) a high pressure gauge (graduated from 0 to 500 psi or 0 to 34 bar);
   (e) three flexible hoses, coloured as follows:
   1 blue 1 yellow or white 1 red
   (f) a hose holder (not shown in drawing)

2. Screw the hose holder and support hook onto the back of the manifold. The hook must be able to move, so that you can hang the analyser from a convenient surface. Make sure that the nut at each end of the manifold bar is tight.

3. Screw the blue (low pressure) gauge tightly into the left connection. Try to position the gauge so that it faces the direction shown on the drawing.

4. Screw the red (high pressure) gauge tightly into the right connection. Position this gauge in the same direction. (Do not use any tools for tightening hose connections. "Hand tight" gives a leak tight connection).

5. Screw the blue (low pressure) hose tightly onto the thread on the left side of the manifold.

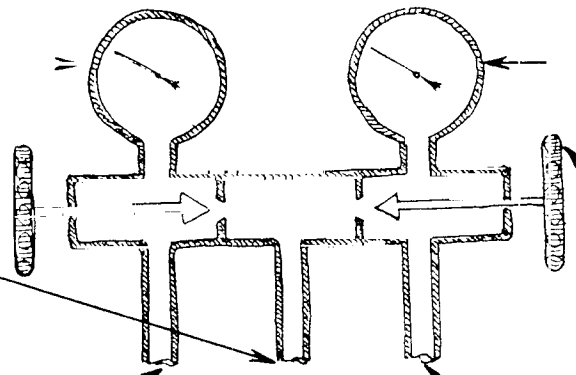
6. Screw the yellow or white (charging) hose tightly onto the thread in the middle of the manifold.

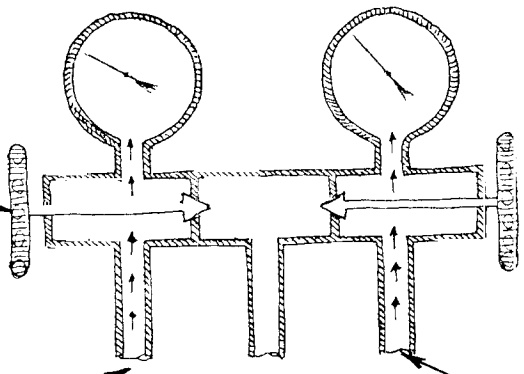
7. Screw the red (high pressure) hose tightly onto the thread on the right side of the manifold.

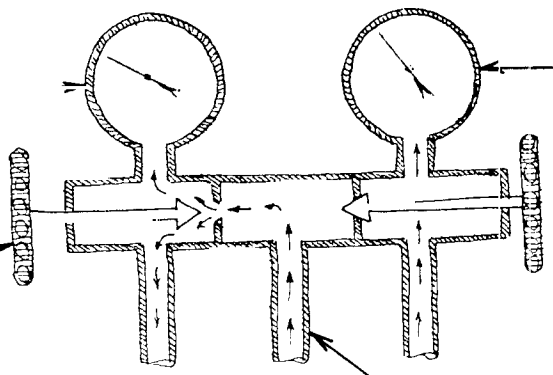
   Note: Always keep the ends of hoses closed when they are not in use.

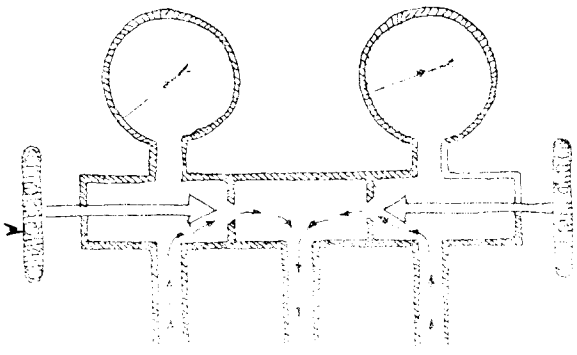
8. Test for leaks below each gauge as follows:

   - fully close both control valves (see page 48 drawing 2)
   - connect the blue hose to a refrigerant R12 service cylinder and fill it with a little gas; carry out the leak test below the blue gauge
   - repeat for the red gauge
   - if you find a leak, seal the connection using thread sealing tape; test for leaks again
   - Unscrew the front cover at each gauge and check that the needle is pointing exactly at zero. If it is not, gently turn the adjusting screw on the dial to "zero" the needle.


Blue
Red
This connection is used for charging and drawing a vacuum.
These valves are fully open. Turn them clockwise to close.
1. This connection (blue) is made to the low pressure part of the system.
This connection (red) is made to the high pressure part of the system.


2. Both valves are fully closed.
The pressure in the suction pipe will be shown on the blue gauge.
The pressure in the delivery pipe will be shown on the red gauge.


Suction pressure shown here when charging.
Delivery pressure shown here when charging.
3. The valve is open to the blue suction gauge when charging.
Refrigerant passes from the middle into the blue connection


4. Both valves are fully open when drawing a vacuum.

How to set the valves on the system analyser

1. The drawings opposite show what happens when the control valves are opened and closed.
   On some system analysers the control valves are on the front of the manifold. This does not affect the valve operation. Follow these instructions whichever arrangement of valves are fitted.

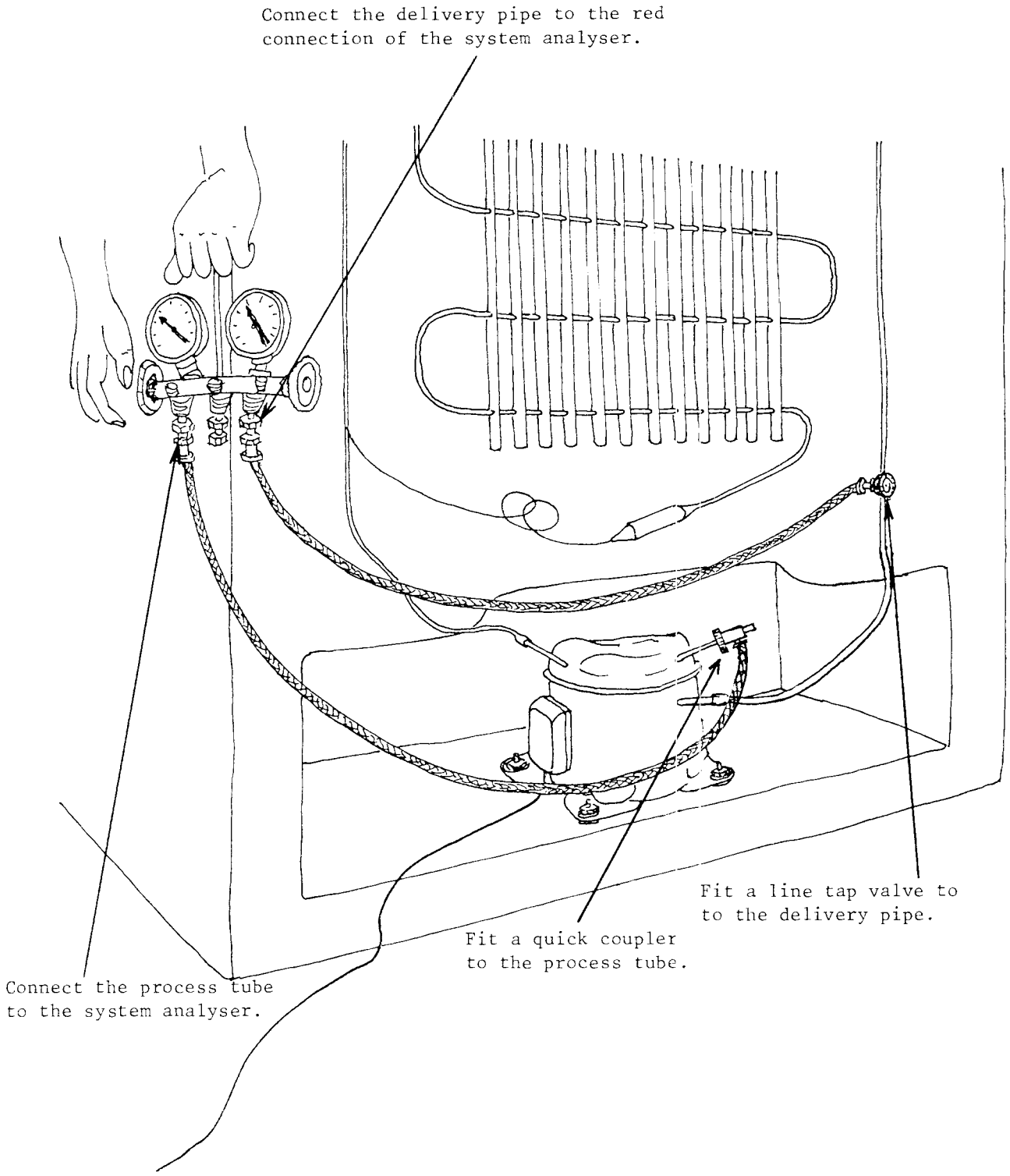
2. Drawing 1 shows the arrangement of the valves inside the manifold. Both valves are in the open position. To open the valves turn the hand controls anti-clockwise. To close the valves turn the hand controls fully clockwise.

3. Drawing 2 shows the valve positions for measuring the system pressures on the gauges.
   Turn both control valves fully clockwise to close them. This stops any flow of refrigerant R12 to the middle connection. The refrigerant R12 flows only to the gauges.
   The pressure in the suction pipe is shown on the blue gauge.
   The pressure in the delivery pipe is shown on the red gauge.

4. Drawing 3 shows the valve positions when charging refrigerant R12 into the system. Refrigerant R12 is passing from a cylinder through the middle connection into the suction pipe. Note that the valve positions enable both the suction and delivery pressures to be measured when charging.

5. Drawing 4 shows the valve positions when drawing a vacuum on the system. The middle connection is joined to a vacuum pump. The other two connections are joined to the system.

6. How to connect the analyser for measuring system pressures: see page 51
   How to connect the analyser for charging refrigerant R12 : see page 65
   How to connect the analyser for drawing a vacuum: see page 59.


Connect the delivery pipe to the red connection of the system analyser.
Fit a line tap valve to to the delivery pipe.
Fit a quick coupler to the process tube.
Connect the process tube to the system analyser.

How to connect the system analyser for measuring pressures

1. Fit a line tap valve (as described on page 28) to the compressor delivery pipe. Fit two line tap valves (as described on page 36) one to the process tube (see drawing) and the other to the delivery pipe. Do not open either valve.

2. Connect a flexible charging hose from each of the following:
   - Process tube to blue connection of system analyser.
   - Delivery pipe to red connection of system analyser.

   Note: "Hand tight" is enough for hose connections. Do not use tools to tighten them.

3. Remove the air in the charging hoses as follows:
   - Connect a refrigerant R12 service cylinder to the middle connection of the system analyzer (see drawing page 64).
   - Fully open the system analyser control valves.
   - Loosen, but do not remove, the charging hose connections at the valves on the process tube and delivery pipe.
   - Open the service cylinder control valve slightly.
   - Allow escaping refrigerant R12 to drive out air at the loose connections.
   - Close the service cylinder control valve and tighten the hose connections.
   - Fully close the system analyser control valves.
   - Remove the refrigerant R12 service cylinder.

4. Fully open the control valves fitted to the top of the line tap valve and at the end of the quick coupler (if used).

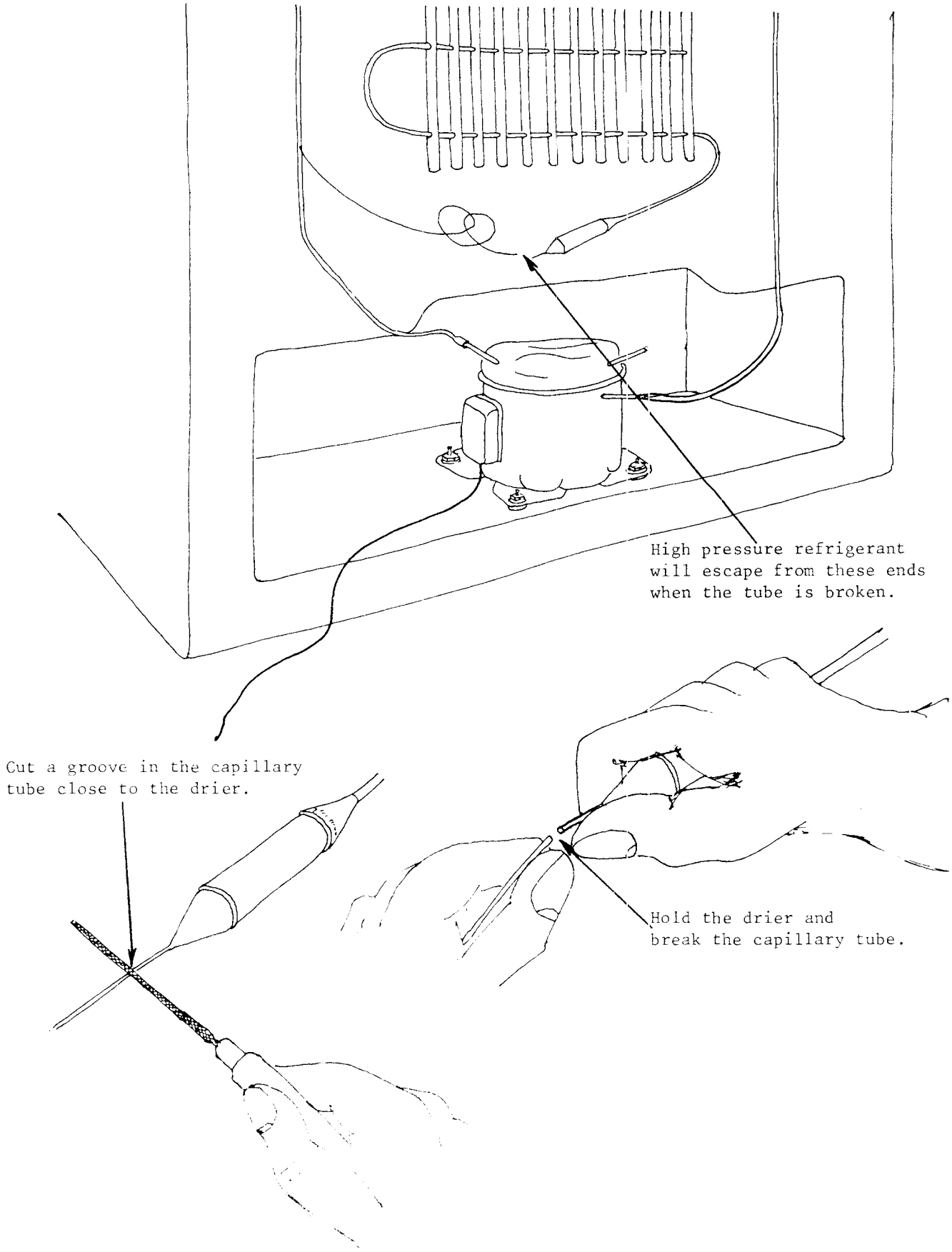
5. The low (suction) pressure can now be read from the blue gauge.
   The high (delivery) pressure can now be read from the red gauge.
   Test for leaks at the gauges and hose connections.

6. On completion, close the line tap control valve(s) and fit a sealing cap (see page 28, drawing 6). If a quick coupler is used, leave it in position on the process tube with the control valve fully closed. Remove the system analyser and flexible charging hoses. Plug the open ends of the charging hoses to stop dirt and moisture entering.

7. Instructions for connecting the system analyser when evacuating (drawing a vacuum on the system) are given on pages 58 and 59.

8. Instructions for connecting the system analyser when charging the system are given on pages 64 and 65.

Note: See also Annex 1, using TEE tubes in a refrigerator system.


High pressure refrigerant
will escape from these ends
when the tube is broken.
Cut a groove in the capillary
tube close to the drier.
Hold the drier and
break the capillary tube.

Section 8: How to release refrigerant R12 from the system

1. To release refrigerant R12 from the system safely (before starting to dismantle or repair) take the following action.

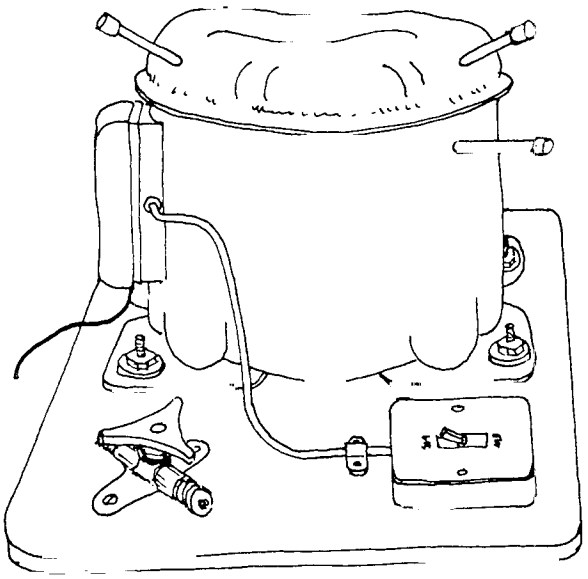
2. Ventilate the room into which the refrigerant R12 will be released. Open a door or window.
   Make sure there are no flames in the room.
   Do not allow anyone to smoke.

3. With a file, cut a groove in one side of the capillary tube, close to the drier (see drawing). File about one quarter of the way through the tube.

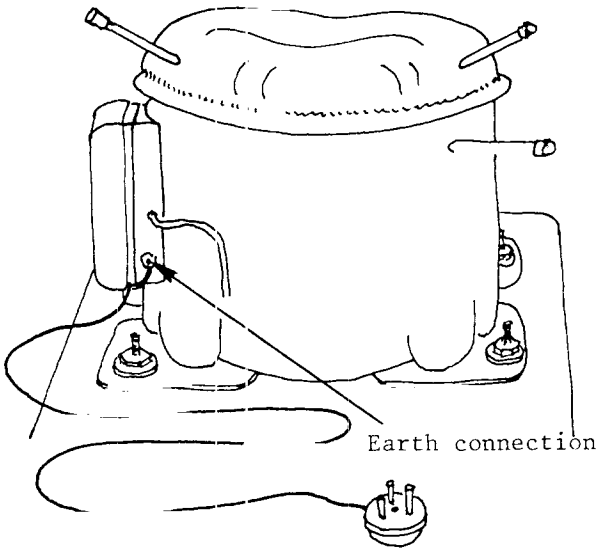
4. Bend the capillary tube gently at the grooves until it breaks. Take care not to damage this tube (see drawing).

5. When you break the capillary tube high pressure refrigerant R12 gas (and some liquid) will escape.
   Bend the tube away from you, so that the escaping refrigerant R12 does not get onto your face.

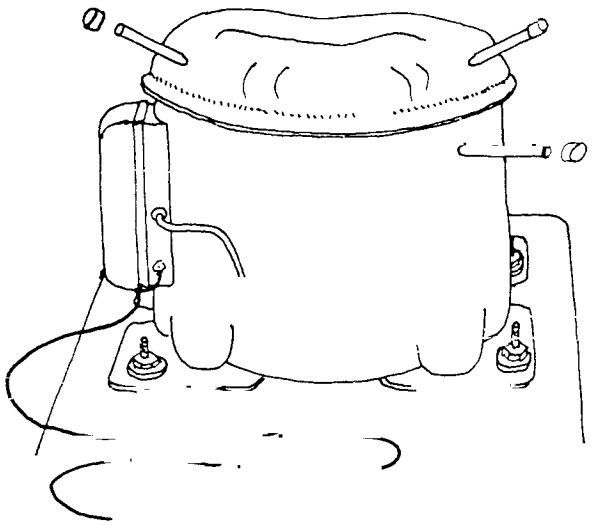
6. Do not start working on the system until all of the refrigerant R12 has escaped.



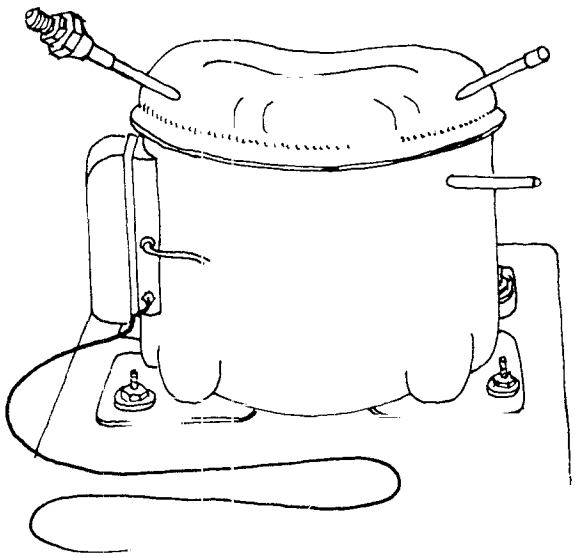
1. Bolt to a piece of strong, hard wood the compressor, line valve and switch. Connect a cable to the terminal box and connect up the switch.



2. Fit a plug to the cable. If there are 3 pins on the plug make the earth connection shown.



3. Remove the sealing caps from the suction pipe and delivery pipe.



4. Extend the suction pipe by brazing on 4 inches (10cm) of 1/4 inch (6mm) copper pipe. Fit a screwed union onto the suction pipe.

Section 9: How to evacuate the system

After carrying out repairs on the system all air and small pieces of dirt must be removed. To do this the system is evacuated with a vacuum pump. If a special-purpose vacuum pump is not available, a spare compressor can be made into a vacuum pump as follows:

How to convert a compressor for vacuum pumping

1. Fix the compressor securely to a piece of strong, hard wood (see drawing 1). Use the bolts and rubber mountings supplied with the compressor. Fit a line valve to the wood, as shown.

2. Connect 6ft (2m) of electric cable to the supply connections inside the terminal box. Follow the instructions provided by the manufacturer. Connect a switch to the terminals normally used for the thermostat (see drawing 1).

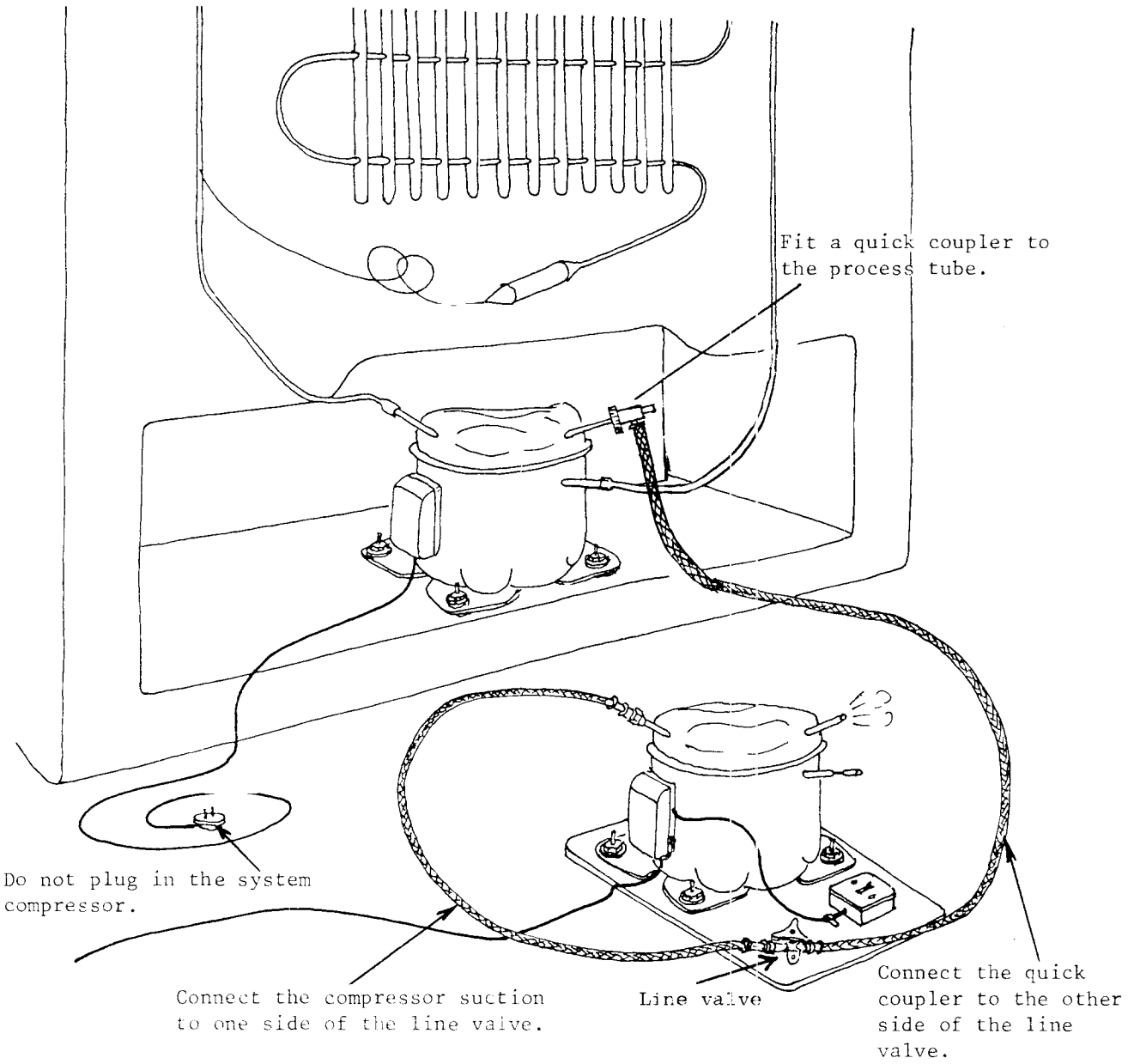
3. Fit a plug onto the other end of the electric cable. If the socket into which the plug fits has 3 holes, the third is for an earth connection. Use a cable which has 3 wires inside and join the earth wire securely to any metal part of the compressor or terminal box (see drawing 2).

4. Identify the compressor pipes (see page 42). Remove the sealing caps from the suction pipe and delivery pipe. Leave the process tube closed.

5. Remove the starting relay and all electrical connections before brazing, to avoid damage by heat. Fit 4 inches (10cm) of 1/4 inch (6mm) copper pipe to the suction pipe and make a strong brazed connection. Remove the starting relay and all electrical connections before brazing, to avoid damage by heat.

6. Fit a screwed union onto the open end of the suction pipe by flaring (page 32) or brazing (page 34). (See drawing 4). Replace the electrical connections.

7. Plug in the compressor. Run it only long enough to make sure that air is being drawn in through the suction. If you draw too much air into the compressor, you may damage it.



'How to connect the vacuum pump'

How to draw a vacuum on the system (using the process tube)

1. Connect the compressor converted for vacuum pumping (pages 55 and 56) to the system, as described below.

2. Connect the compressor suction to the line valve (see drawing).

3. Join one end of another charging hose to the line valve.

4. Fit a quick coupler to the process tube of the system compressor (as described on pages 36 and 37).

5. Connect the other end of the charging hose to the quick coupler (see drawing).
   Make sure that the charging hoses are connected tightly at both ends.

6. Open the control valve on the quick coupler fully (turn it anti-clockwise). Open the line valve fully.

7. Plug in the vacuum pump to the nearest electrical socket: (do not plug in the system compressor).

8. Warm the condenser carefully with the blow torch to help remove moisture.

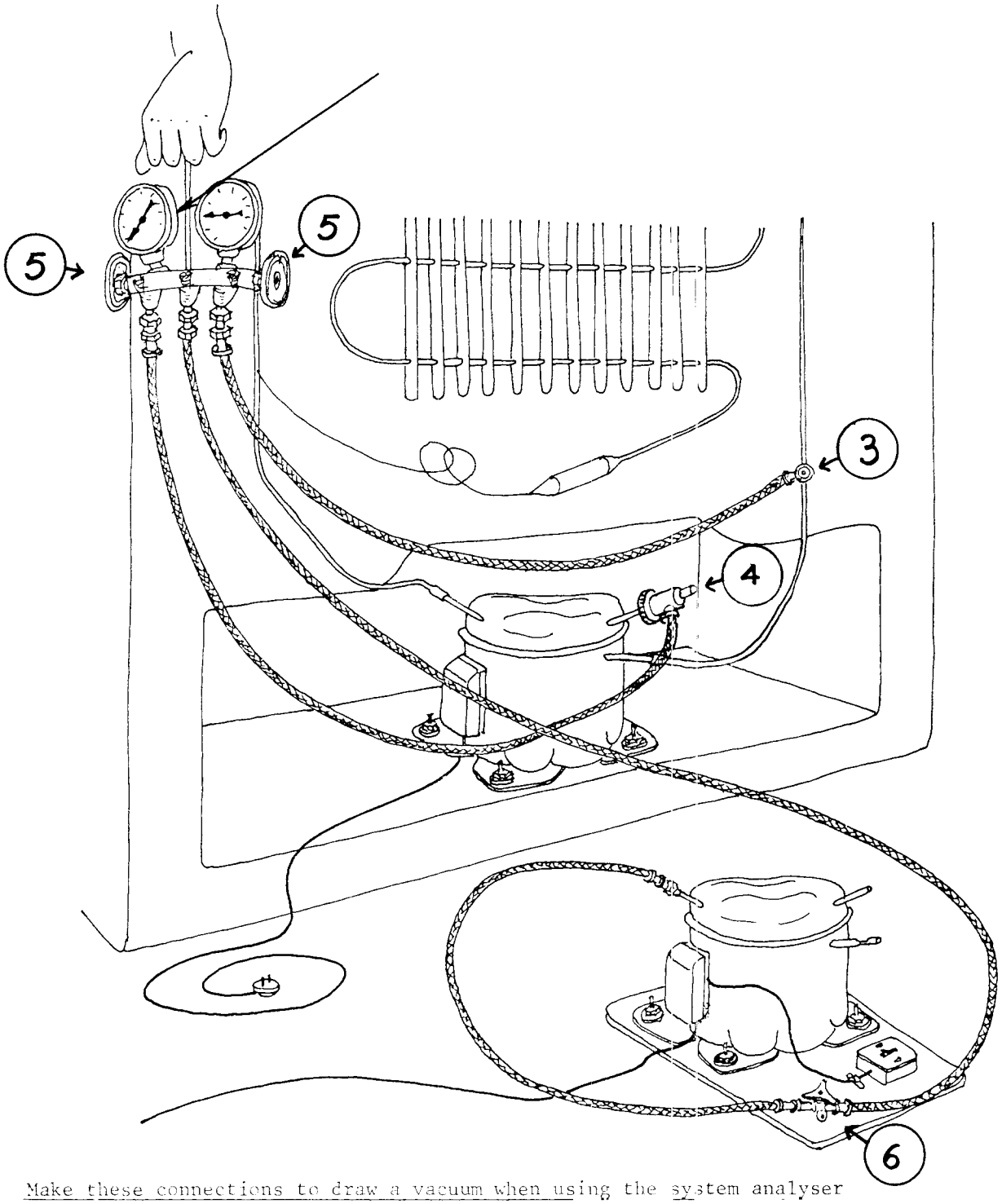
9. Switch the vacuum pump on (to remove air from the system) for 5 minutes.

10. Try to keep the vacuum pump cool. If it stops the vacuum pump motor is getting too hot. Take the action described below.

11. Fully close the quick coupler control valve. Switch off the vacuum pump and allow it to cool for 15 minutes. Re-start at 6 above.

12. After the vacuum pump has run for 5 minutes all air will have been removed. Fully close the quick coupler control valve to maintain the vacuum in the system. Unplug the vacuum pump and remove it.

13. Leave the quick coupler and charging hose in position.



Make these connections to draw a vacuum when using the system analyser

How to draw a vacuum on the system (using the system analyser)

1. If you have made connections to the system for measuring pressures with the system analyser (as described on pages 50 and 51) use these connections to draw a vacuum.

2. Make the connections shown in the drawing with charging hoses. The blue hose may be connected to the process tube by either a quick coupler or a line tap valve.

3. Fully open the control valve fitted to the top of the line tap valve on the delivery pipe (as described on pages 28 and 29).

4. Fully open the quick coupler control valve (or line tap valve) on the process tube.

5. Fully open the control valves on the system analyser (see page 48, drawing 4).

6. Fully open the line valve fitted to the vacuum pump.

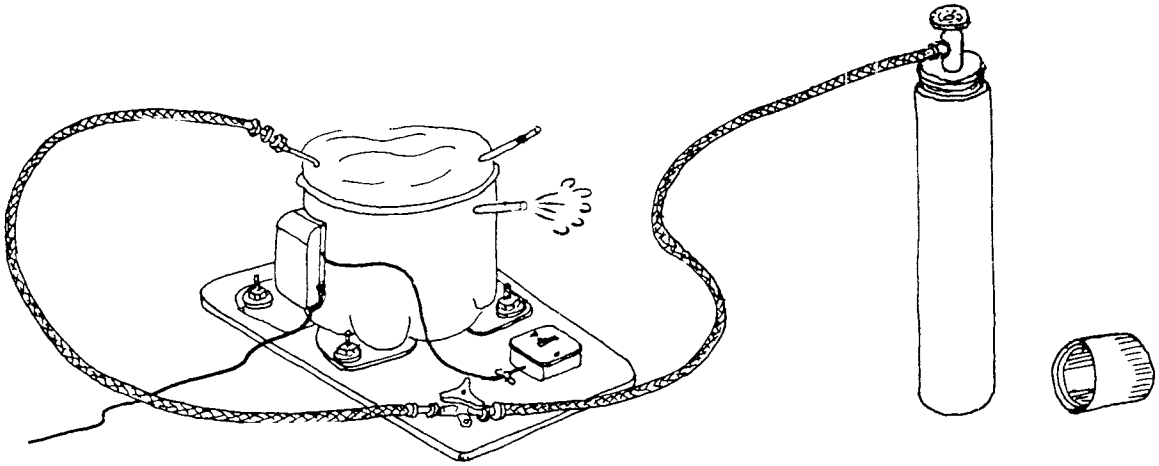
7. Draw a vacuum on the system (as described on page 57, paragraphs 7 to 10). Watch the blue gauge and draw the best possible vacuum (you should be able to get a vacuum of at least 28 inches (710mm or -0.9 bar)

8. Fully close the control valves on the system analyser. Switch off the vacuum pump and watch the blue gauge. The vacuum obtained must be maintained. If the pressure on this gauge rises, there is an air leak into the system. Tighten all the hose connections and re-start at paragraph 5.

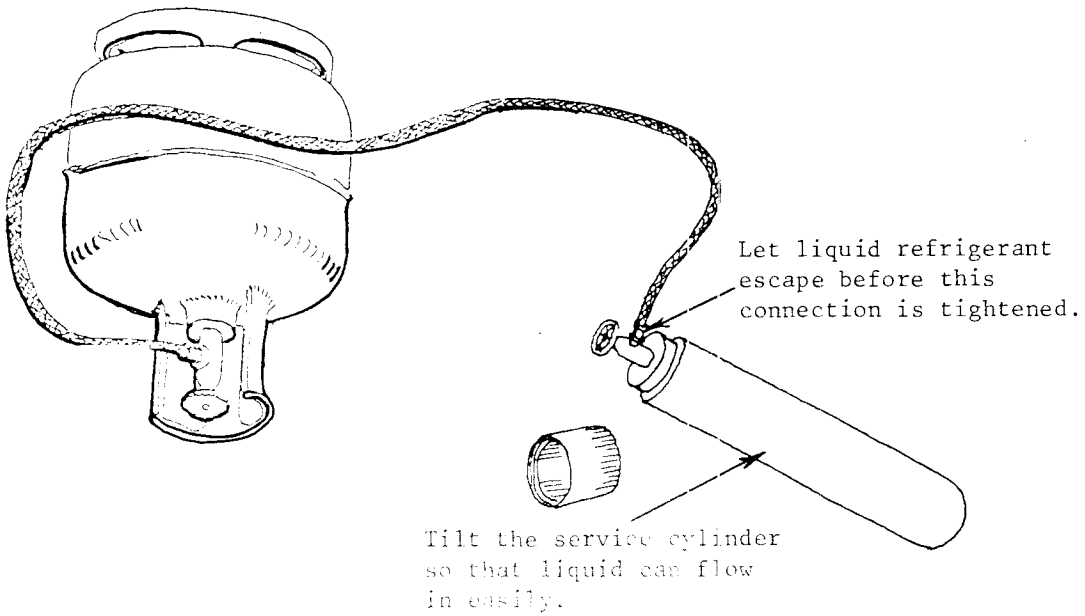
9. Remove the hose connecting the vacuum pump to the middle connection of the system analyser. Fit a cap nut to this connection and plug the open end of the hose.

10. Remove the vacuum pump.

11. Leave the system 'under vacuum'. It is ready for charging with refrigerant R12.



Use the vacuum pump to draw a vacuum inside the service cylirder.



Transfer liquid refrigerant from the supply cylinder into the service cylinder.

Section 10: How to charge the system with refrigerant R12

Refrigerant R12 is supplied in large cylinders. It must be transferred into smaller cylinders (that are easier to carry) before charging the refrigerator system. It is transferred as a liquid.

Read Section 5: 'refrigerant R12 safety precautions' before starting any of the work in this Section, (pages 38 to 41).

How to transfer liquid refrigerant into a small service cylinder

1. Connect the service cylinder to the vacuum pump with a charging hose. Make sure that the hose connections are tight. (See drawing).

2. Fully open the line valve fitted to the vacuum pump and the service cylinder control valve.

3. Switch the vacuum pump on and let it draw a vacuum on the service cylinder for 5 minutes.

4. Fully close the service cylinder control valve. Switch off the vacuum pump and remove it.

5. Cool the charging cylinder before filling it with liquid refrigerant R12. Cool it with ice or in a spare refrigerator. When cold it will hold more refrigerant R12.

6. If the refrigerant R12 supply cylinder has two outlets, connect the charging hose and service cylinder to the outlet marked "liquid". Keep the supply cylinder upright.
   If the supply cylinder has one outlet, connect the charging hose and service cylinder and turn the supply cylinder upside down (see drawing).

7. Remove all air from the charging hose as follows:

   (a) loosen the connection at the service cylinder;
   (b) open the supply cylinder valve very slightly;
   (c) allow escaping liquid refrigerant R12 to remove air at the loose connection;
   (d) tighten the hose connection at the service cylinder.

8. Open fully the supply cylinder valve then the service cylinder control valve. Keep the service cylinder lower than the supply cylinder. Liquid will flow from the supply cylinder into the service cylinder. The flow can be felt by touching the charging line.

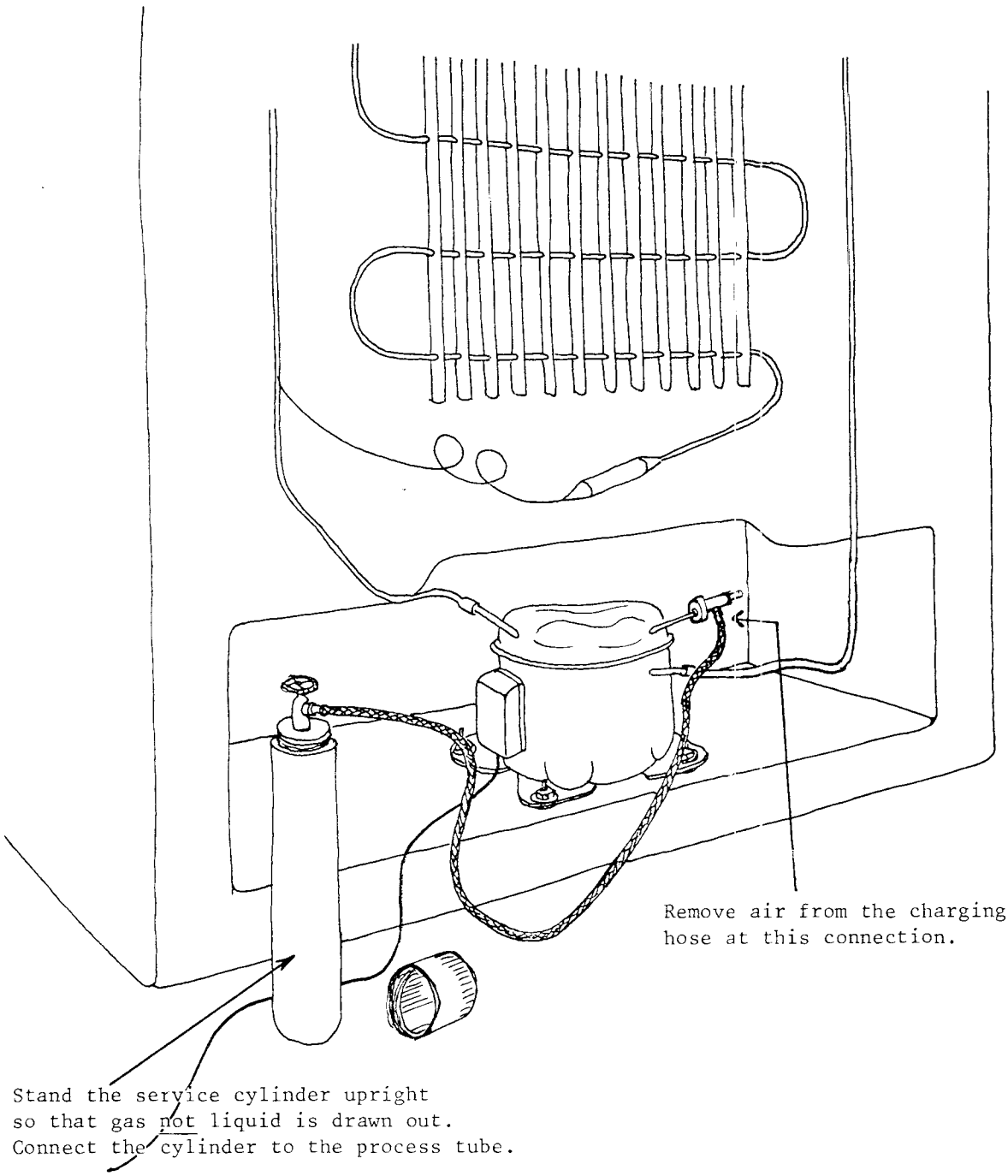
9. When this flow stops, take the following action:

   (a) close the supply cylinder valve fully;
   (b) close the service cylinder valve fully;
   (c) remove the charging hose carefully. It will have high pressure liquid refrigerant R12 inside.
   (d) re-fit any sealing caps to the connections on both the supply and service cylinders.

10. To fill the service cylinder with a measured amount of refrigerant R12:

   - weigh the empty cylinder with a weighing scale;
   - check the weight as the cylinder is filled. Do not exceed the weight of charge marked on the cylinder.

11. Caution. Make quite sure that you have not overfilled the service cylinder (see page 40, drawing 1 and page 41, instruction 10).



'Make these connections when charging into the process tube'

How to charge the system (without using the system analyser)

1. The system is charged with refrigerant R12 after a vacuum has been drawn on it. The necessary connections for charging were left after the vacuum was drawn (see page 57, paragraph 13).

2. Find the type of refrigerant R12 from the manufacturer's data plate.

3. Transfer refrigerant R12 into the service cylinder (as described on pages 60 and 61).

4. Connect the charging hose to the service cylinder control valve (see drawing). Keep the service cylinder upright so that refrigerant R12 gas not liquid flows into the system.

5. Remove air which will have entered the charging hose as follows:

   - Loosen, but do not remove, the hose connection at the quick coupler.
   - Open the service cylinder control valve slightly.
   - Allow escaping refrigerant R12 to remove air at the loose connection.
   - Close the service cylinder valve and tighten the hose connection.

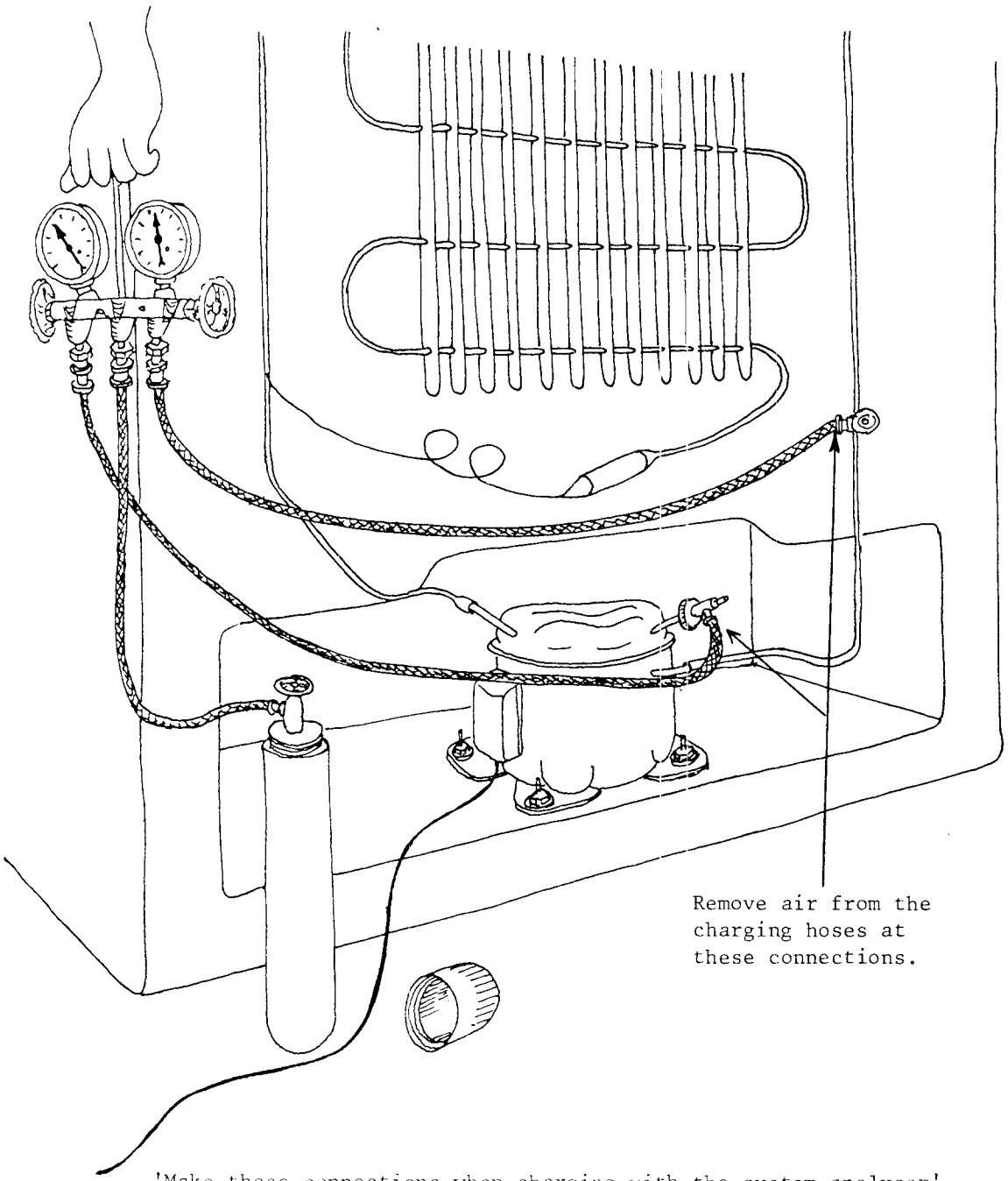
6. Fully open the control valve on the quick coupler. Open the service cylinder valve slightly and control the flow of refrigerant R12 into the system carefully. If the system is under vacuum, you must let some refrigerant R12 into the system before starting the compressor.

7. Start the compressor and draw refrigerant R12 into the system. Stop the compressor when ice or frost starts to form on the suction pipe where it comes out of the evaporator.

8. Fully close the quick coupler control valve and the service cylinder valve. (See page 1 in booklet "Faults and faults finding").

9. Check all the symptoms at correct charge and check for leaks.

10. Crimp the process tube twice. Remove the quick coupler and braze the end of the process tube. Check again for leaks.



'Make these connections when charging with the system analyser'

How to charge the system (using the system analyser)

1. This method of charging is used when the pressures in the system are measured to find the correct charge (this is explained in Part B).

2. Find the type of refrigerant R12 from the manufacturer's data plate and transfer it into the service cylinder (as explained on pages 60 and 61).

3. Make the hose connections shown in the drawing. The blue hose may be connected to the process tube by either a quick coupler or a line tap valve.

4. Remove air from the hoses (following the procedure explained on page 63, paragraph 5) at the points shown on the drawing.

5. Fully open the following valves: line tap control; quick coupler (or line tap valve).

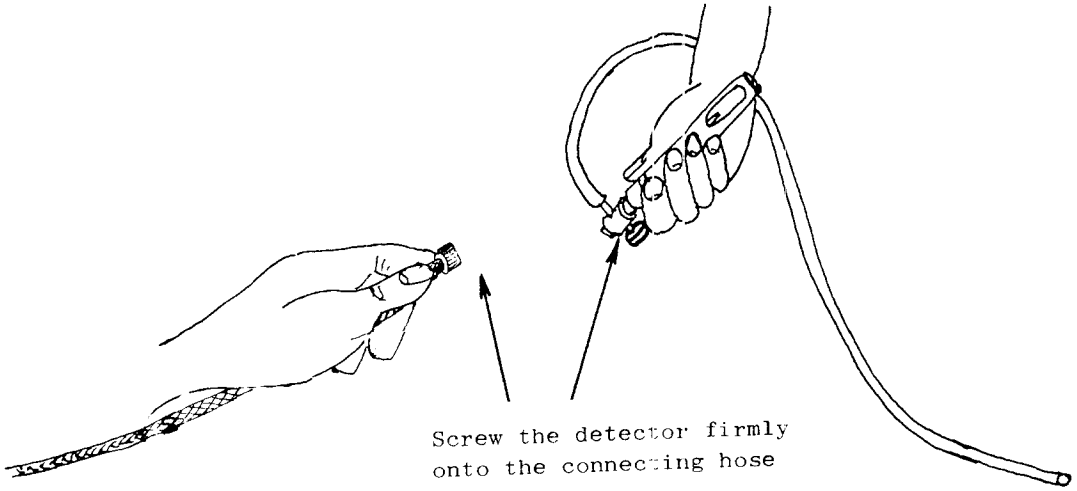
6. Set the system analyser valves to the charging position. (Page 48, drawing 3). Blue gauge valve open. Red gauge valve closed.

7. Open the service cylinder valve slightly. Control the flow of refrigerant R12 into the system carefully. You must let some refrigerant R12 into the system before starting the compressor.

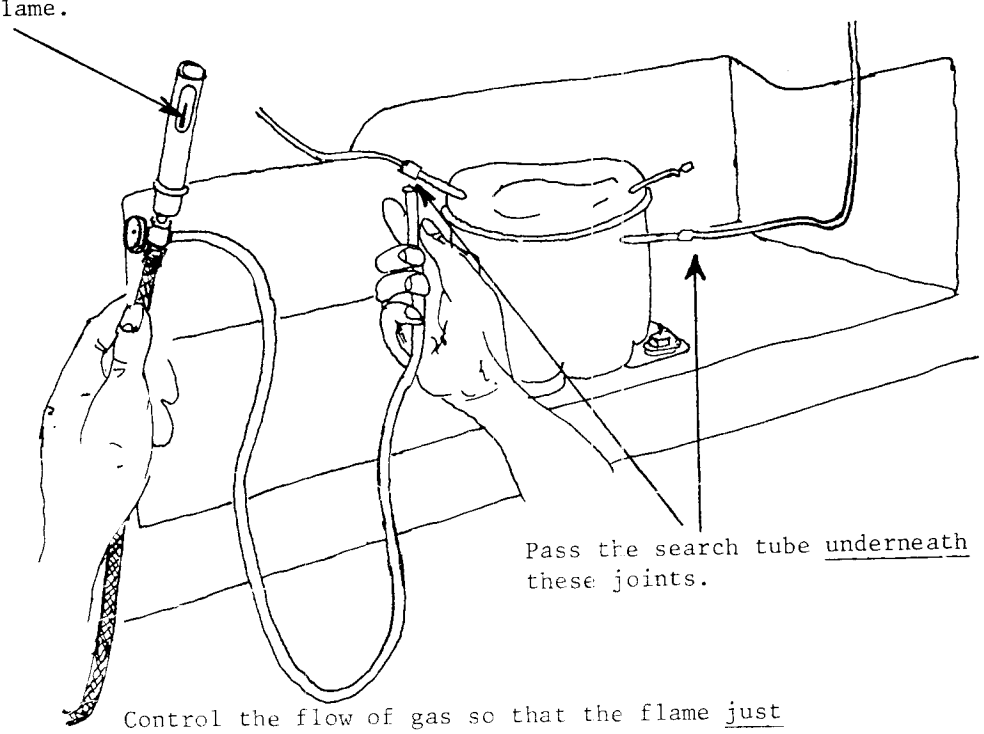
8. Start the compressor and draw refrigerant R12 into the system. Stop the compressor when ice or frost starts to form on the suction pipe where it comes out of the evaporator, or when the system analyser gauges are showing normal running pressures. To get a correct reading on the blue gauge, close the blue gauge valve for 5 minutes and then read the pressure.

9. Fully close the following valves: service cylinder control; line tap control; quick coupler (or line tap control).

10. Leave the line tap and quick coupler connection in position. Remove the service cylinder, system analyser and charging hoses.



Hold the detector so that
you can see the flame.



Section 11: How to find refrigerant R12 leaks

1. Refrigerant R12 leaks are found either with a propane type detector (see drawing), or with a spirit type detector.

   Note: If you do not have a leak detector, make a check at the places where leaks may occur using soapy water. If there is a leak, the soapy water will bubble.

2. If your leak detector is the propane type, screw the detector firmly onto the connecting hose and turn it so that you can see the flame tube when working. Check all connections for gas leaks with soapy water before lighting the leak detector.

3. Open the regulating valve sufficiently to get a small flow of gas.

4. Light the gas and control the size of flame with the regulating valve.

5. On both types of detector the flame must be large enough to fill the opening in the flame tube (so that it can be easily seen).

6. The flame must not be large enough to cause the detector to get hot.

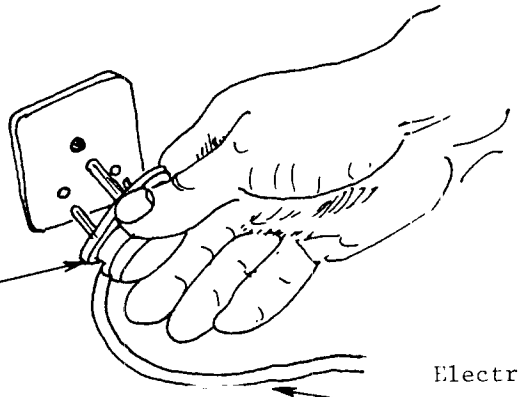
7. The detector should burn with a pale blue flame. If it is green refrigerant R12 gas is present in large quantities. You cannot find leaks in the system if there is a lot of gas in the room.

8. Turn off the detector flame. Ventilate the room thoroughly. Open all windows and doors. When you think all of the gas has gone start again.

9. When the detector is burning with a pale blue flame, pass the open end of the search tube underneath joints where leaks are likely to occur (see drawing).

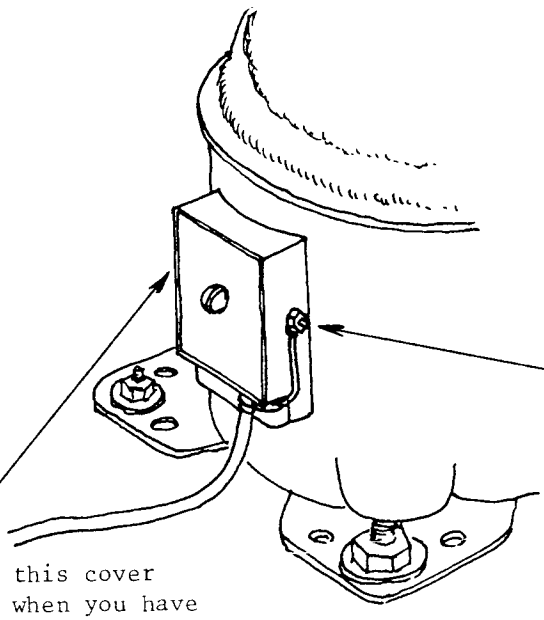
10. When the flame turns green refrigerant R12 gas is being drawn up the search tube. This is the point where refrigerant R12 is leaking.

    Remember: The leak detector can only find the leak if there is some refrigerant R12 gas in the system.



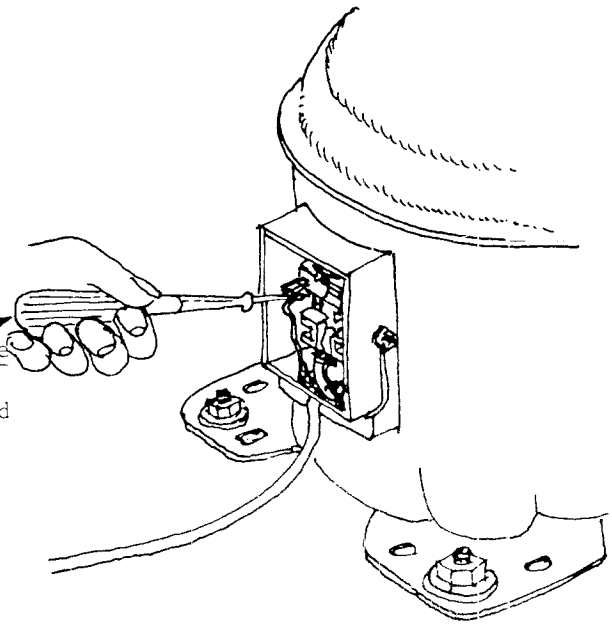
Pull out the plug before starting any electrical repair.

Electric cable that is worn, damaged or brittle is dangerous. Replace it.



An earthing wire must be connected tightly to a metal part of the compressor.

Replace this cover tightly when you have finished work.



Use a screwdriver with an insulated handle when you have to work with the refrigerator plugged into its socket.

Section 12: Electrical safety precautions

1. Make quite sure that the refrigerator plug is removed from its socket before starting any electrical repair.

2. If an earthing connection is provided make sure that it is connected tightly to the following:

   Refrigerator plug earth pin. A metal part of the compressor or refrigerator.

3. Discharge capacitors (as explained on page 95) before handling them.

4. Handle old brittle electric cable carefully. Replace it as soon as possible.

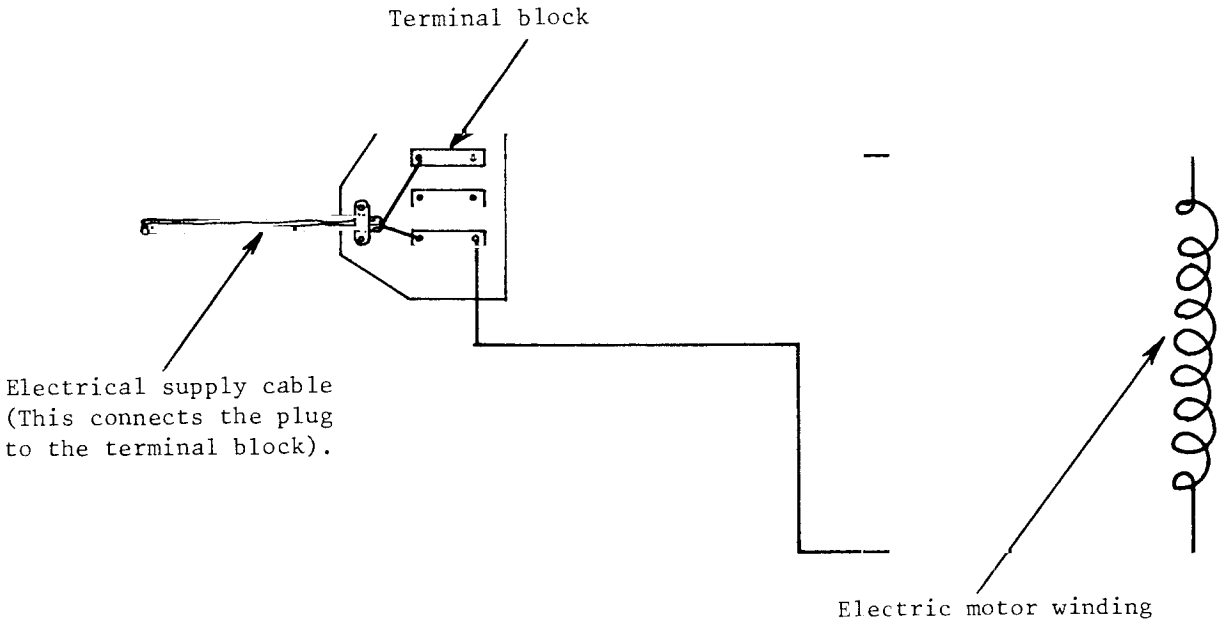
5. Always replace terminal box covers tightly, so that they will not work loose.

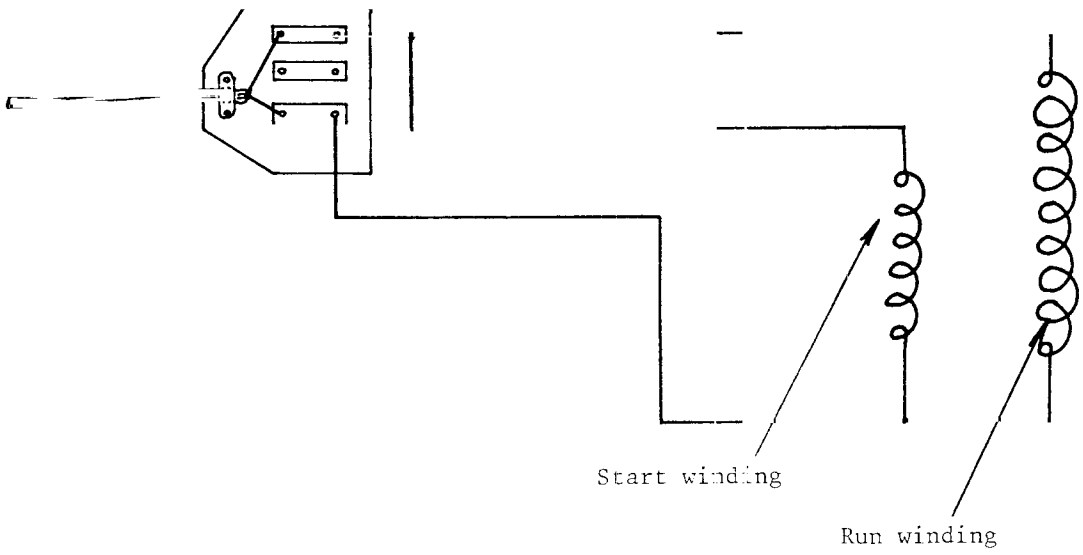
6. Make sure that a fuse of the correct capacity is used.

When you have to work with the refrigerator plugged into its socket:

A. Do not stand on wet ground. Use a piece of wooden board or rubber sheet to stand on.

B. Use insulated tools and rubber-soled shoes.


Terminal block
Electrical supply cable
(This connects the plug
to the terminal block).
Electric motor winding


Start winding
Run winding

Section 13: Electric circuits and wiring diagrams

1. The electric power required to drive the compressor motor is connected as shown in the top drawing.

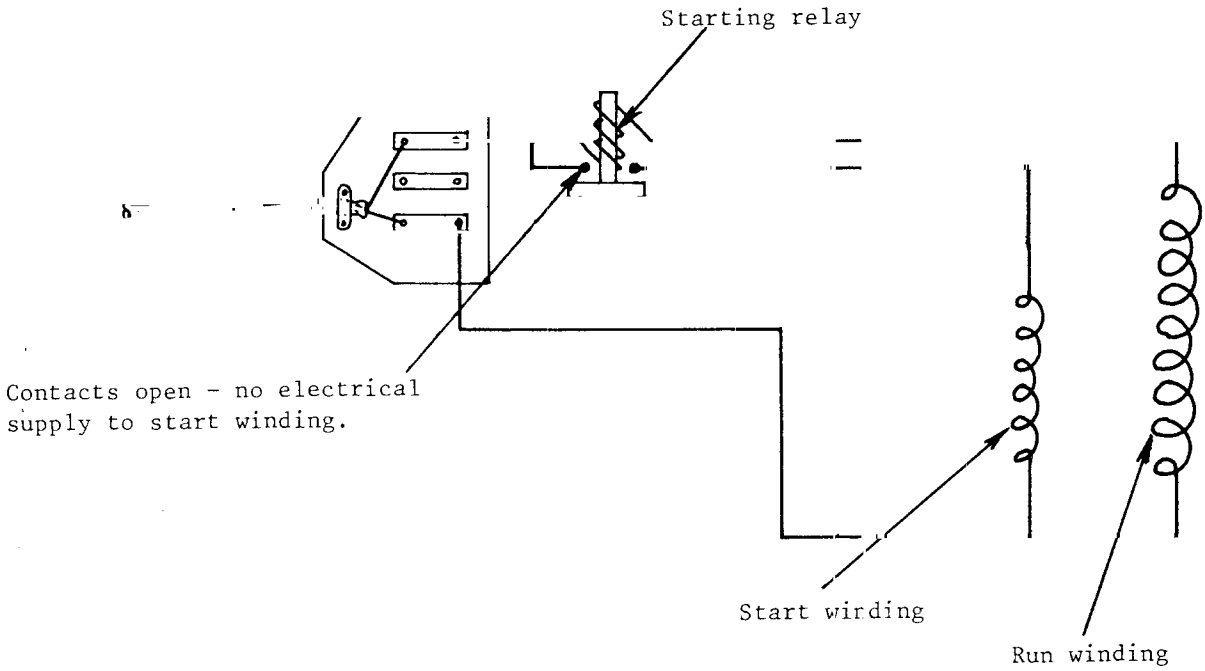
2. The refrigerator plug is connected to a terminal block. This block is joined to a coil of wire inside the motor. This coil of wire is called a winding.

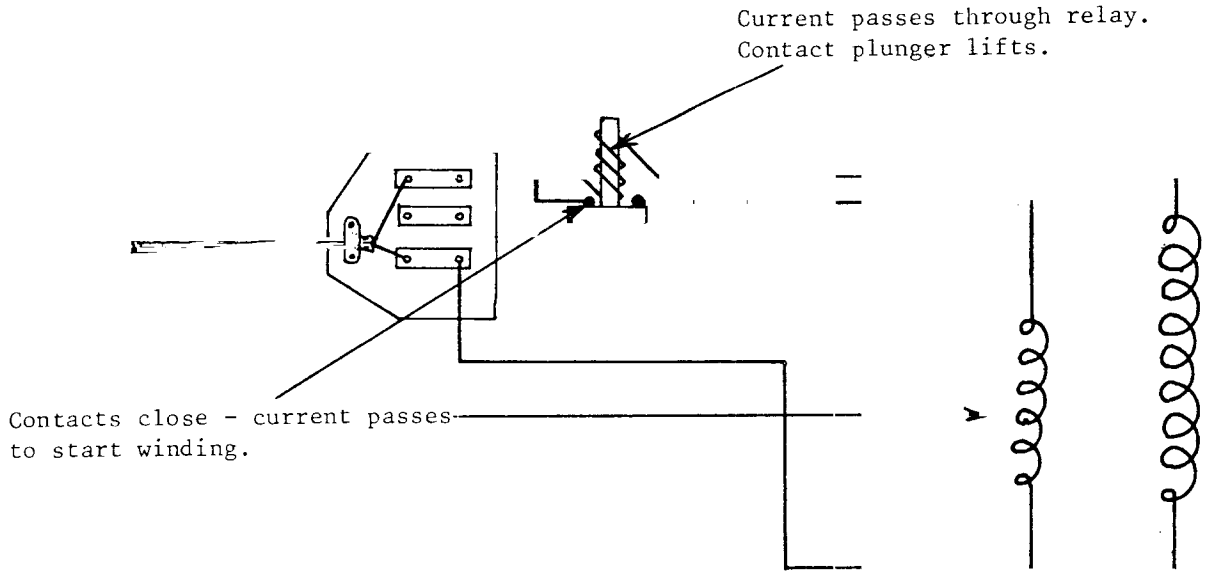
3. To help the compressor motor start a second winding is required. This is shown in the bottom drawing.

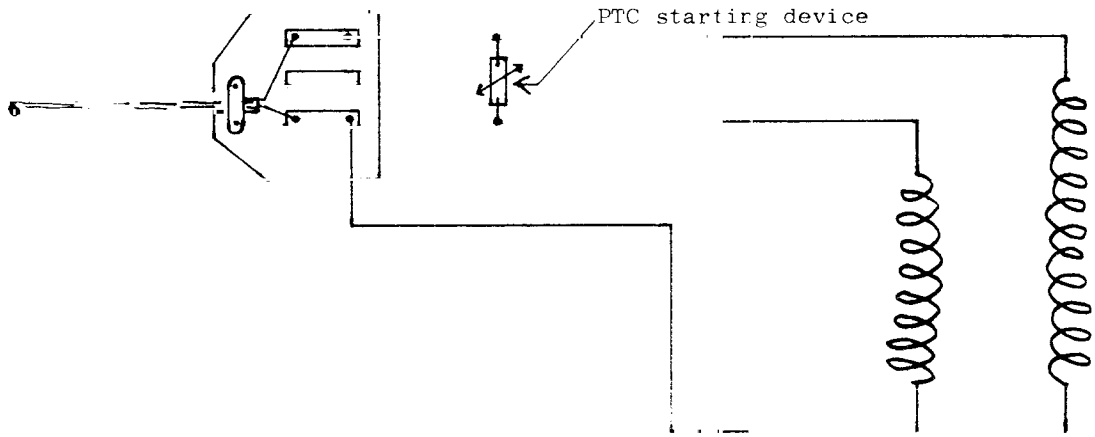
4. The electrical coil which drives the compressor motor is called the 'run winding'. Electric current will always flow through this winding when the compressor is running.

5. The electrical coil which helps the compressor to start is called the 'start winding'. Electric current flows through this winding for a few seconds only. (When the compressor first starts).

6. Not all refrigerators are fitted with a terminal block. The connections from the plug are often joined to other components inside the terminal box. The principle of 'run winding' and 'start winding' is always the same, whichever method of connection is used.


Starting relay
Contacts open - no electrical supply to start winding.
Start wirding
Run winding


Current passes through relay. Contact plunger lifts.
Contacts close - current passes to start winding.


PTC starting device

The starting circuit

The starting circuit is controlled by either a magnetic (or solenoid) starting relay or a 'PTC' starting device. These are shown at the bottom of page 82.

Starting with a magnetic (sofenoid) starting relay

1. The electrical supply to the run winding passes through a small coil (called a starting relay) as shown in the top drawing.

2. When the compressor is switched on a very high electric current passes through the relay coil. The magnetic action of this current lifts a contact plunger.

3. When these contacts close electric current passes to the start winding as shown in the middle drawing.

4. The compressor motor now starts, with electric current going to the start winding and the run winding.

5. As soon as the compressor reaches its running speed the relay contacts open again, to disconnect the start winding. The compressor motor runs with electric current going to the run winding only.

Starting with a 'PTC' starting device

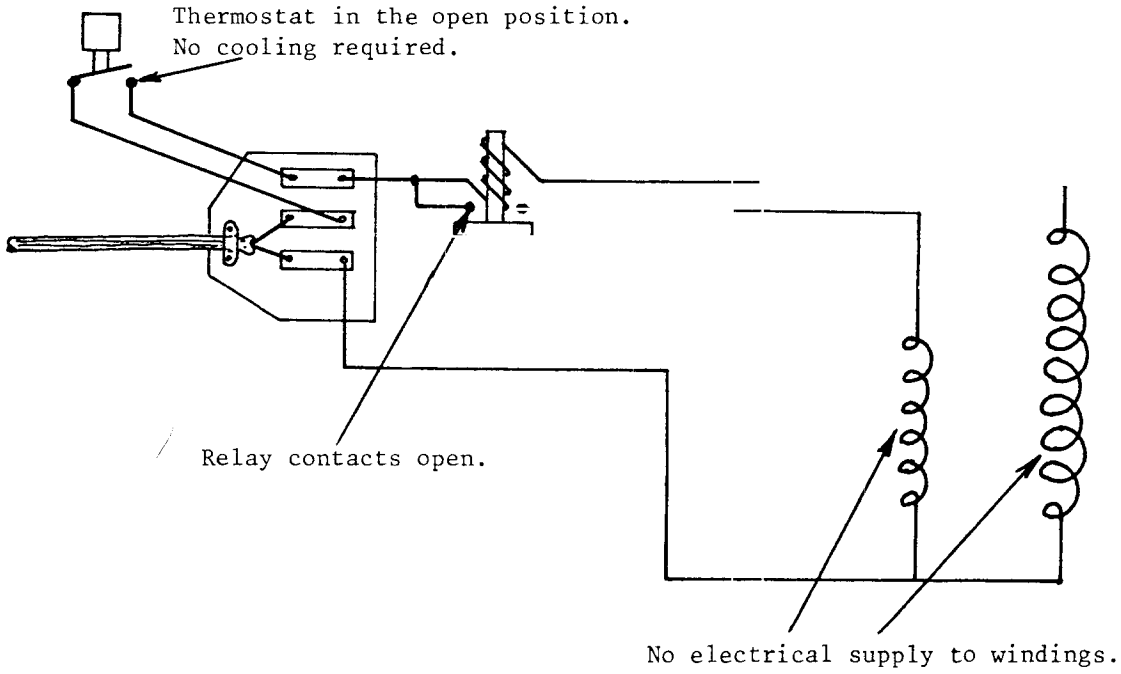
1. 'PTC' stands for 'Positive Temperature Coefficient'. When this device is cold, it has a low resistance and current can flow through it. When it is hot, it has a high resistance and current cannot flow through it.

2. When the compressor is switched on electric current passes to the run winding and to the start winding (through the cold PTC device). The motor starts.

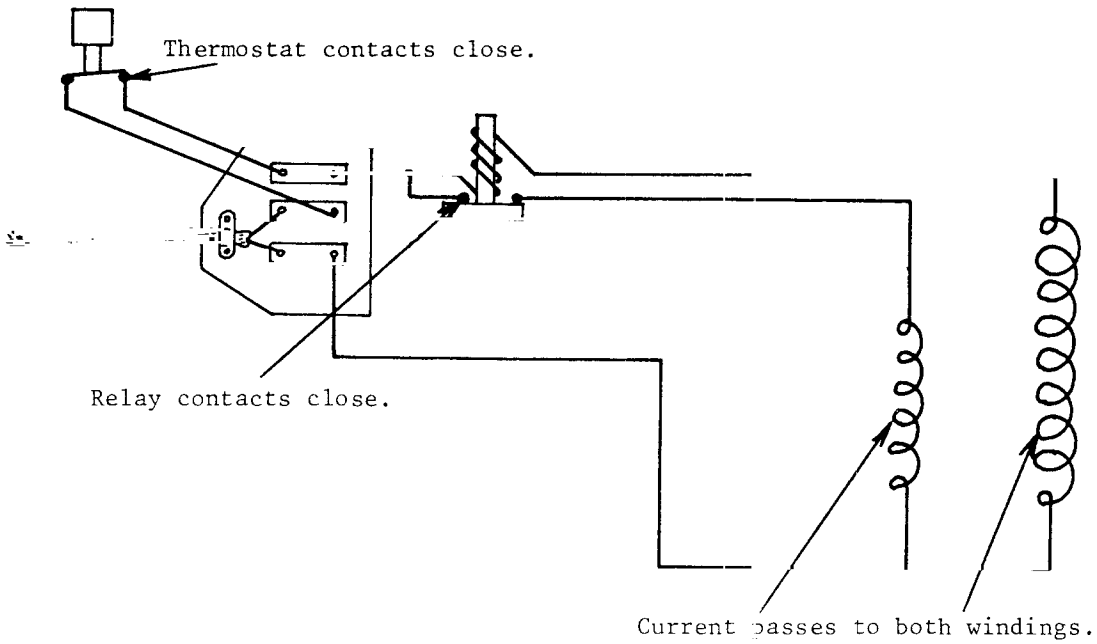
3. The PTC device heats up very quickly and its resistance gets very high. The current cannot flow through it now and so the start winding is disconnected.

4. The compressor motor runs with electric current going to the run winding only. A very small current through the PTC device keeps it warm enough so that the start winding stays disconnected.

   Note: After the compressor stops, the PTC device needs a few minutes to cool off before the compressor can start again.



Circuit position when compressor is 'not' running.



Circuit position when compressor 'starts'.

The control circuit

1. The temperature inside the refrigerator is controlled by a switch called the thermostat.

2. The thermostat is connected at the terminal block between the supply and the connection to the compressor winding (see drawings).

3. When the temperature inside is low enough (+4°C to +8°C for vaccine) the thermostat contacts are open and the electric current is not connected to the compressor motor.

4. The compressor motor will not run. This position is shown in the top drawing.

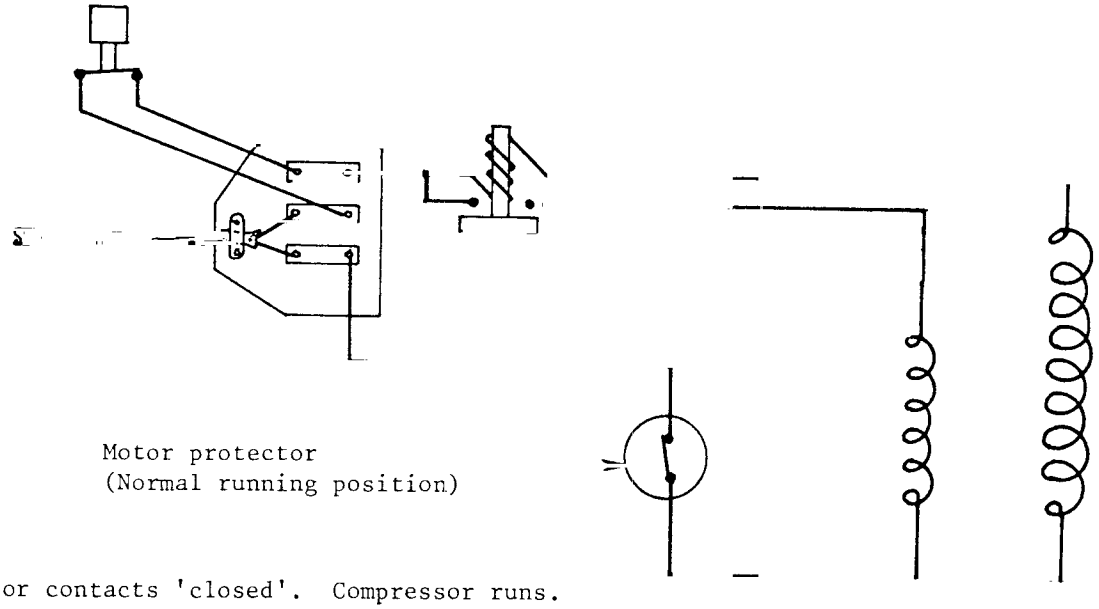
5. When the temperature inside the refrigerator rises the following action occurs:

   - the thermostat contacts close;
   - electric current flows to the main winding through the relay coil;
   - the relay plunger lifts;
   - the relay contacts close;
   - electric current passes to the start winding;
   - the compressor motor starts.

   This position is shown in the bottom drawing.

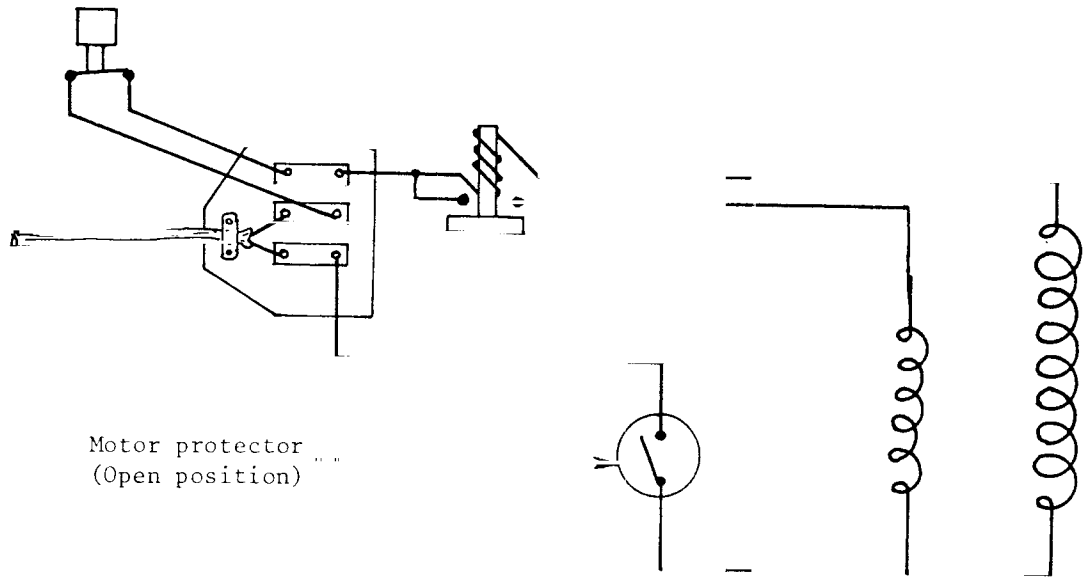
6. The running of the compressor cools the inside of the refrigerator until the temperature is low enough for the thermostat contacts to open. The compressor then stops.

7. The temperature at which the thermostat starts and stops the compressor is fixed by the position of the thermostat control knob.



Motor protector
(Normal running position)

Protector contacts 'closed'. Compressor runs.



Motor protector
(Open position)

Protector contacts 'open'. Compressor stops.

The motor protector circuit

1. An electric motor 'protector' is fitted to the electrical circuit in the position shown in the drawings. Note that the total current (start and run windings) passes through the protector.

2. The protector reduces the possibility of damage to the electric motor.

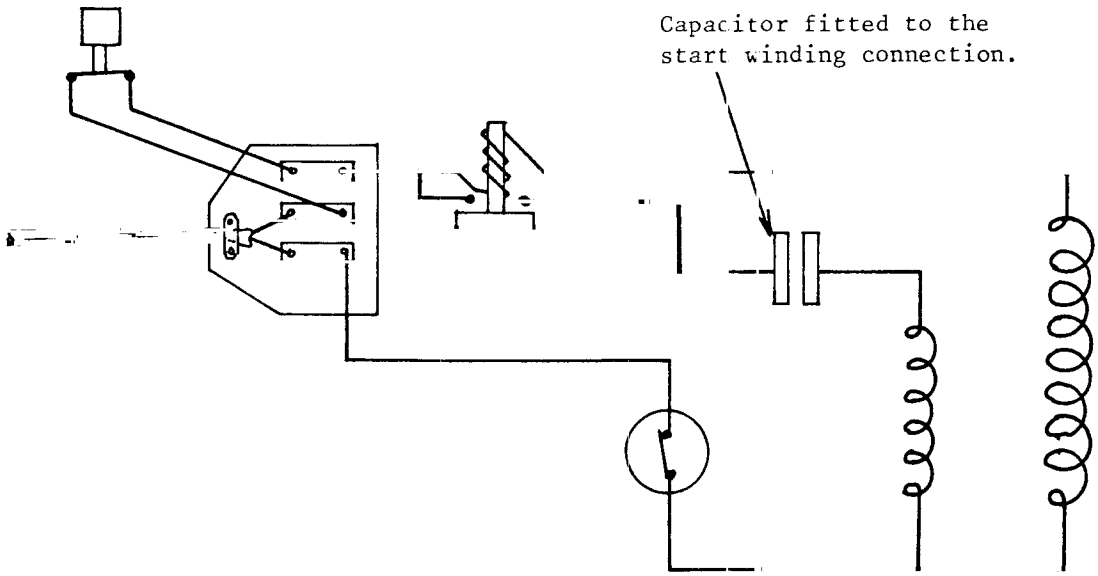
3. During normal running the protector contacts are closed, as shown in the top drawing.

4. If either of the following conditions occur heat will cause the contacts to open.

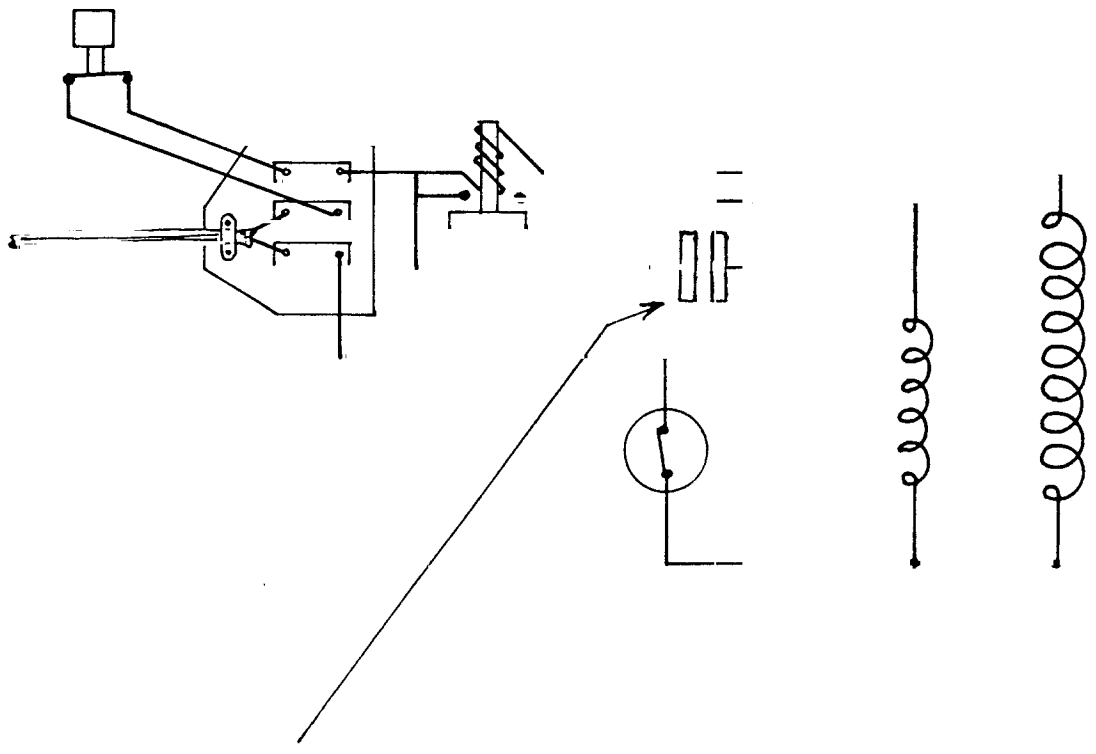
   (a) too high an electrical current flowing through the motor windings;
   (b) compressor overheating.

5. When the protector contacts open, the electrical circuit is disconnected and the compressor stops. This is shown in the bottom drawing.

6. After the protector has cooled (5 to 10 minutes) the contacts close and the compressor starts again.



Circuit position for a 'starting' capacitor.



Circuit position for a 'running' capacitor.

The capacitor circuit

1. Capacitors are sometimes fitted to compressors if higher starting power is needed or if the cooling requirement is very high (large refrigerators or high air temperatures).

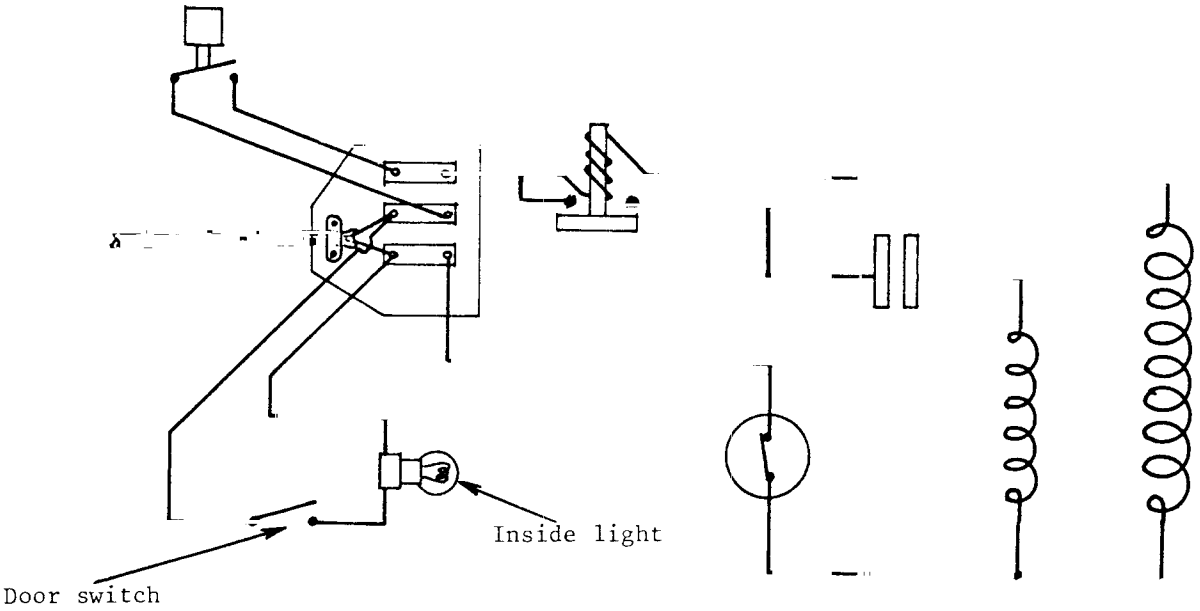
2. Capacitors 'store' electrical energy.

3. In some copmpressors starting capacitors are fitted in the connection to the start winding, as shown in the top drawing.

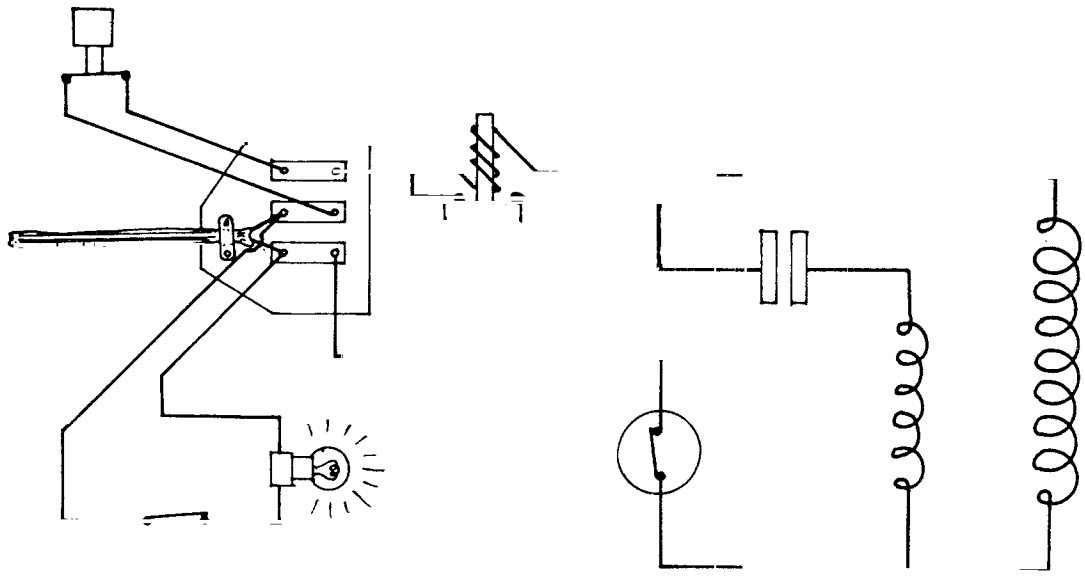
4. This capacitor makes it easier for the compressor to start. It reduces the possibility of damage to the start winding.

5. In some compressors, running capacitors are fitted in series with the start winding, as shown in the bottom drawing.

6. This capacitor allows the start winding to take some of the load of the run winding. This reduces the possibility of damage to the run winding.



Circuit position with door closed - light 'off'.



Circuit position with door open - light 'on'.

The inside light circuit

1. Some refrigerators are fitted with a small light inside.

2. This light produces heat and it must be switched off unless the user needs to see inside the refrigerator.

3. A pressure operated switch is controlled by the opening and closing of the refrigerator door.

4. When the door is closed the switch contacts are open and the inside light is off, as shown in the top drawing.

5. When the door is opened the switch contacts close and the light comes on, as shown in the bottom drawing.

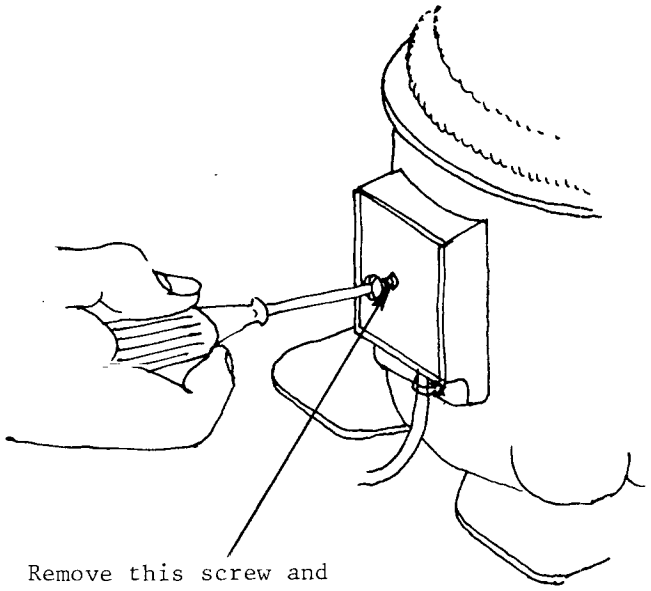
6. Note that the drawings opposite show the complete electrical circuit for a refrigerator fitted with the following components:

Terminal block.	Magnetic type starting relay.
Thermostat.	Motor protector.
Starting capacitor.	Inside light.

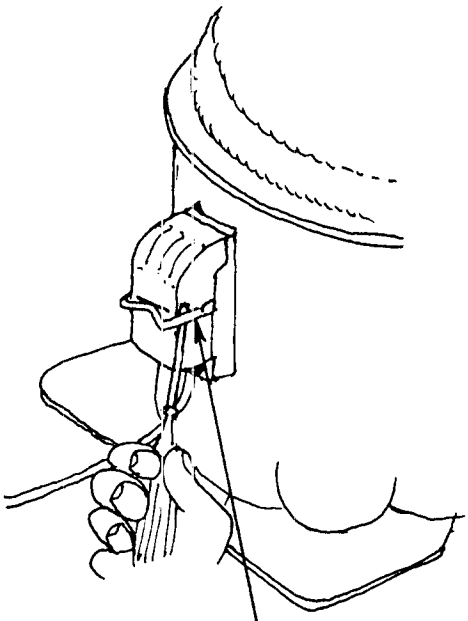
7. The top drawing shows the thermostat and relay contact positions with the compressor switched off.

8. The bottom drawing shows the same contact positions when the compressor first starts.

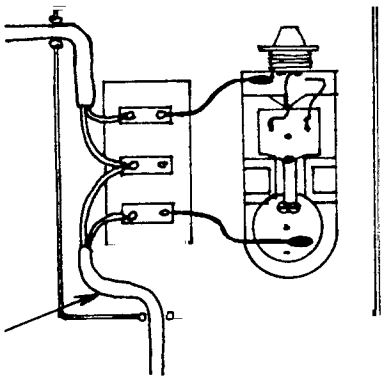
9. Note: The inside light circuit is separate from the circuit to the compressor motor. It does not matter whether the compressor is running or not for the light to go on and off.



Remove this screw and
keep it in a safe place.

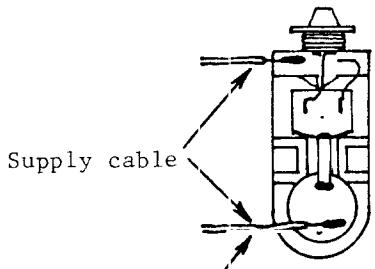


Try to lift up this clip by hand.
If it is tight and you have to use
a screwdriver, be careful not to
damage the cover.



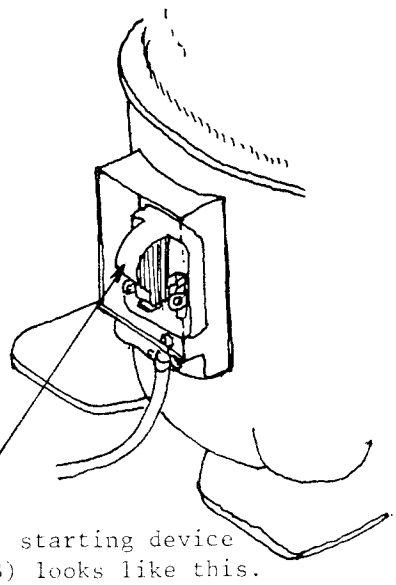
Supply cable

If a terminal block is used, the wiring
will look like this.

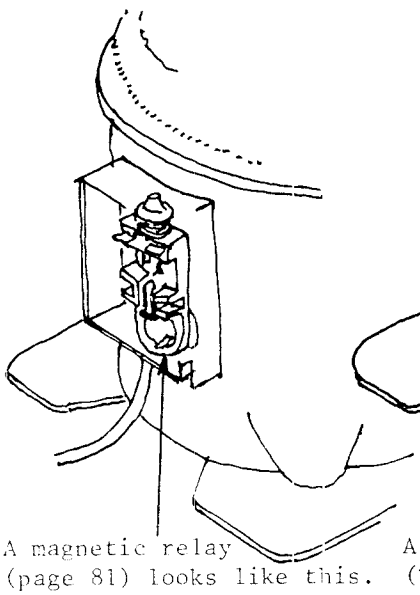


Supply cable

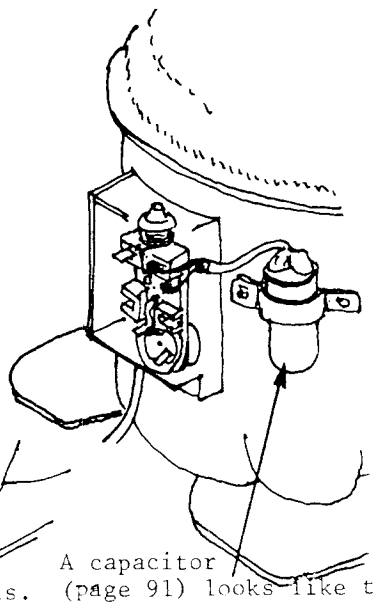
If a terminal block is not used,
the wiring will look like this.



A P.T.C. starting device
(page 83) looks like this.



A magnetic relay
(page 81) looks like this.



A capacitor
(page 91) looks like this.

Section 14: Electrical testing

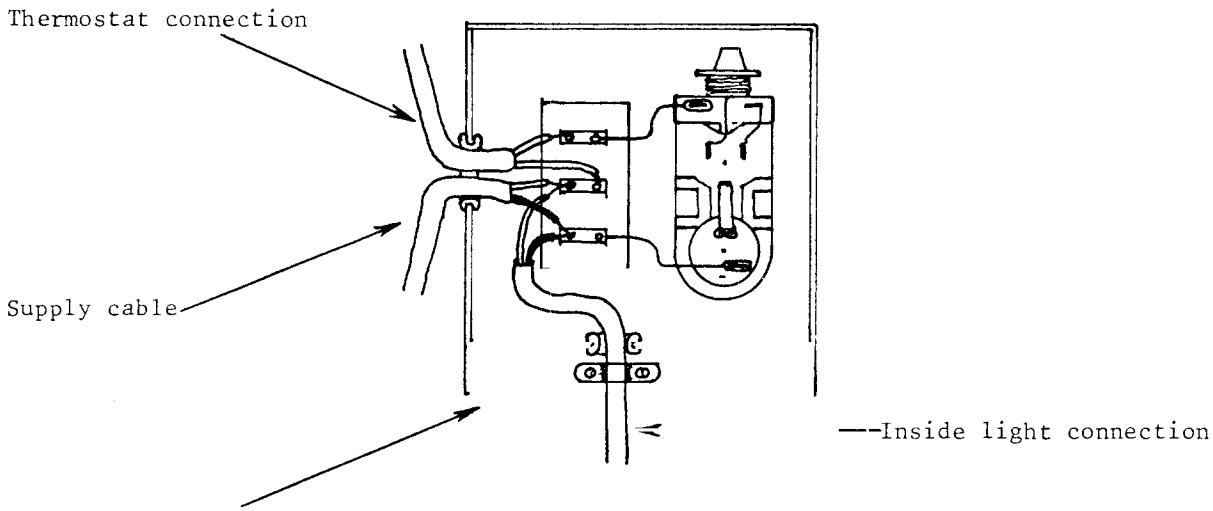
1. Before attempting any electrical test or repair you must find which of the components described in this Section are fitted to the refrigerator.

2. Pull the plug out of the electrical socket.

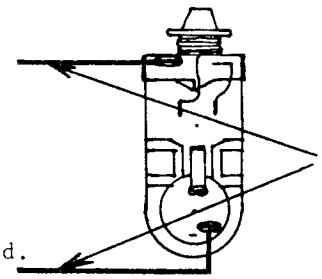
3. Remove the terminal box cover. It is usually held in place by either a screw or a clip, as shown in the top drawings.

4. Examine the parts inside the terminal box. Compare them with those shown in the middle and bottom drawings.

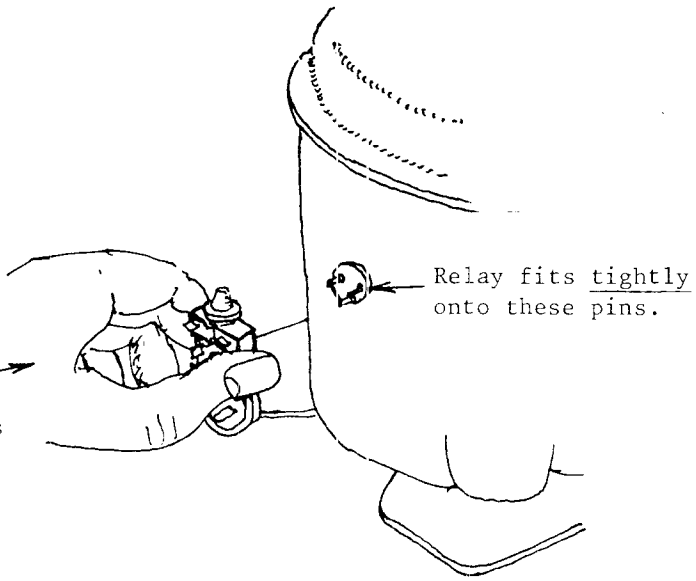
5. Make quite sure that you know what components are fitted before you remove them.



These are the usual connections when a terminal block is fitted.



These are the usual connections when a terminal block is not fitted.



After removing the connecting wires hold the relay tightly and pull it off the compressor motor pins.

How to remove a magnetic (solenoid) type relay

1. This type of relay is often fitted with the motor protector (see page 77) as one component.

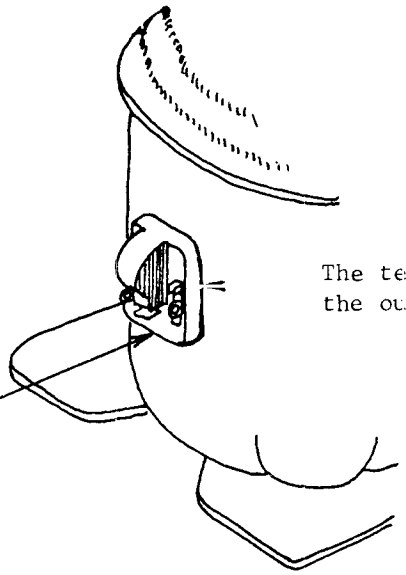
2. If a terminal box is fitted the usual arrangement for the electrical connections will be as shown in the top drawing.

3. If a terminal box is not fitted the usual arrangement for the electrical connections will be as shown in the middle drawing.

4. If the electrical connections are not as shown in these drawings, make a drawing on a piece of paper of the position of each connection to the relay.

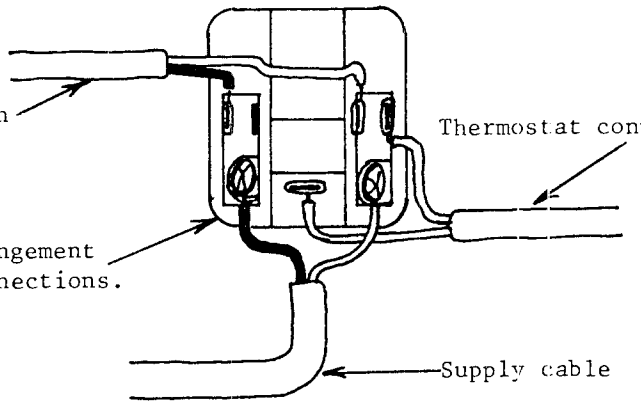
5. Remove the connections to the relay. Make quite sure that you know how to re-fit them correctly.

6. Pull the relay carefully off the compressor motor pins as shown in the bottom drawing.



The terminal box fits over the outside.

This is a P.T.C. type starting device fitted to a compressor.

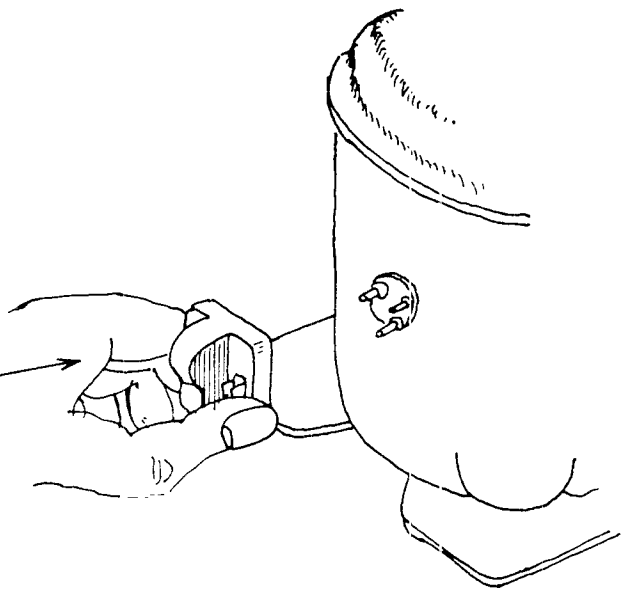


Inside light connection

Thermostat connection

This is the usual arrangement for the electrical connections.

Supply cable



Remove the starting device in the same way as a relay (page 84).

How to remove a PTC type starting device

1. If the component attached to the outside of the compressor looks like the top drawing it is a PTC type starting device (see page 77). The motor protector with this type of starting device is usually inside the compressor casing.

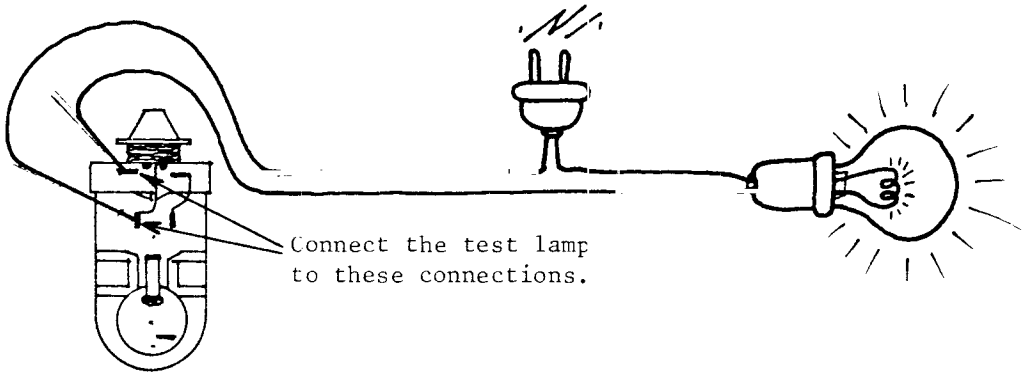
2. The usual arrangement for the electrical connections will be as shown in the middle drawing.

3. If the electrical connections are not as shown in these drawings, make a drawing on a piece of paper of the position of each connection to the starting device.

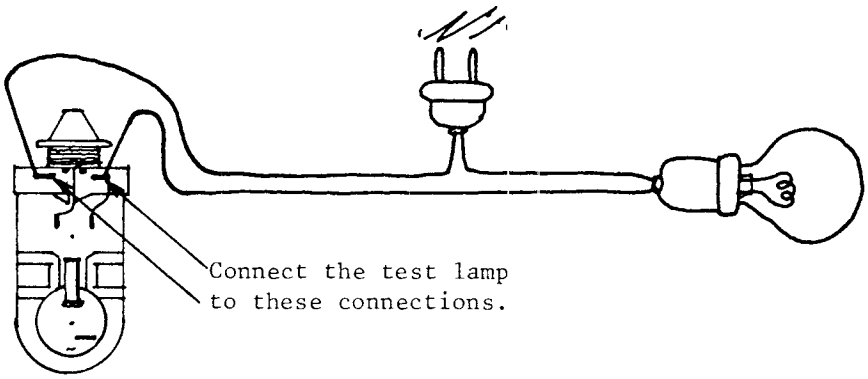
4. Remove the connections to the starting device. Make quite sure that you know how to re-fit them correctly.

5. Pull the starting device carefully off the compressor motor pins, as shown in the bottom drawing.

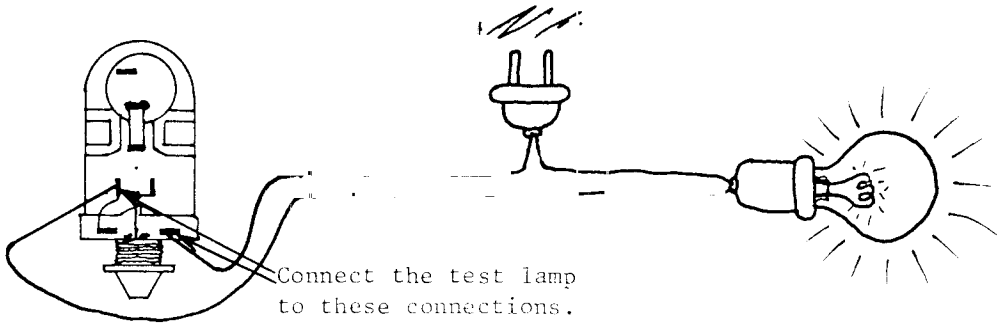
Note: A starting capacitor can not be fitted if the compressor has a PTC type starting device.



Stand the relay upright for this test.



Stand the relay upright for this test.



Stand the relay upside down for this test.

How to test a magnetic relay

1. Remove the relay from the compressor (see pages 84 and 85).

2. Plug a powered test lamp into a socket. Touch the two bare ends together and make sure that the lamp lights. Always test the lamp in this way before using it. Repair it if necessary.

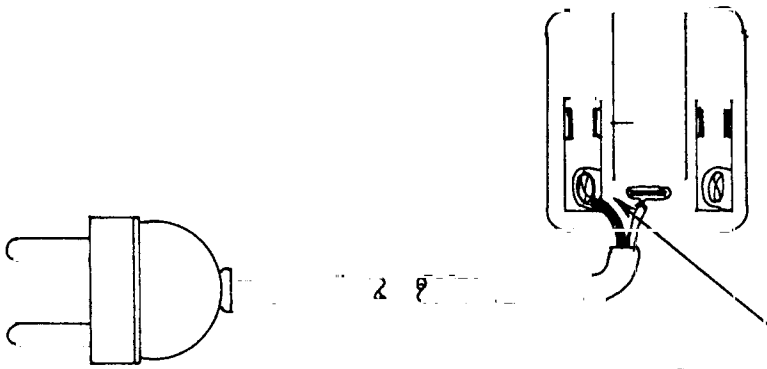
3. Work carefully and carry out the following tests.

4. Connect the bare ends to the connections shown in the top drawing. The lamp should light, to show that there is continuity in the relay coil. If it does not, the relay is faulty.

5. Connect the bare ends to the connections shown in the middle drawing. The lamp should not light, to show that the relay contacts are open. If the lamp does light the relay is faulty.

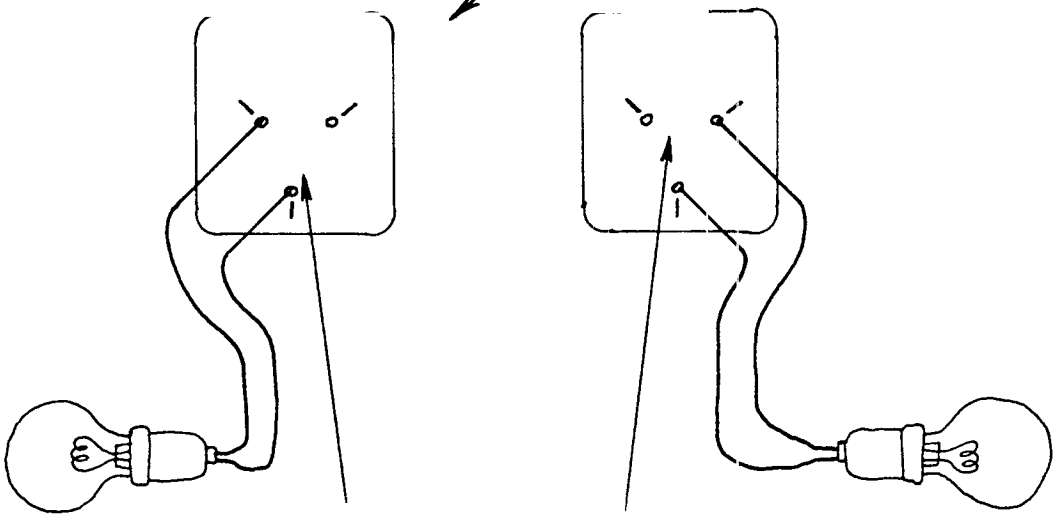
6. Turn the relay upside down. Shake it until you hear the plunger move. Connect the bare end to the connections shown in the bottom drawing. The lamp should light, to show that the relay contacts are now closed. If it does not light, the relay is faulty.

7. How to repair a faulty relay is described in Part C.



Connect these terminals to a plug.

Turn the starting device round.



These are the usual connections to the start winding. The lamp should light but after a few minutes it should go out.

These are the usual connections to the run winding. The lamp should light and stay alight.

How to test a PTC type starting device

1. Remove the starting device from the compressor (see pages 86 and 87).

2. Connect the terminals shown in the top drawing to an electric plug.

3. Turn the starting device into the position shown in the bottom drawings then plug into an electric socket. Be careful.

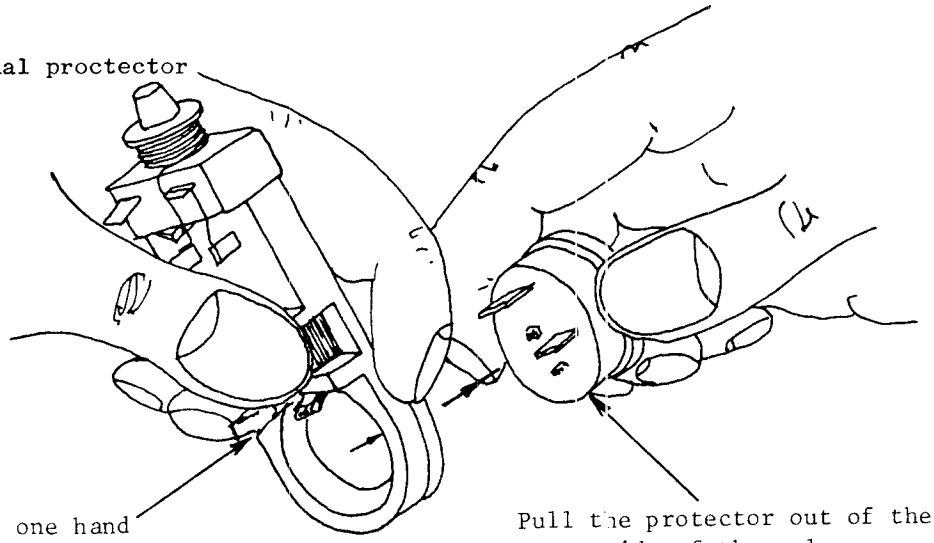
4. Fit a test lamp into the compressor pin sockets as shown in the bottom drawings.

5. Test the connection to the run winding. The lamp should light immediately and stay alight. If the lamp does not light the starting device is faulty.

6. Test the connection to the start winding. The lamp should light immediately but after a few minutes it should go out. (Using a 100-watt lamp and a 240-volt electrical supply the time is about 3 minutes). This shows how the PTC cuts off the electrical supply to the start winding when it gets hot (see page 73). If the lamp does not light or if the lamp lights and stays alight, the starting device is faulty.

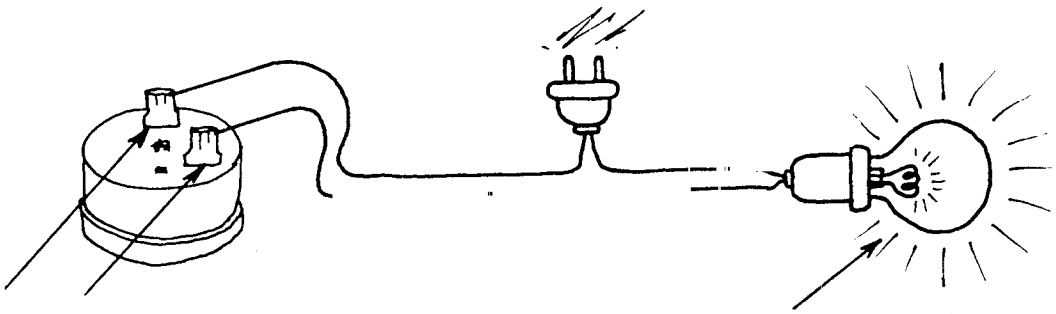
7. How to repair a faulty starting device is described in Part C.

External proctector



Hold the relay in one hand with the electrical connections facing the way shown.

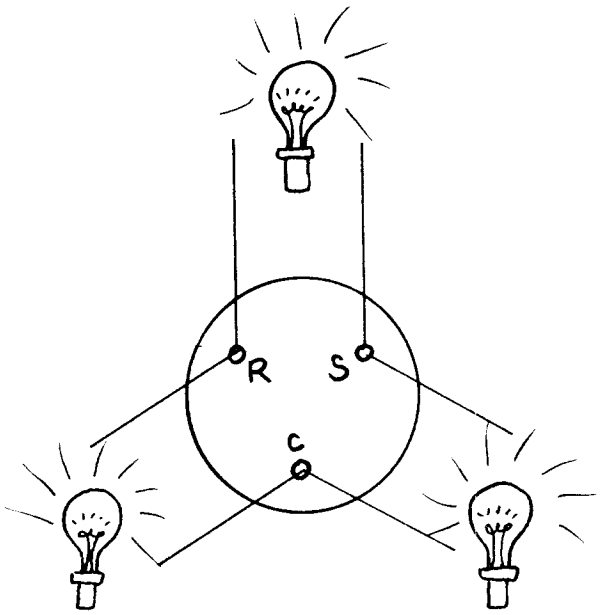
Pull the protector out of the other side of the relay.



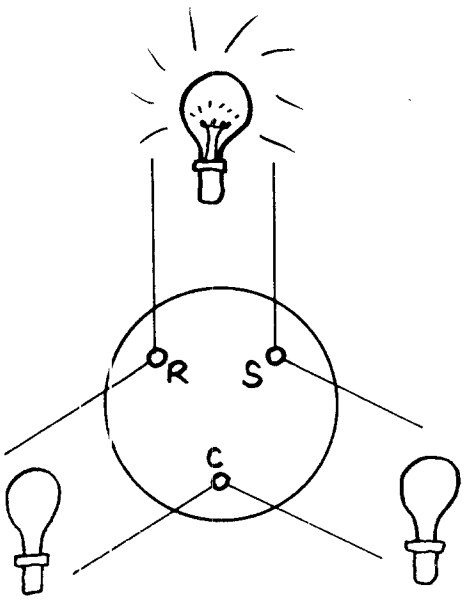
Connect the test lamp to these connections on the protector.

The test lamp should light. If it does not light, the protector is faulty.

Internal protector



Lamp always lights - protector is good



Lamp only light on 'run' to 'start' protector is faulty

How to test a motor protector

The motor protector is either external (filled outside the compressor) or internal (fitted inside the compressor). See part B page 22.

If it is an external protector

1. Disconnect the protector. If it is fixed to the relay assembly, remove it as shown in the top drawing. Take care not to damage the connections.

2. Connect the bare ends of a powered test lamp to the connections shown in the middle drawing. The lamp should light.

3. If the lamp does not light the protector is faulty.

4. If the protector is not faulty replace it carefully in the relay assembly.

If it is an internal protector

1. Identify the compressor motor terminals (pins). See pages 96 and 97. The protector is connected inside at the "common" terminal.

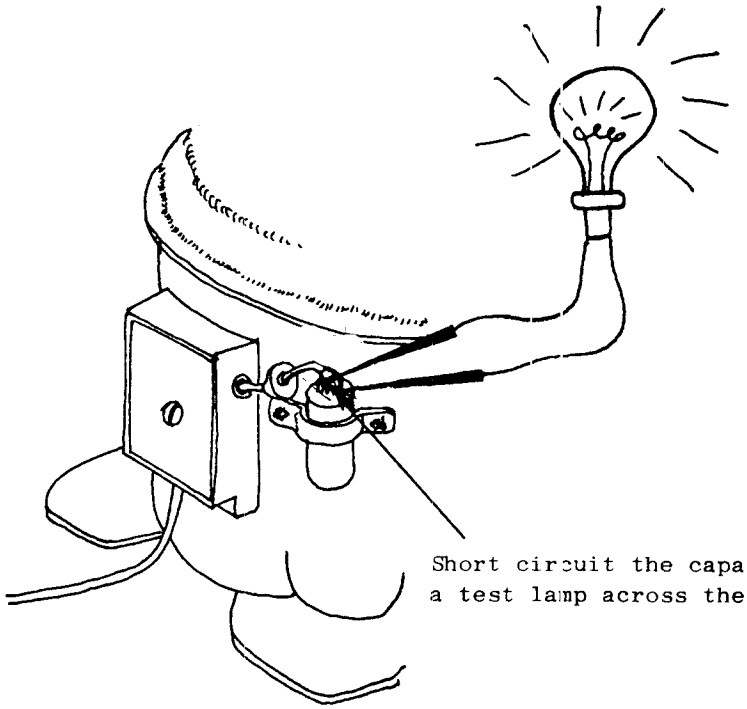
2. Test each pair of terminals in turn with a powered test lamp. See bottom drawing and Part B page 24. The lamp should light every time.

3. If the lamp only lights on "run-to-start"

   and does not light on "common-to-start"

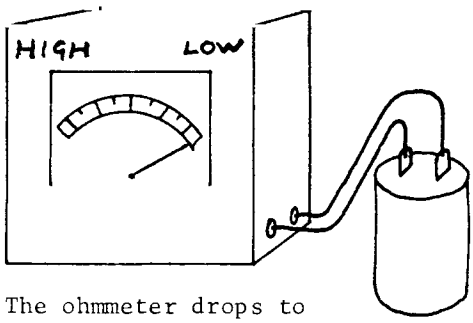
   and does not light on "common-to-run"

   The internal protector is faulty.

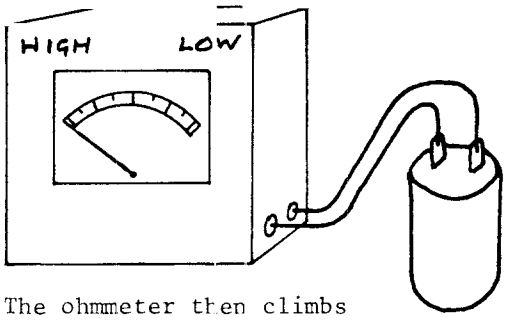


Short circuit the capacitor by placing a test lamp across the two terminals

These drawings show what happens when a capacitor is good:

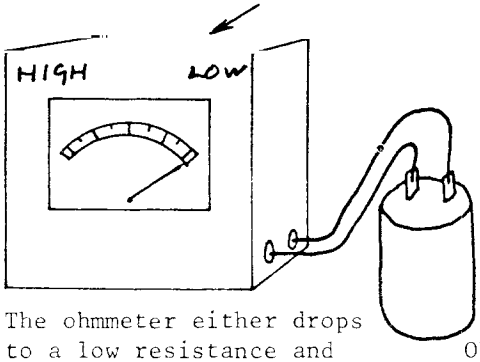


The ohmmeter drops to a low resistance.



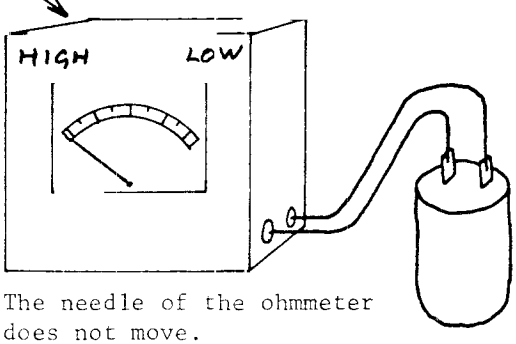
The ohmmeter then climbs to a high resistance.

These drawings show what happens when a capacitor is faulty:



The ohmmeter either drops to a low resistance and stays there.

OR



The needle of the ohmmeter does not move.

How to test a capacitor

1. Caution: Handle capacitors very carefully. They 'store' electricity and even though the refrigerator is not plugged into its socket if you touch the capacitor contacts you will probably get a severe electric shock.

2. Before removing a capacitor:

   (a) Unplug the refrigerator

   (b) 'Short circuit' the contacts with a simple test lamp. This will discharge the capacitor and make it safe to handle;

   (c) Examine the outside. If it is leaking or bulging outwards, handle it very carefully (wear goggles and place a shield around the outside). It may burst.

3. First test:

   (a) Read paragraphs 1 and 2 above, then remove the capacitor.
   (b) Examine the capacitor. If any of the following symptoms are present, replace the capacitor with a new one:

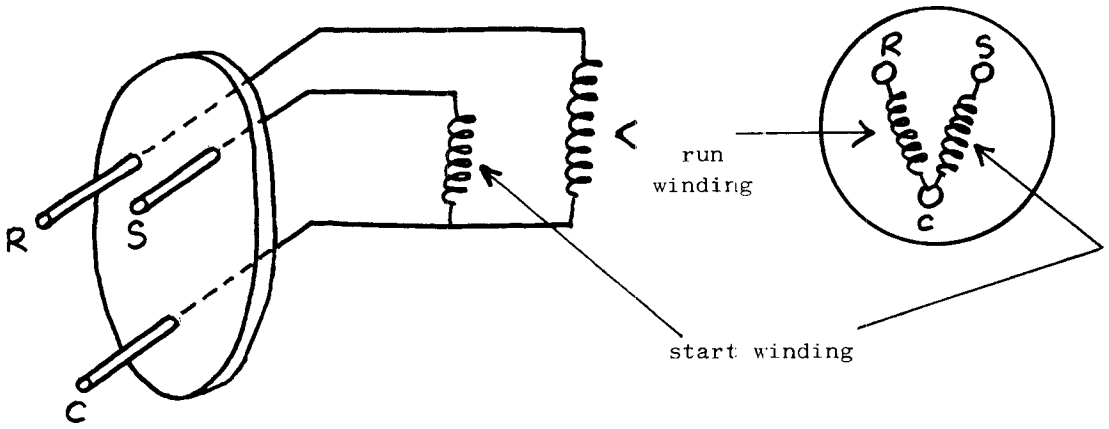
   Leakage of fluid - Plug or disc on top missing - Outside surface distorted (indicating pressure on the inside).

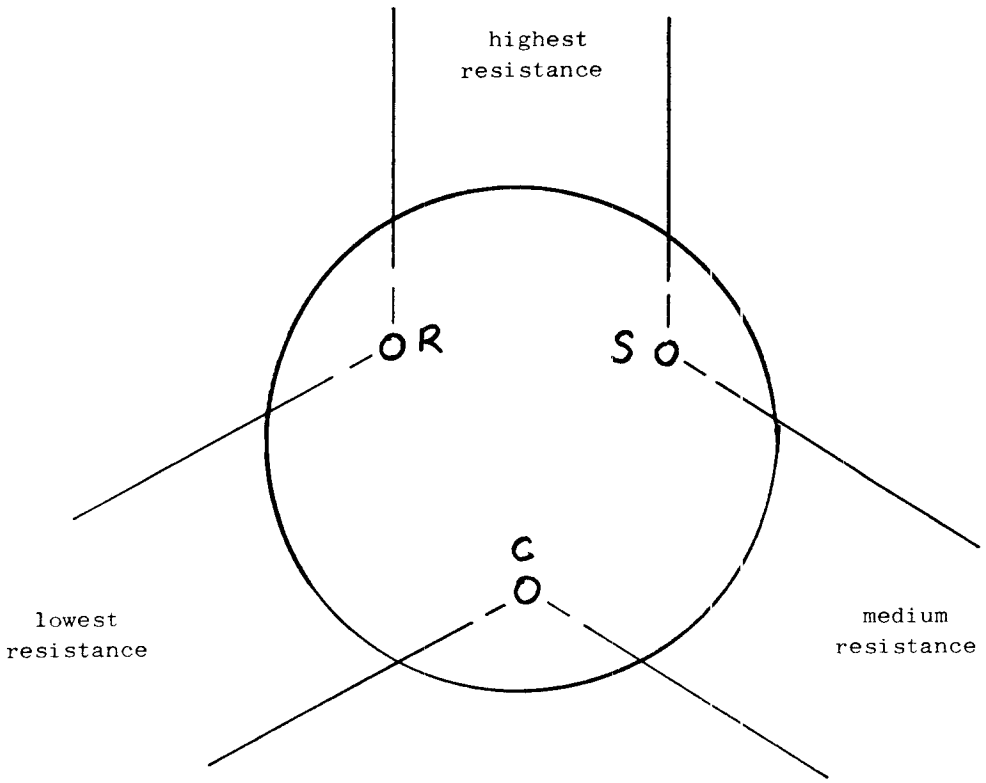
   (c) If the capacitor looks good, carry out the next test.

4. Second test:

   If you have an ohmmeter select the highest range and connect it as shown in the bottom drawings.

   (a) If the needle goes up to show a low resistance then comes back to show a high resistance, the capacitor is good.
   (b) If the needle either goes to a low resistance and stays there, or the needle does not move, the capacitor is faulty.


R
S
C
run
winding
start winding


highest
resistance
lowest
resistance
medium
resistance

How to identify the compressor motor terminals

1. The three terminals (pins) on the compressor are connected to the motor windings as shown in the top drawing.

   One terminal is connected to the run winding (R)

   One terminal is connected to the start winding (S)

   One terminal is connected to both windings; it is known as common (C).

2. The terminals may be in different positions on different compressors. If they are not marked, R, S and C, they can be identified as follows:

3. With an ohmmeter set at its lowest range, R x 1, measure the resistance between each pair of terminals in turn.

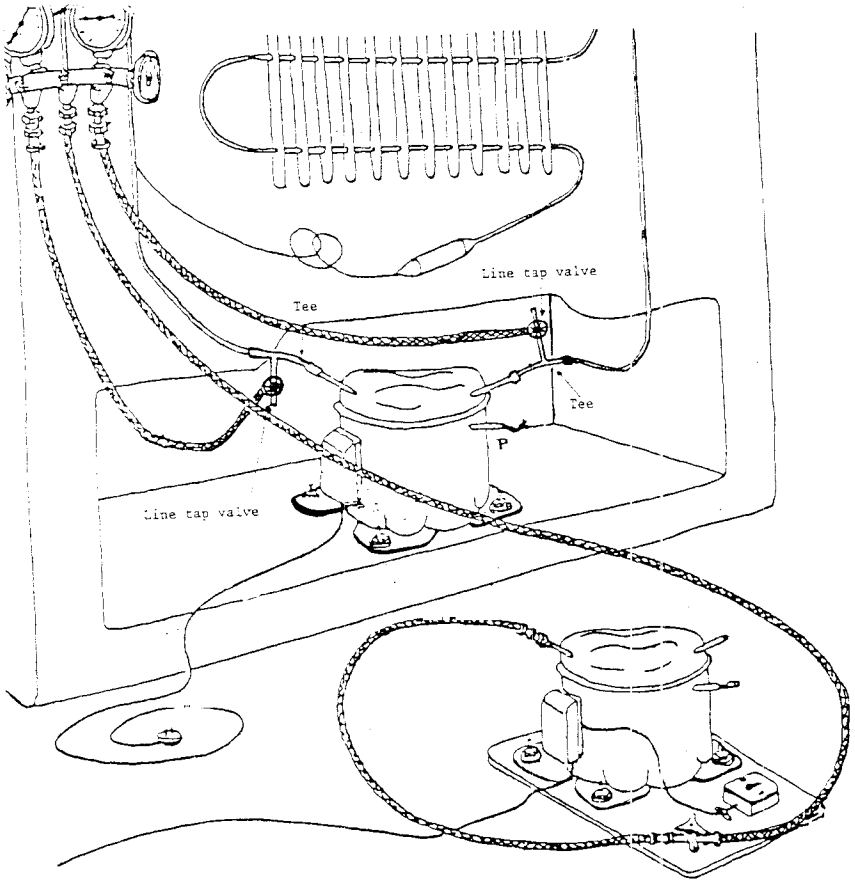
4. Identify the terminals as follows (see bottom drawing):

   The lowest resistance is between common and run;

   The medium resistance is between common and start;

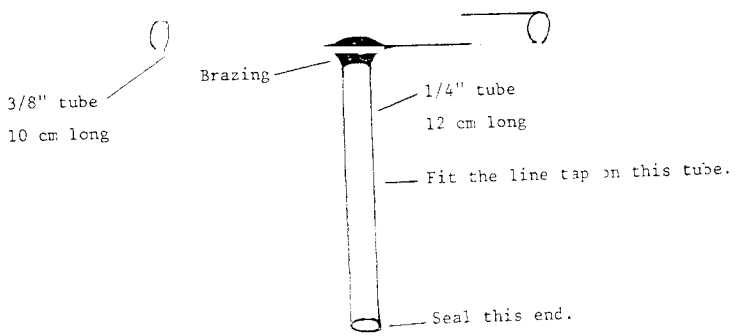
   The highest resistance is between start and run; it is the total of the lowest and medium resistances.

USING TEE TUBES IN THE REFRIGERATOR SYSTEM



The purpose of this annex is to demonstrate how to fit TEE tubes on the suction and delivery tube. This enable the technician to remove and reuse the line valves again.

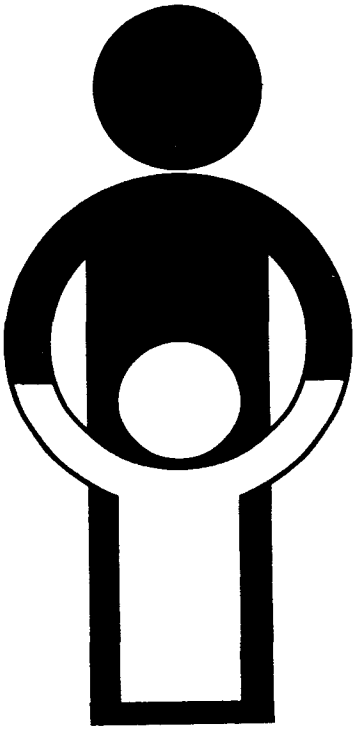
This can be done by either using a standard TEE 1/4 OD copper TEE or by forming one as shown below.



Approximate conversion tables

Bar	Pounds per square inch	Kilograms per square centimetre
-	in HG	Kg/cm^2 abs
BAR	PSI	Kg/cm^2
		Kp/cm^2
-1	30 inches mercury	-1
-0.5	15 inches mercury	-0.5
0	0	0
0.5	7	0.5
1	14	1
1.5	21 (Normal low pressure)	1.5
2	28	2
2.5	35	2.5
3	43	3
3.5	50	3.5
4	58	4
5	72	5
6	87	6
7	102	7
8	116	8
9	131	9
10	140	10
15	220	15
20	292	20
25	362	25

Abbreviations



WORLD HEALTH ORGANIZATION ORGANISATION MONDIALE DE LA SANTE

1211 GENEVA 27-SWITZERLAND Telegr.: UNISANTE-GENEVA Telex: 27821 OMS 1211 GENEVE 27-SUISSE Télégr.: UNISANTE-GENEVE